

汽车电子与控制
Automotive Electronics and Control Systems (AECS)

李建秋、徐梁飞

清华大学车辆与运载学院

汽车安全与节能国家重点实验室

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第三章：传感器、执行器及基本接口电路
Sensors, actuators and basic interface circuits



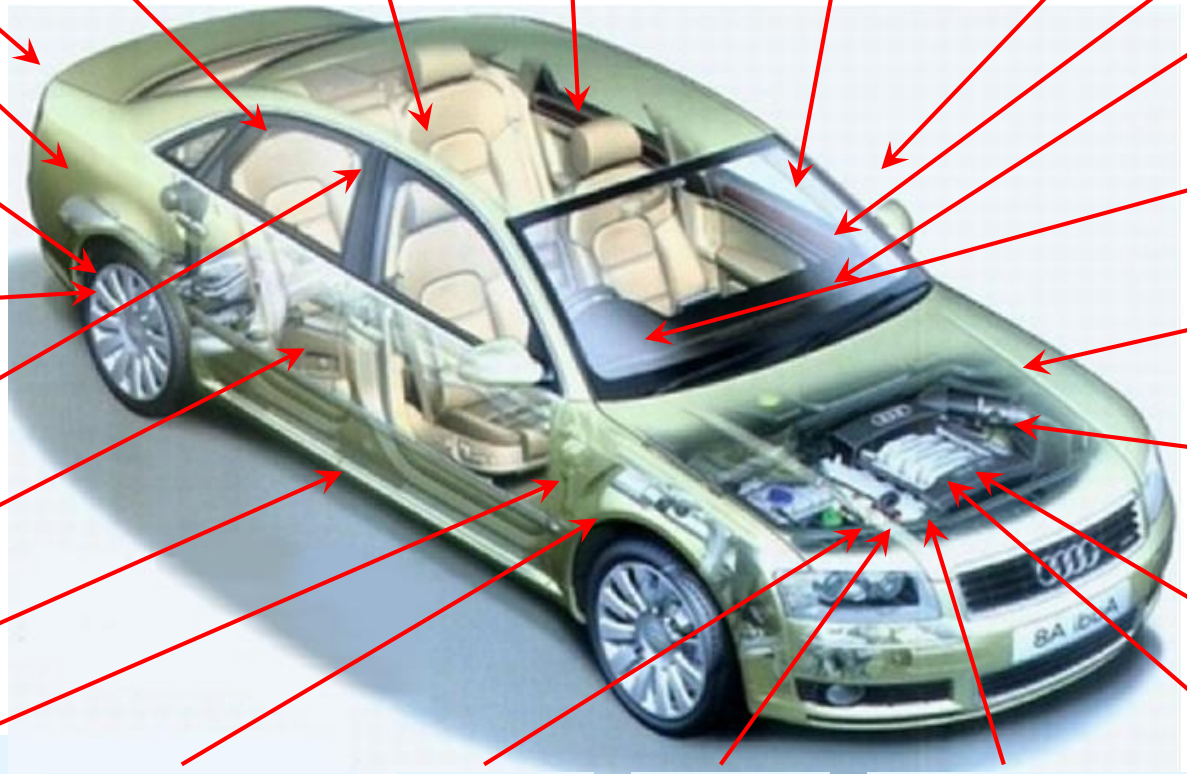
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1. 传感器、信号接口电路及运算放大器
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   - ② 传感信号接口电路 (Interface circuits of sensor signals)
   - ③ 典型运算放大器 (Typical Operational Amplifier)
2. 执行器、开关器件及驱动接口电路
   - ① 执行器及四种功率开关器件
   - ② 驱动信号接口电路：四拓扑结构
   - ③ 驱动电路的续流回路
   - ④ 驱动电路的故障诊断和容错控制
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重点：运算放大器(Operational Amplifier)和MOSFET的使用

§1 传感器工作原理/The Principle of Sensors in AECs





PDC Ultrasound

Passenger Comp. Protection, Ultrasound

Seat

Over roll Sensor

Rain Sensor, Optical

Mirror Position

Suspension Resistive

Fuel Level Potentiometer

Airbags μ Mechanics

Type Pressure μ mechanics

Steering Angle Position

ABS Inductive

HFM

Anti Pinch Protection

Level Sensor Hall

Power Window Hall

Temperature, Inside, Outside, Water, Air, Oil

Lambda Sensor λ

Accelerator Pedal Position, Hall

Brake Pedal Position, Hall

Camshaft, Hall

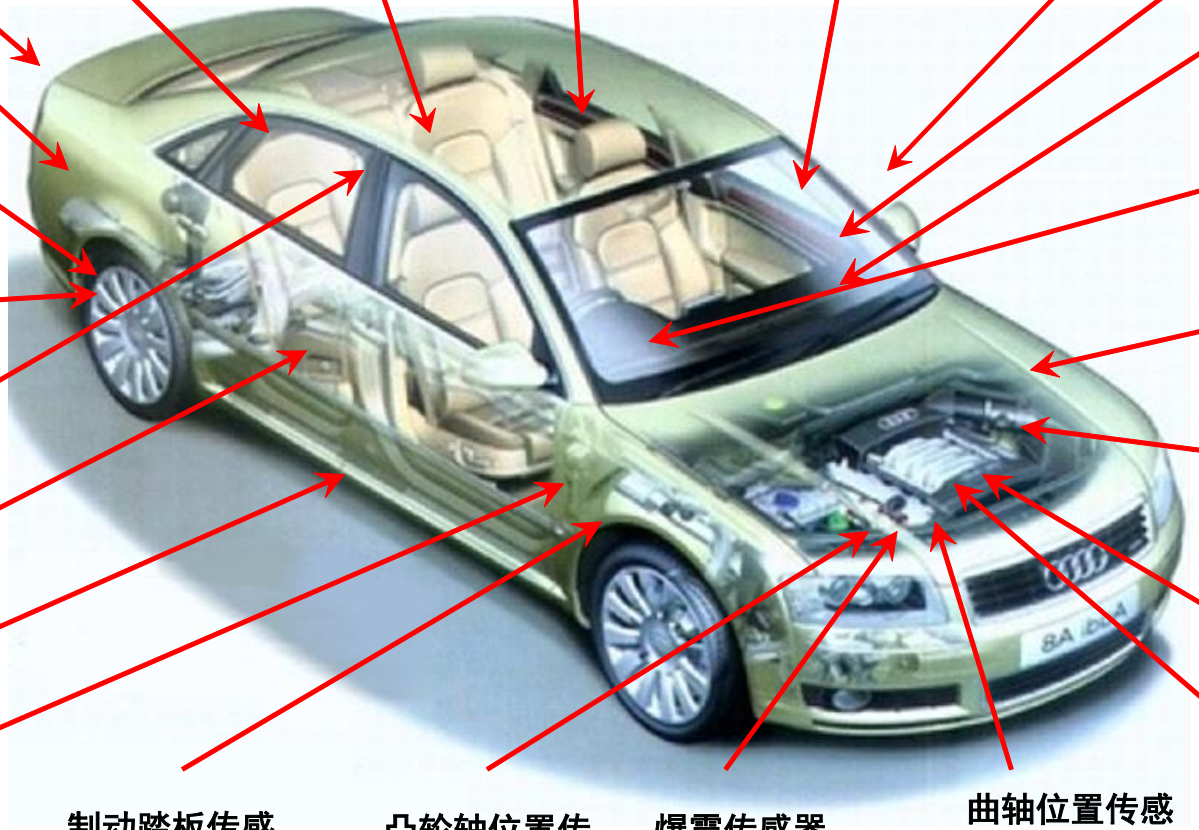
Knock Sensor

Crankshaft, Inductive, Hall

Pressure Sensor- Intake Manifold



§1 传感器工作原理/The Principle of Sensors in AECs





§1 传感器工作原理/The Principle of Sensors in AECs



Crankshaft Speed Sensor



Camshaft Speed Sensor



Pressure Sensor



Mass Airflow Sensor



Crankshaft & Wheel Speed Sensor



Gear-box Speed Sensor



Knock Sensor

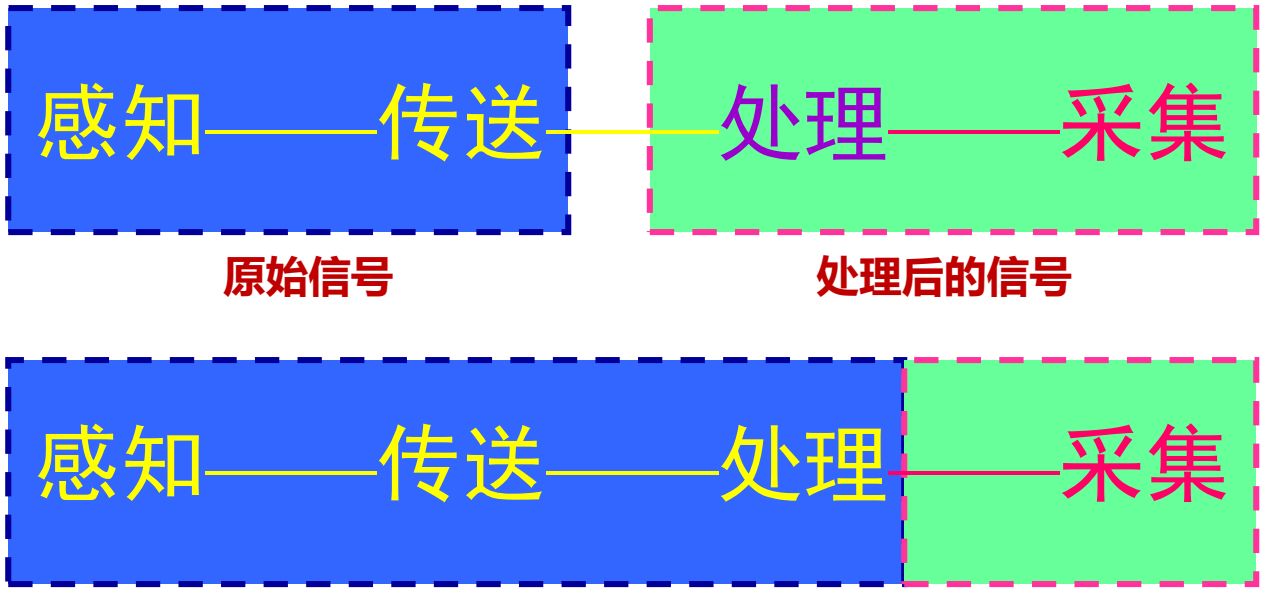


Coolant Temperature Sensor



§1 传感器工作原理/The Principle of Sensors in AECs

- **传感器信号输出/The output signal of sensor**
  - **直接输出原始信号 / Direction outputs original signals**
  - **处理后的信号/ Outputs unified signals**





● 传感器分类/Sensor classification

- **被测物理量**：位置/角度、速度/角速度、力/压力、加速度、气体浓度、流量、光强、温度
- **测量原理**：电磁、霍尔效应、微机电系统（MEMS）、科里奥利力、电化学、光的干涉/反射
- **转化电量**：**电压**、电流、电阻、电容/电感

感知 | 转化 | 传送 | 处理

结构 | 原理 | 冗余 | 诊断

学习目的

第一阶段：**理解**传感器原理

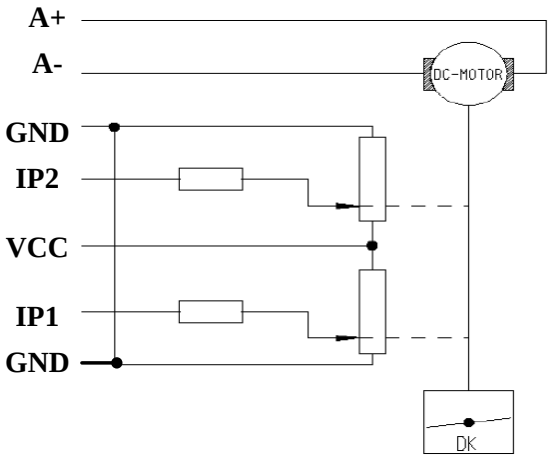
第二阶段：**设计**新型传感器

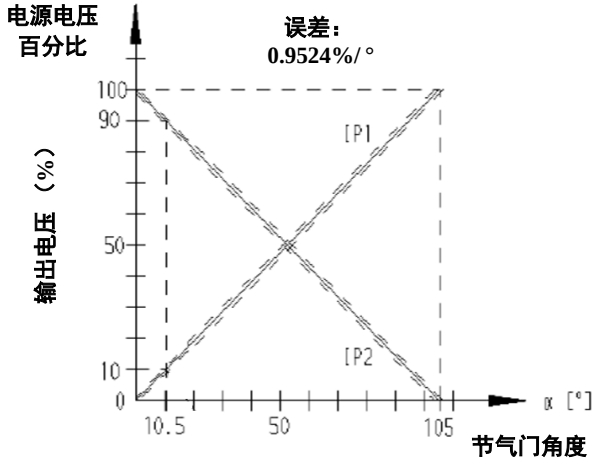


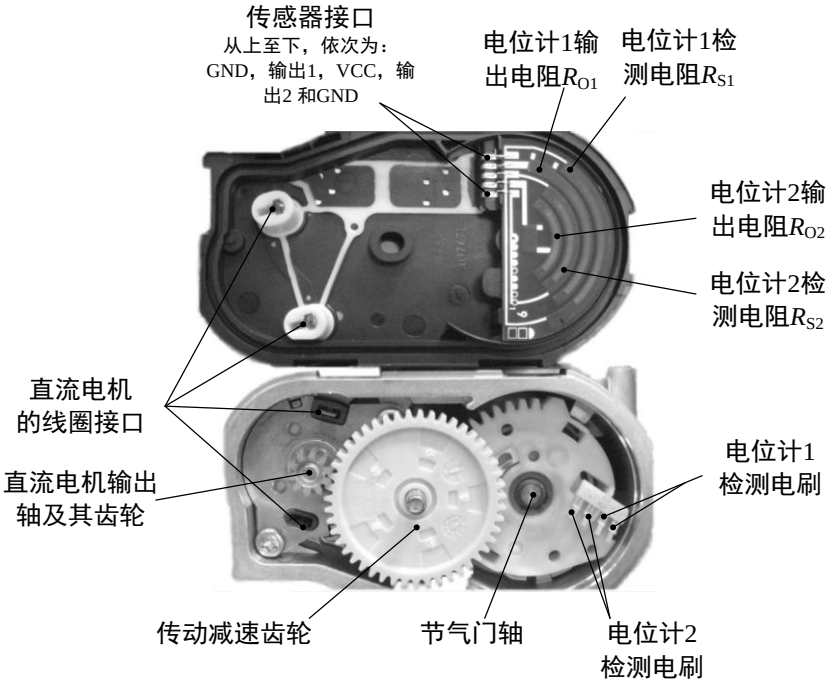
§1 传感器工作原理/The Principle of Sensors in AECs

● 位置/角度传感器/Linear or Angular Position Sensor

◆电位计/Potentiometer





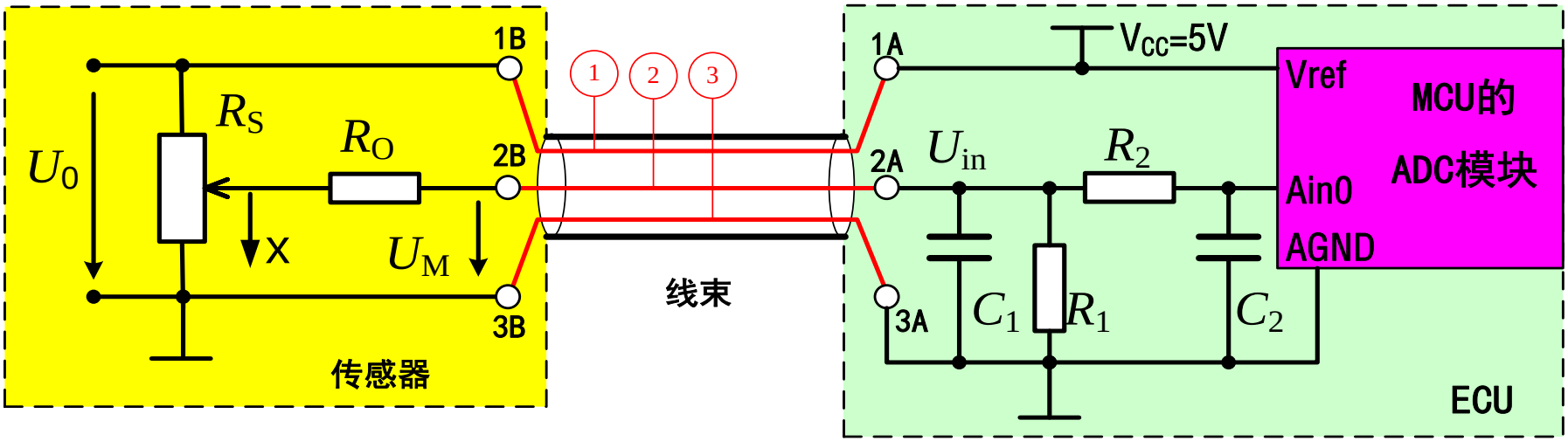


$$\begin{cases} v_1 = v + \varepsilon_1 \\ v_2 = v_{max} - v + \varepsilon_2 \\ v_{mes} = \dfrac{v_{max} - v_2 + v_1}{2} = v + \dfrac{\varepsilon_1 - \varepsilon_2}{2} \end{cases}$$



§1 传感器工作原理/The Principle of Sensors in AECs

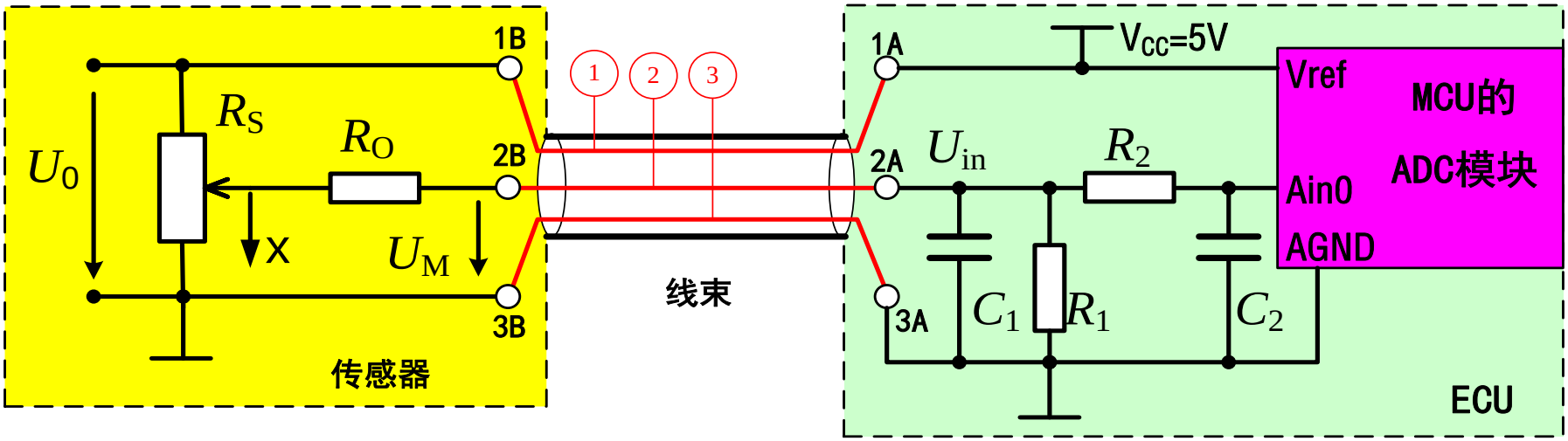
- 位置/角度传感器/Linear or Angular Position Sensor
  - 传感器的信号处理和故障诊断/Signal process and diagnosis
  - 节气门位置/Mechanical angle:10.5°~94.5°
  - 相应的输出/Output voltage:10%~90% of Vcc
  - 即对应为[0.5V~4.5V]之间/ The voltage =[0.5V to 4.5V];
  - 输出带有0.5V的诊断区域/ Output Safety Margin is 0.5V



结构 原理 冗余 诊断

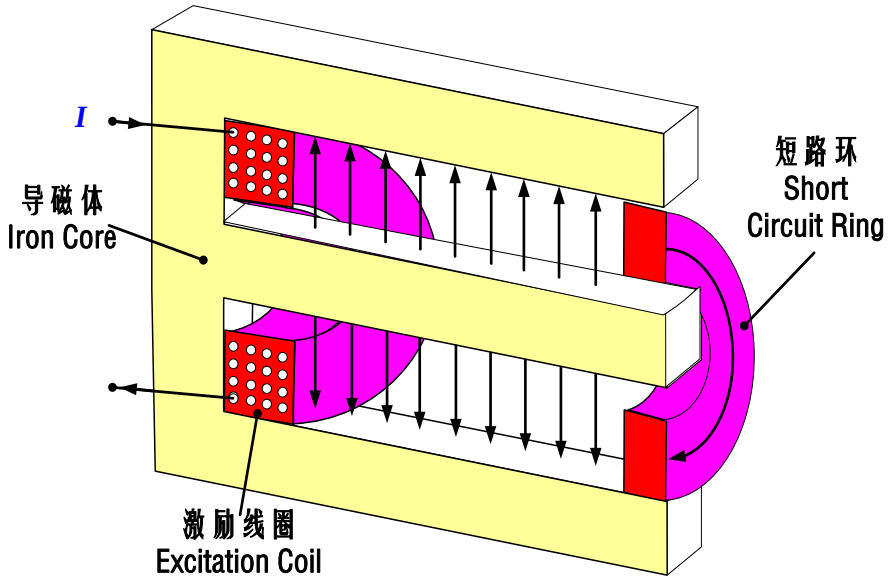
- 位置/角度传感器/Linear or Angular Position Sensor
  - 如果电源线1A和1B断路/ If power supply line open circuit: U_{in}=0；
  - 如果信号输入2A和2B断路/If signal line broken: U_{in}=0；
  - 如果2A对电源短路/If signal line short circuit to V_{CC}： U_{in}=V_{CC}；
  - 如果2A对地短路/ If signal line short circuit to GND： U_{in}=0；
  - 如果地线3A和3B断路/ If GND line broken：

$$U_{in}(\min)=\frac{R_1}{R_S(\max)+R_O+R_1}\cdot Vcc>4.5V$$



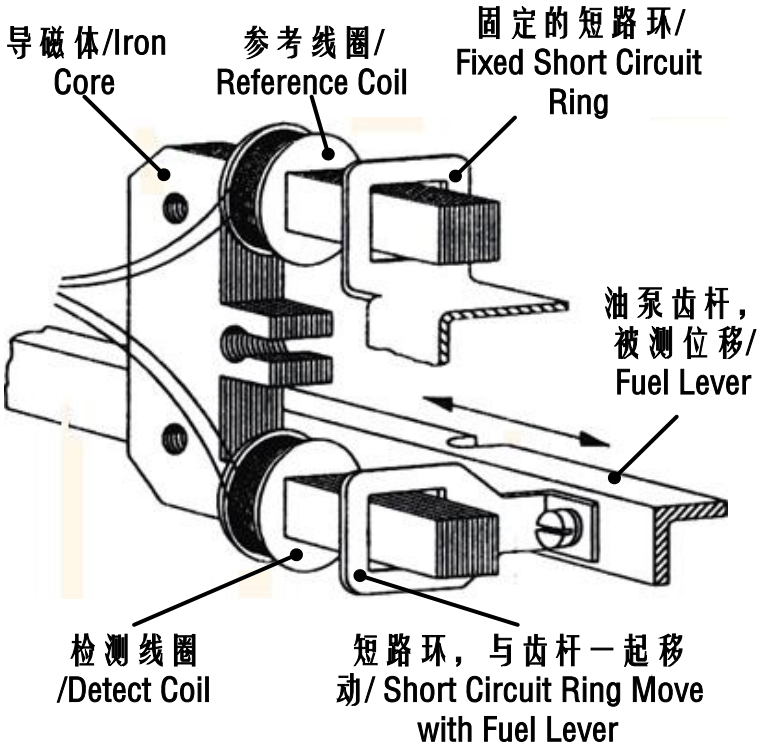
● 位置传感器/Position sensor

◆电磁感应



电磁感性的短路环测量原理

磁场被约束在导磁体和短路环之间

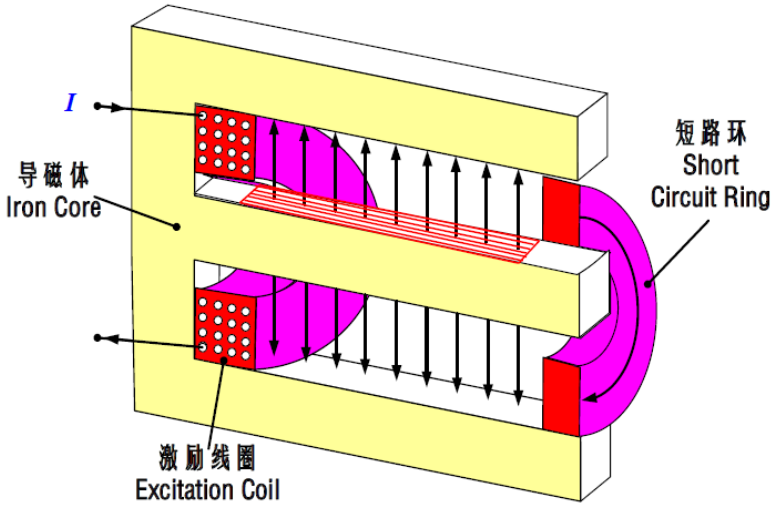


柴油机齿条位置反馈传感器
(德国BOSCH)

磁电式传感器原理：在变化的磁场中放置垂直磁场方向的线圈，线圈将产生感应电动势



§1 传感器工作原理/The Principle of Sensors in AECs



电流 $I(t)=I_0\sin(\omega t)$

磁动势 $MNF=NI_0\sin(\omega t)$

磁通量 $\Phi=\dfrac{MNF}{R_m}=\dfrac{NI_0\sin(\omega t)}{R_m}$

磁链 $\Psi=N\Phi$

磁感应强度 $B=\dfrac{\Phi}{S}$

磁阻 $R_m=\dfrac{k}{S_{磁截}}$

基本定义

感应电动势

$$E=\frac{d\Psi}{dt}=N\frac{d\Phi}{dt}=\frac{N^2I_0\omega\cos(\omega t)}{R_m}$$

与磁路有关，导磁体内近似0，其他通道与介质特性（正比）、磁路途径长度（正比）、横截面积（反比）相关

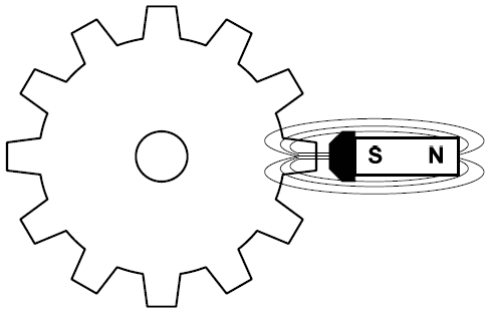
感应电动势 $E=\dfrac{d\Psi}{dt}=N\dfrac{d\Phi}{dt}=\dfrac{N^2I_0\omega\cos(\omega t)}{R_m}\propto S_{气隙}$

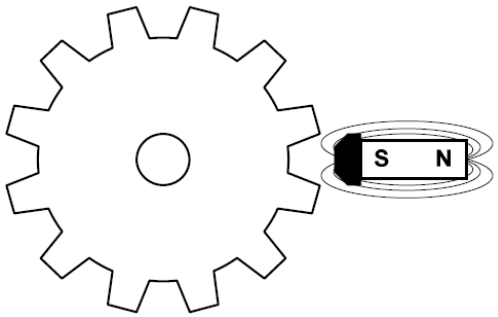


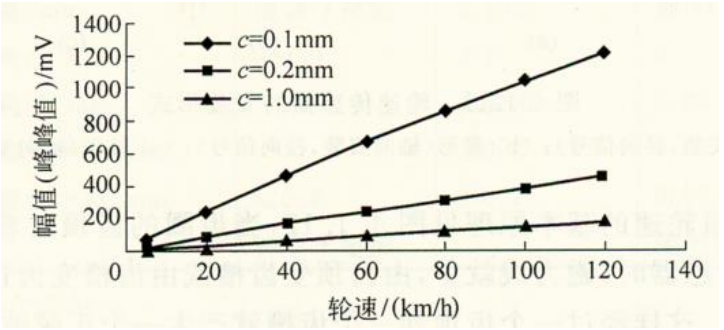
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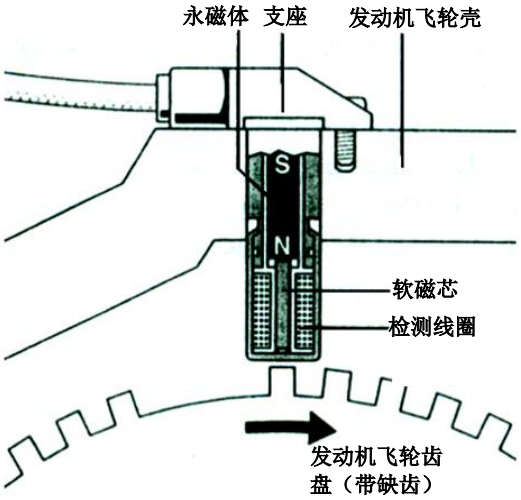
● 角速度传感器/Angular Speed Sensor

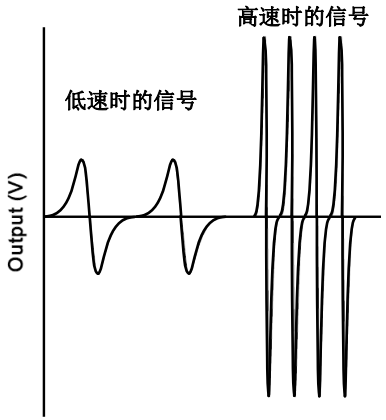
◆ 磁电式/ Magnetic type











$$n \approx \frac{60}{Z\Delta t}$$

旋转速度n(rpm, r.min^{-1})
沿圆周均匀分布的齿数Z
每个齿的间隔时间Δt(s)

$$U_M = -N\frac{d\Phi}{dt}$$

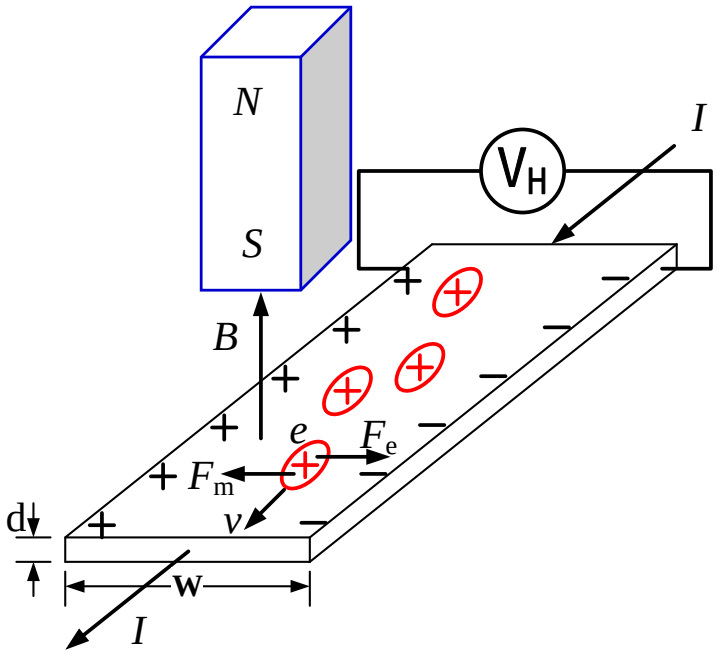
$$U_{Mavg} = -N\frac{\Delta\Phi}{\Delta t} = -\frac{NZ\Delta\Phi}{60} n \propto n$$

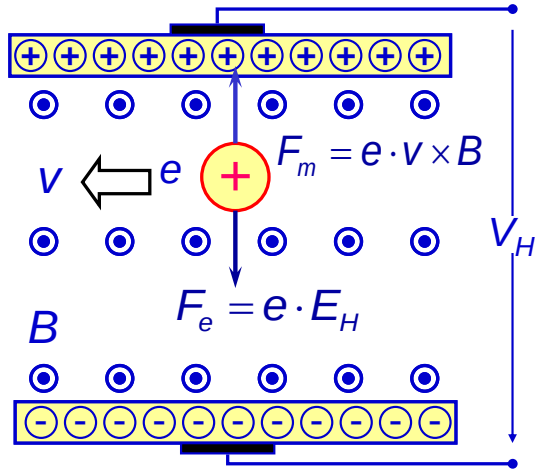
转过一个齿在线圈中感应出来的平均电压

磁电式传感器原理：在变化的磁场中放置垂直磁场方向的线圈，线圈将产生感应电动势

- 位置/角度传感器/Linear or Angular Position Sensor
  - ◆霍尔式/ Hall effective sensor

$$V_H = R_H \cdot I \cdot B \cdot d^{-1}$$

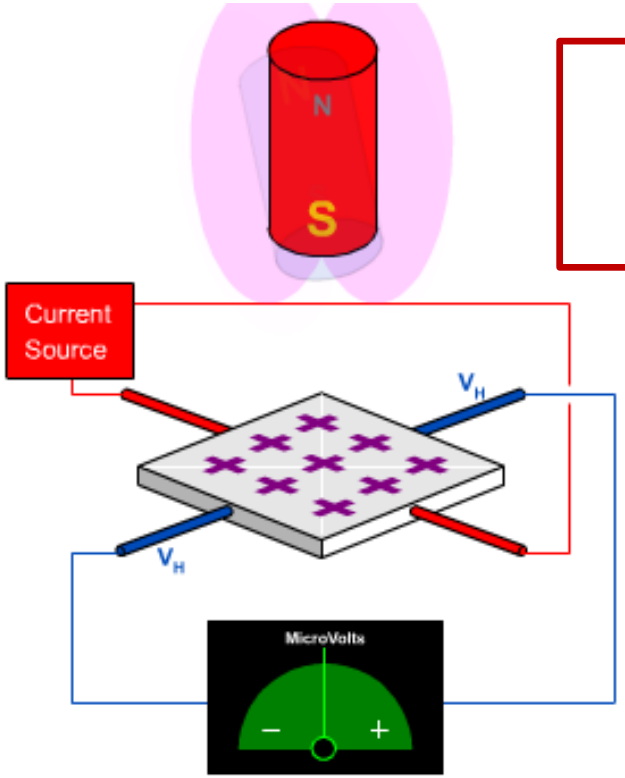




输出V_H取决于材料常数R_H、材料厚度d，激励电流I和电磁场强度B
/ The output voltage depends on material constant R_H, thickness d, the input current I and flux strength B



- 位置/角度传感器/Linear or Angular Position Sensor
  - ◆霍尔式/ Hall effective sensor



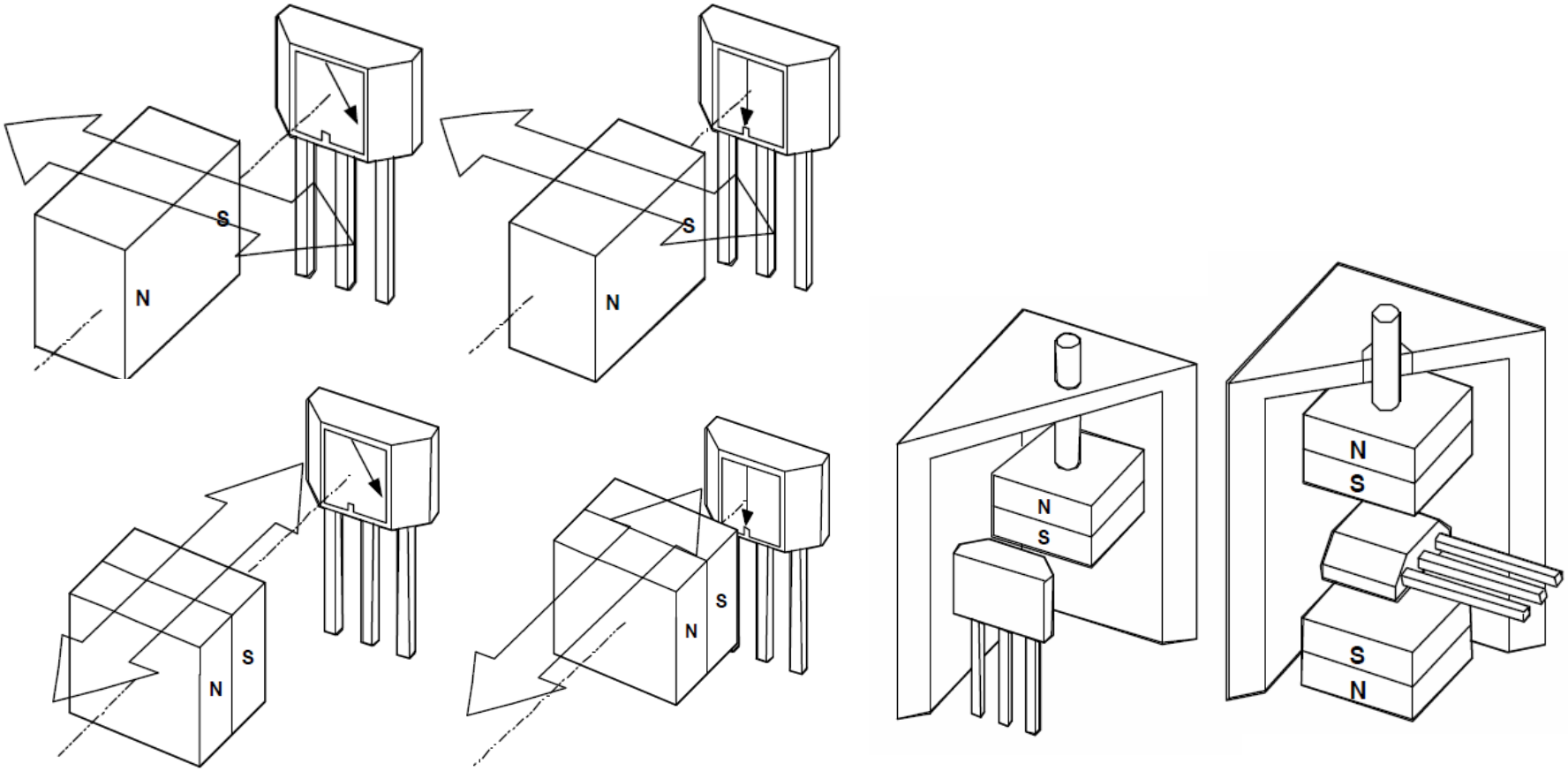
电荷受力符合左手定律
手心：正面穿过磁力线
手指：电流方向
大拇指：正电荷受力方向

霍尔原理：在垂直于磁场方向施加电流，则会在垂直于磁场、电流方向产生感应电压，该电压与磁场强度、电流大小成正比



§1 传感器工作原理/The Principle of Sensors in AECs

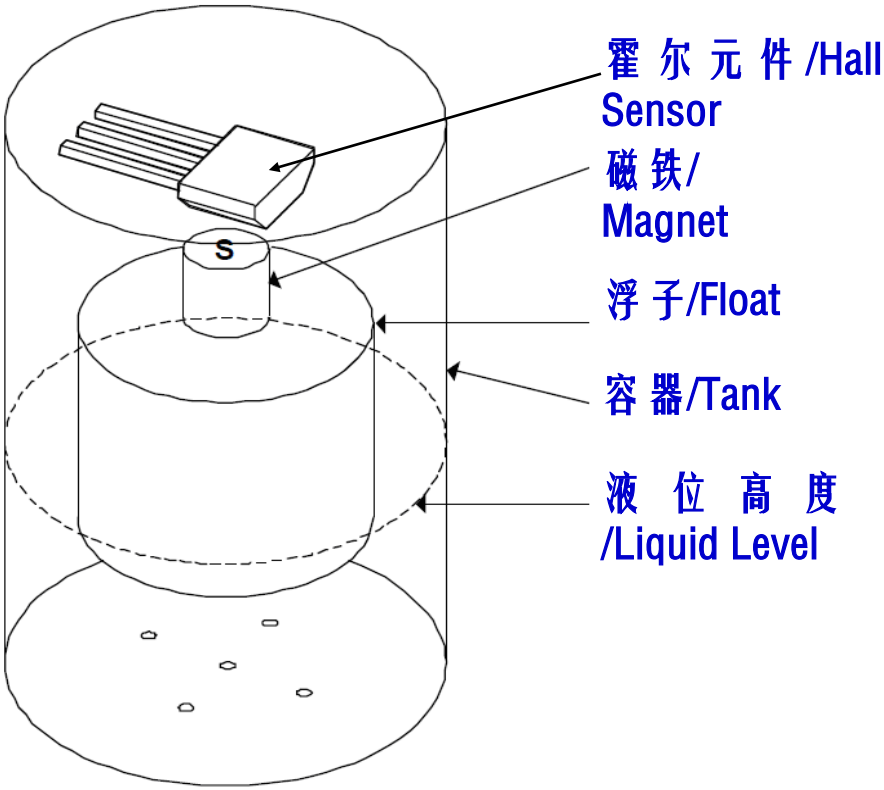
- 位置/角度传感器/Linear or Angular Position Sensor
  - ◆开关式/连续式霍尔传感器/ On/off type Hall sensor

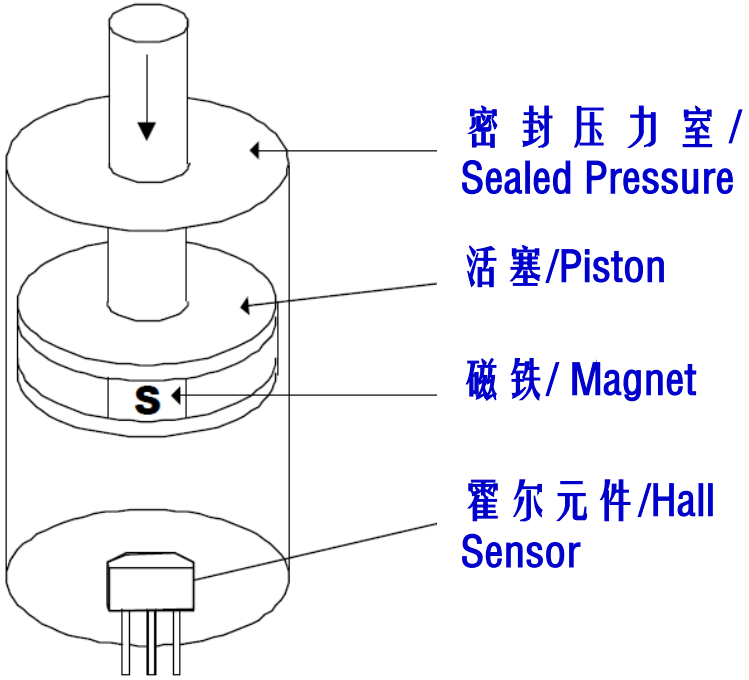




§1 传感器工作原理/The Principle of Sensors in AECs

- 位置/角度传感器/Linear or Angular Position Sensor
  - ◆连续式霍尔传感器/ On off type Hall sensor

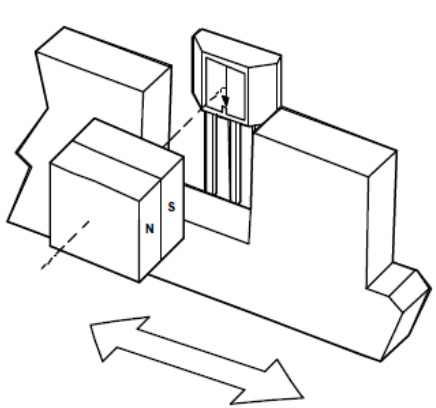


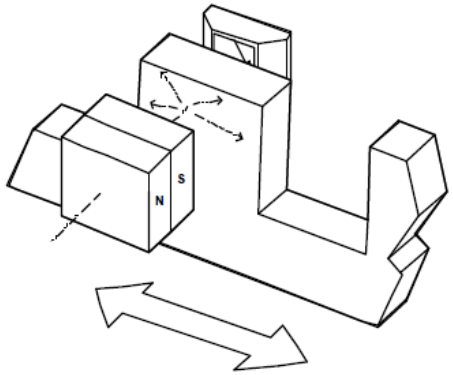


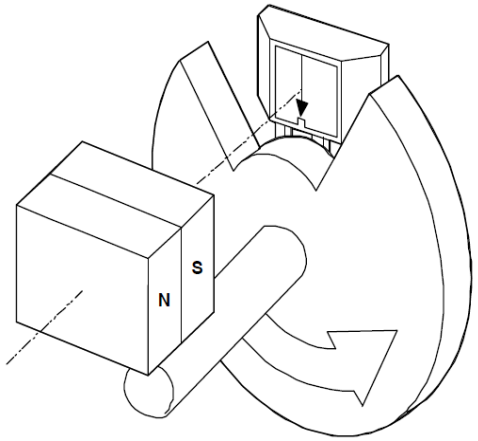


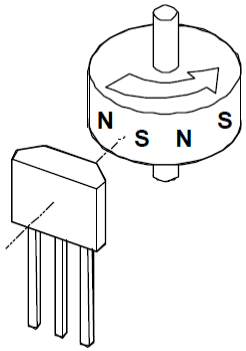
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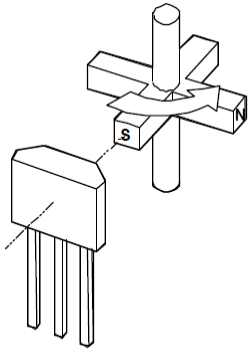
- 位置/角度传感器/Linear or Angular Position Sensor
  - ◆开关式霍尔传感器/ On off type Hall sensor

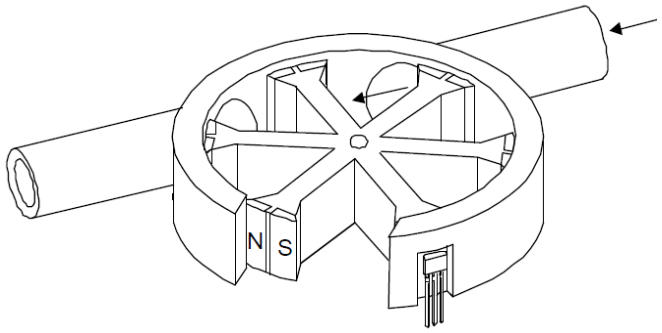








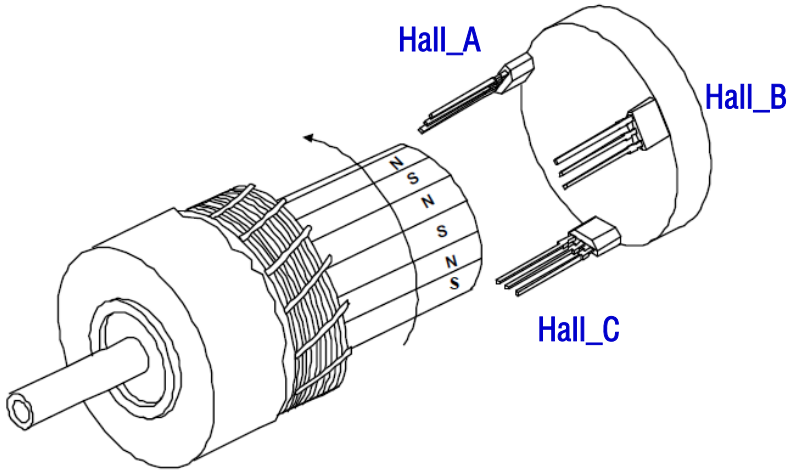


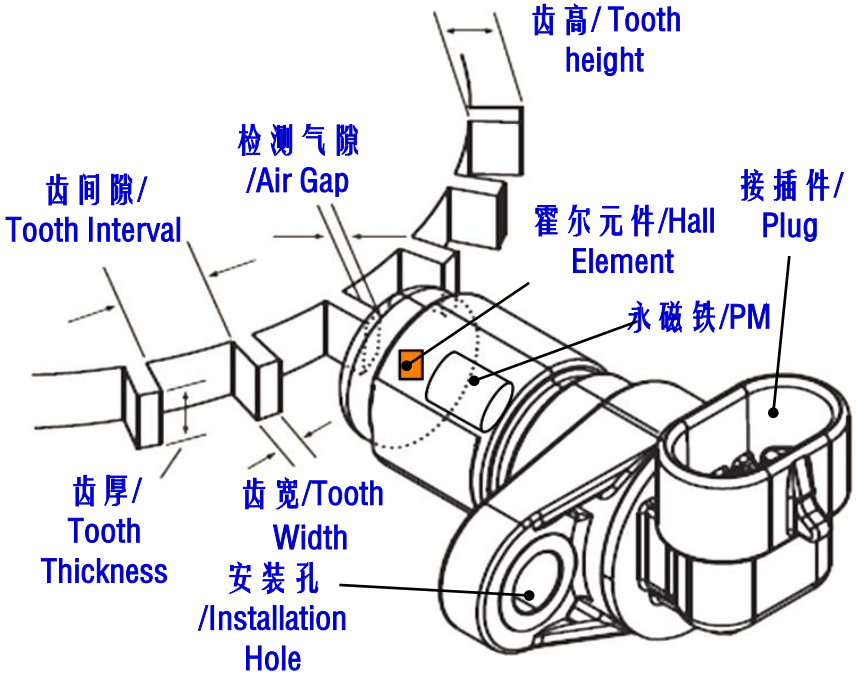




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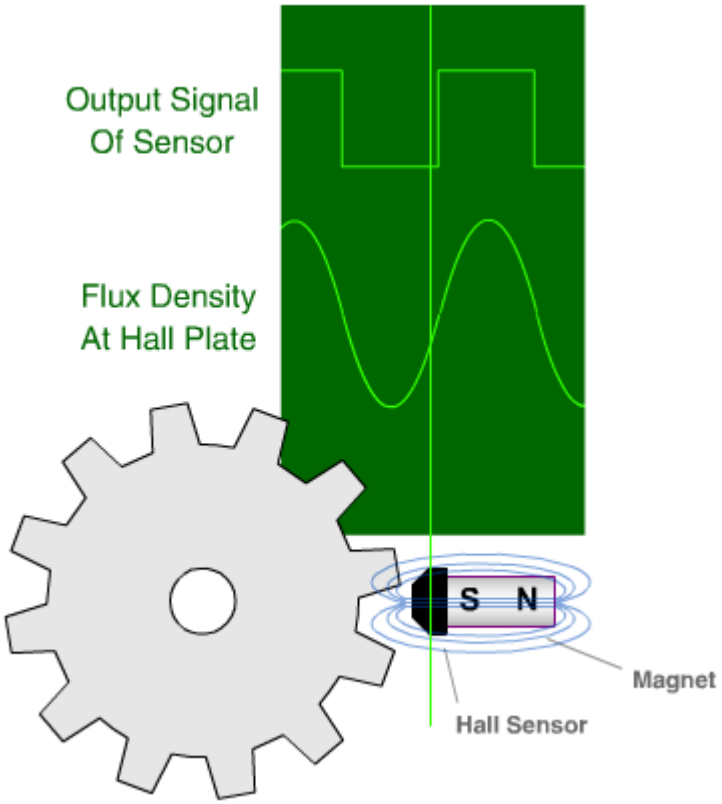


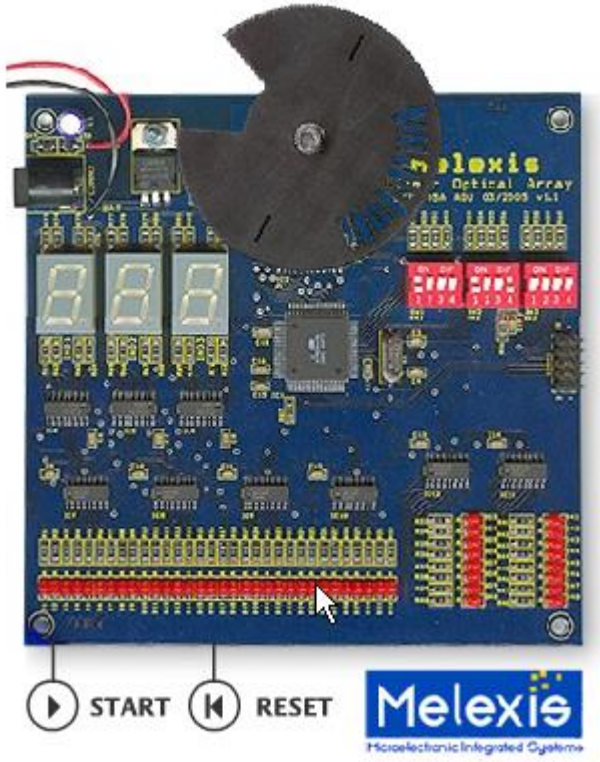




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- 位置/角度传感器/Linear or Angular Position Sensor
  - ◆开关式霍尔传感器/ On off type Hall sensor



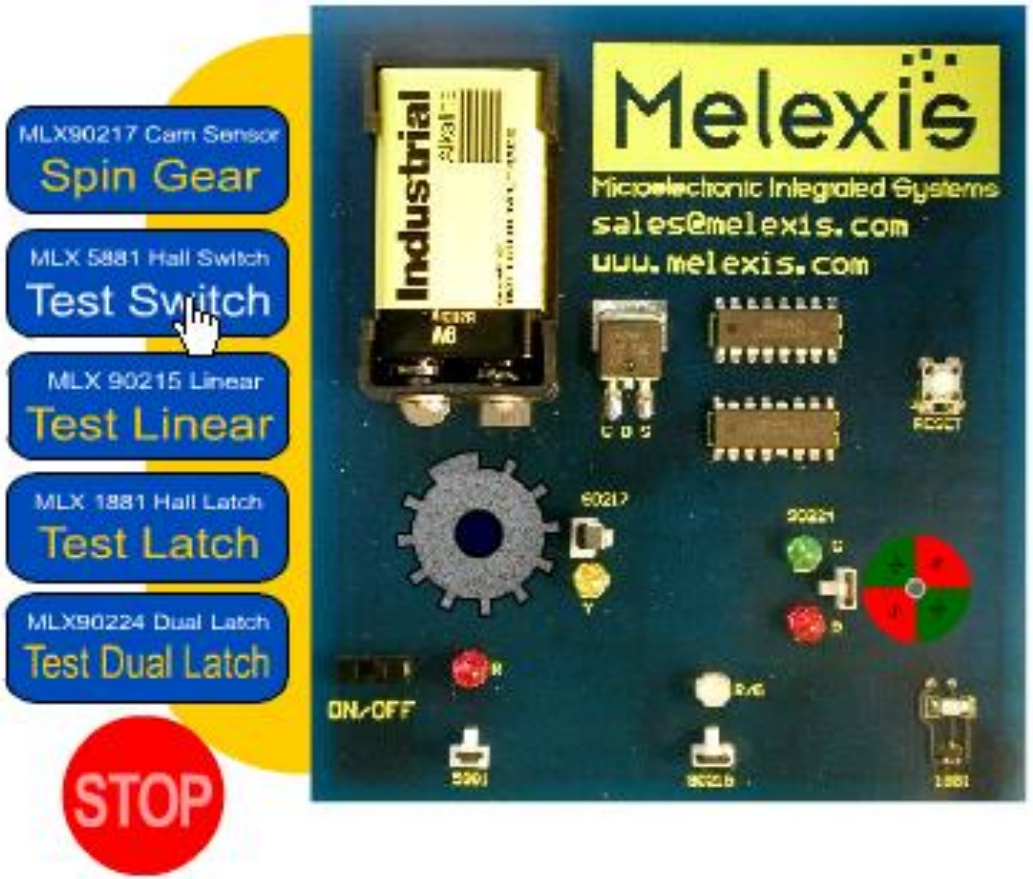


德国Melexis公司



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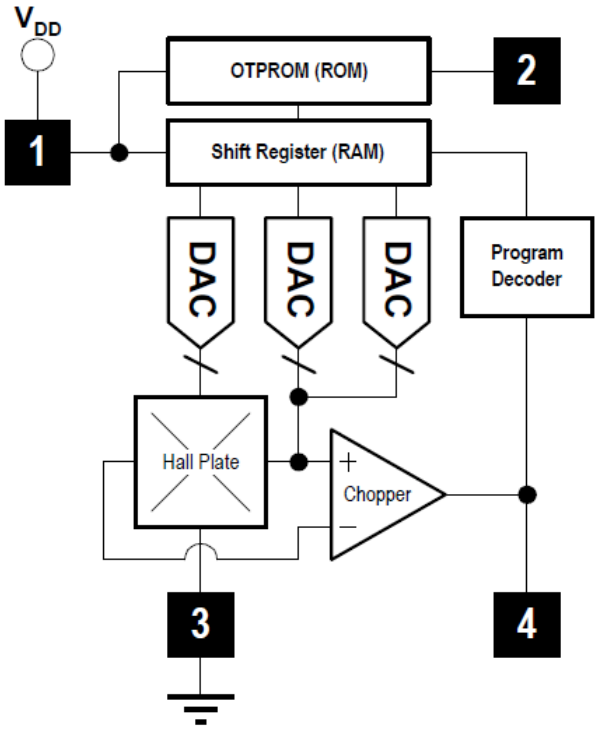
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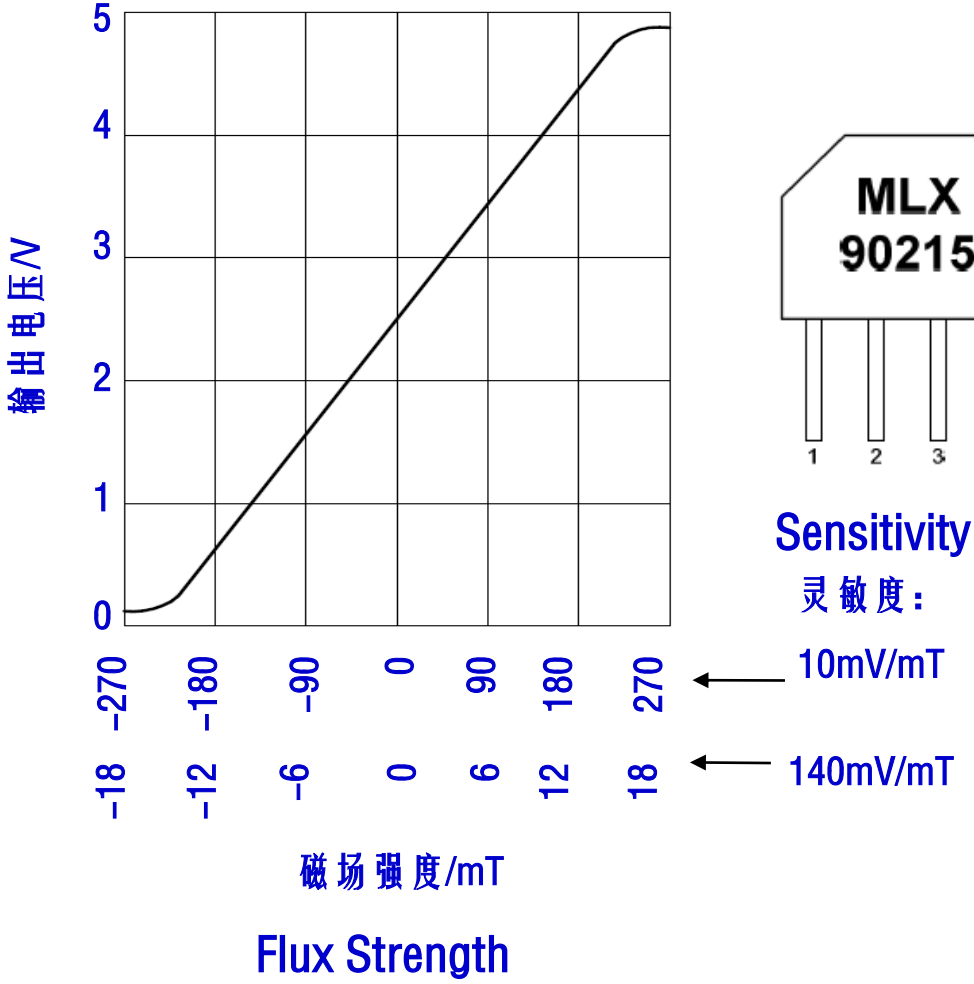


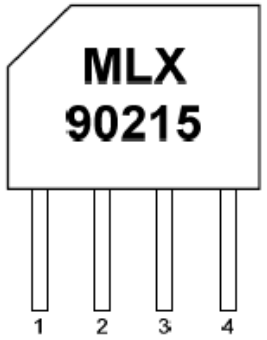


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- 位置/角度传感器/Linear or Angular Position Sensor
  - ◆连续输出的线性霍尔传感器/ Linear Hall sensor







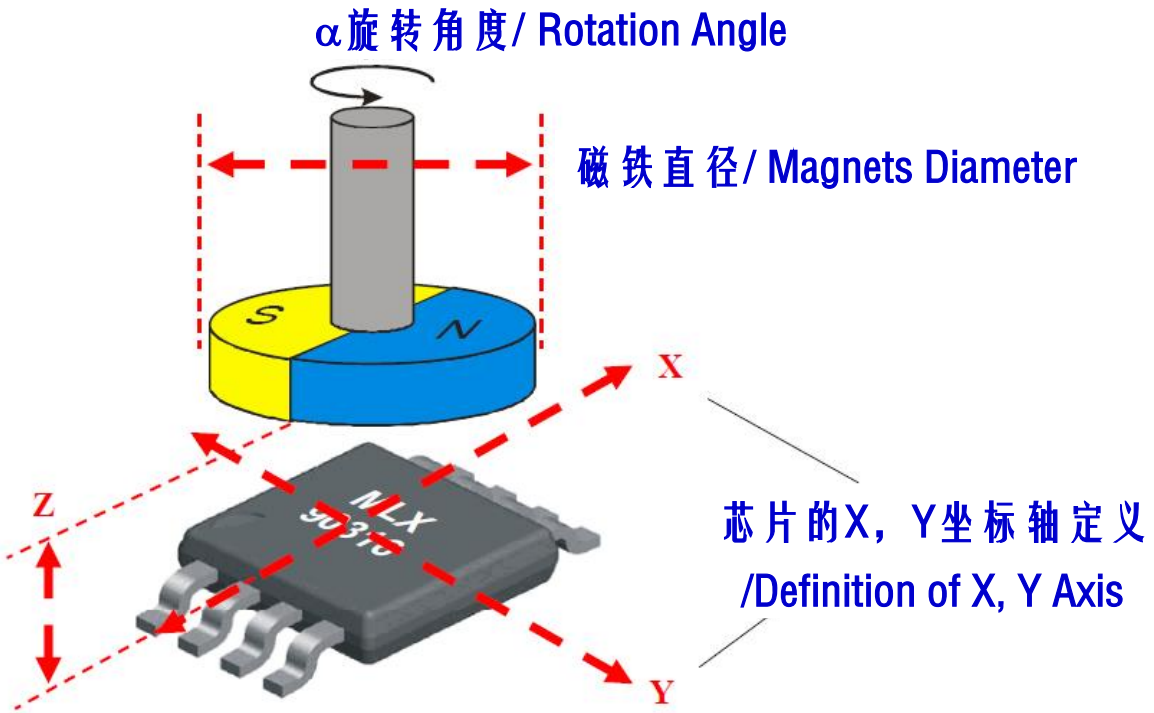
Sensitivity

灵敏度：

- 位置/角度传感器/Linear or Angular Position Sensor
  - ◆角位移传感器/Angular position sensor MLX90316

Melexis
Microelectronic Integrated Systems

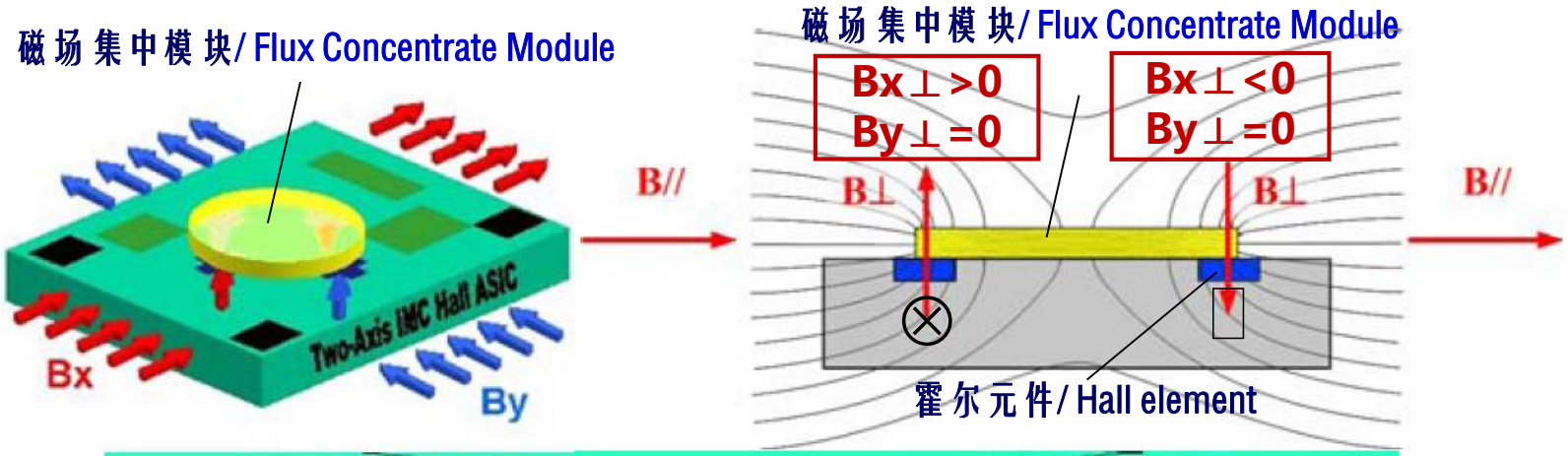
TriaXis™

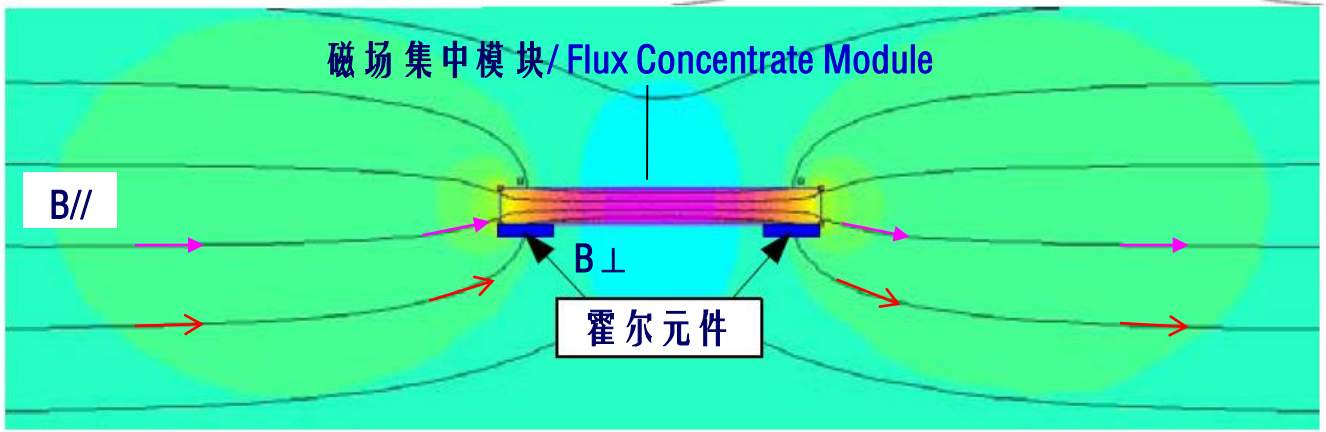




§1 传感器工作原理/The Principle of Sensors in AECs

- 位置/角度传感器/Linear or Angular Position Sensor
  - ◆角位移传感器/Angular position sensor MLX90316



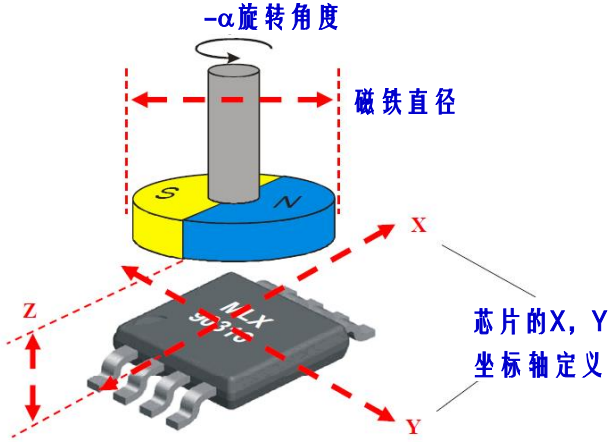


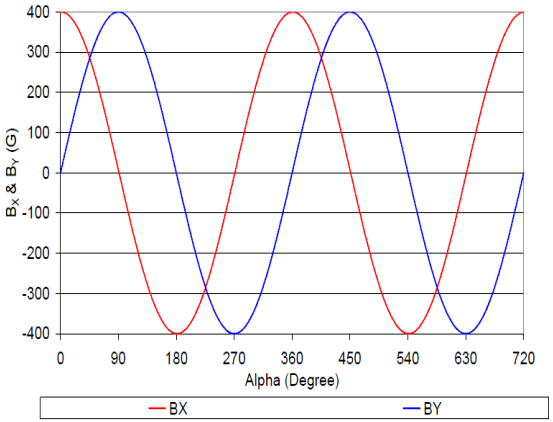
在平行于磁场的集中模块中线方向，B⊥=B//=Bx or By;
在垂直于磁场的集中模块中线方向， B⊥=0

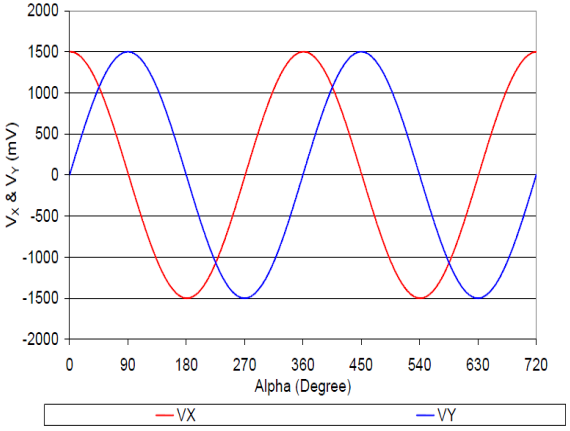
● 位置/角度传感器/Linear or Angular Position Sensor

◆角位移传感器/Angular position sensor MLX90316

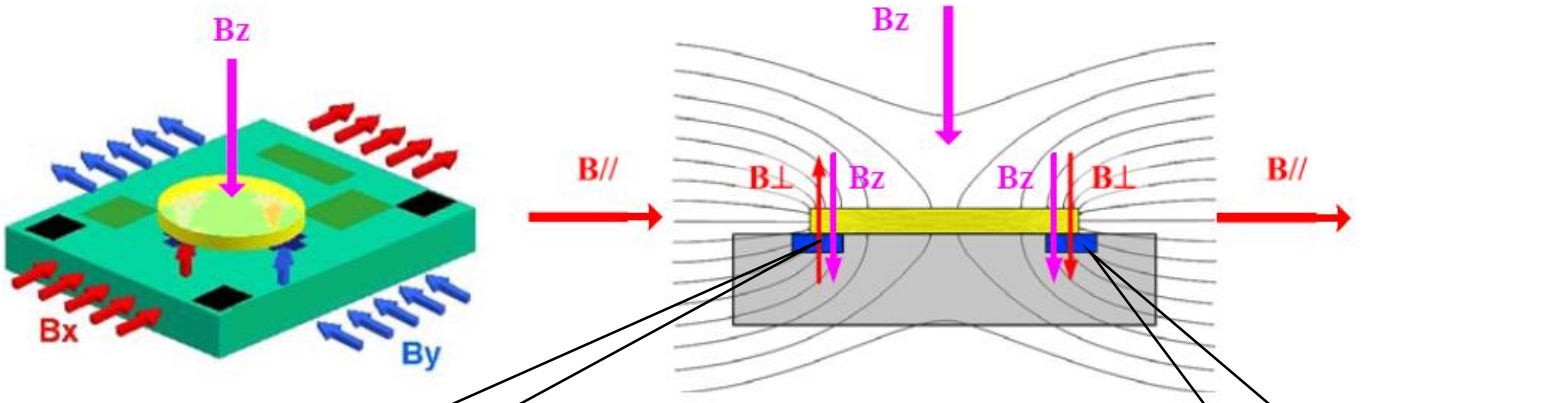
$$V_x \propto B_x \propto \cos(\alpha),\quad V_y \propto B_y \propto \sin(\alpha),\quad \Rightarrow \alpha = \arctan(\frac{V_y}{V_x})$$







- 位置/角度传感器/Linear or Angular Position Sensor
  - ◆三坐标磁场矢量芯片/3D Flux Strength Sensor MLX90333



左/Left:

$B_{//}=B_{\perp}$

$B_L=B_{\perp}-B_Z$

右/Right:

$B_{//}=B_{\perp}$

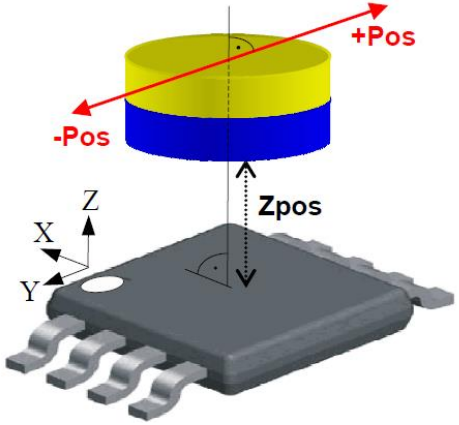
$B_R=-B_{\perp}-B_Z$

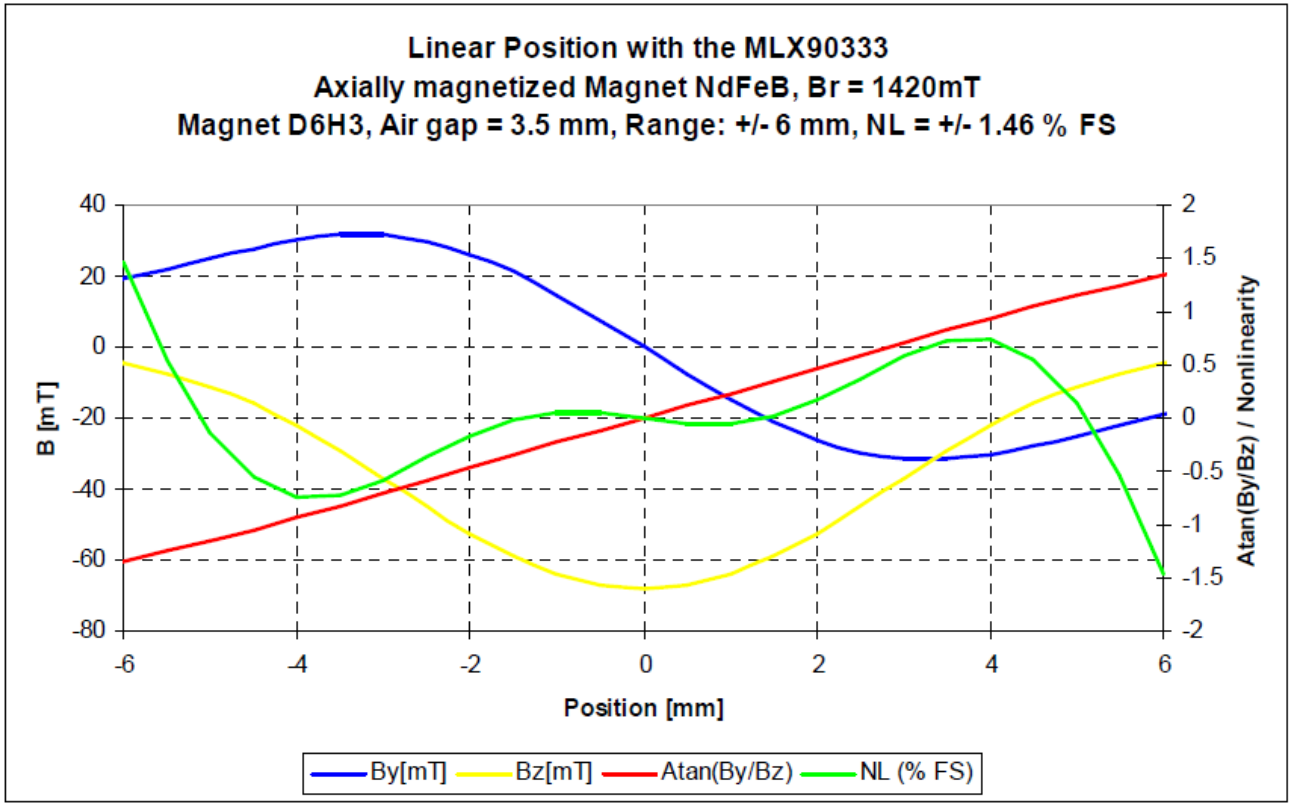
$$B_x=B_{//}=B_{\perp}=(B_L-B_R)/2$$
$$B_z=-(B_R+B_L)/2$$



● 位置/角度传感器/Linear or Angular Position Sensor

◆三坐标磁场矢量芯片/3D Flux Strength Sesnor MLX90333

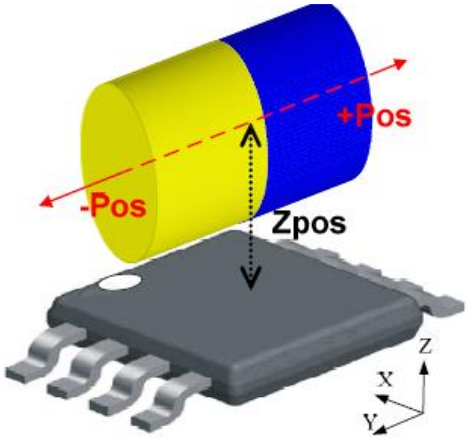


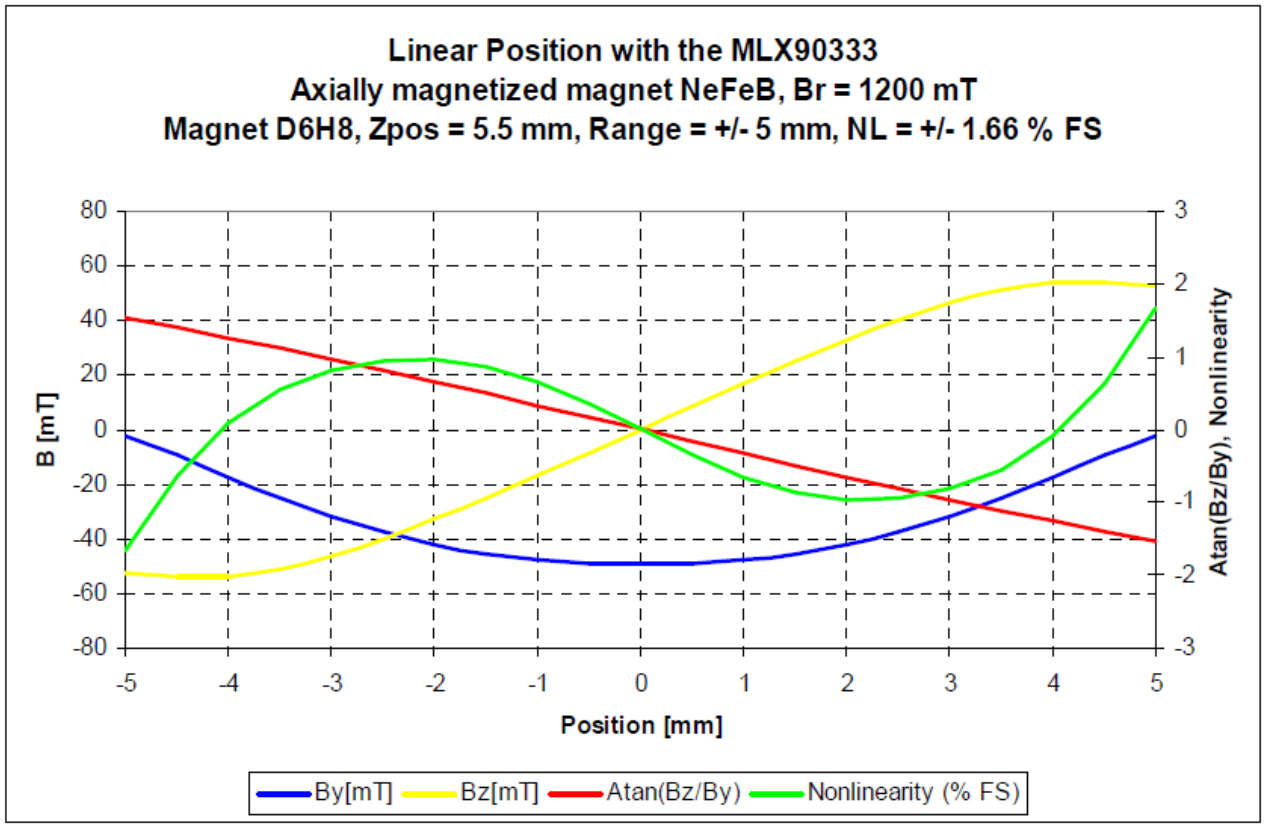




● 位置/角度传感器/Linear or Angular Position Sensor

◆三坐标磁场矢量芯片/3D Flux Strength Sesnor MLX90333

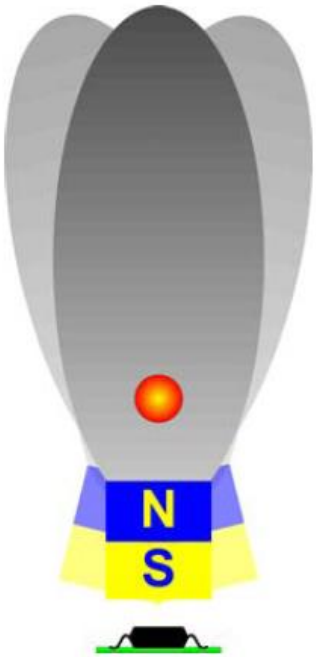






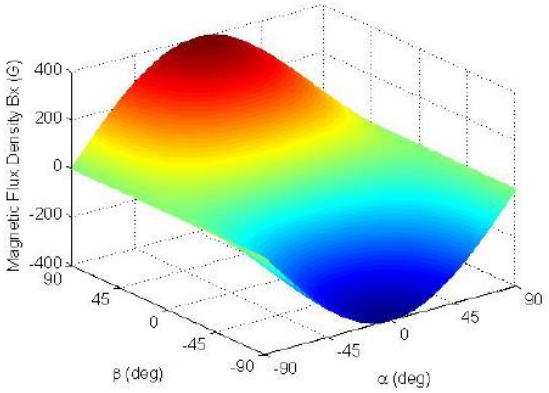
- 位置/角度传感器/Linear or Angular Position Sensor
  - ◆三坐标磁场矢量芯片/3D Flux Strength Sesnor MLX90333
  - ◆ 3D游戏杆位置传感器/Triaxis 3D–Joystick Position Sensor

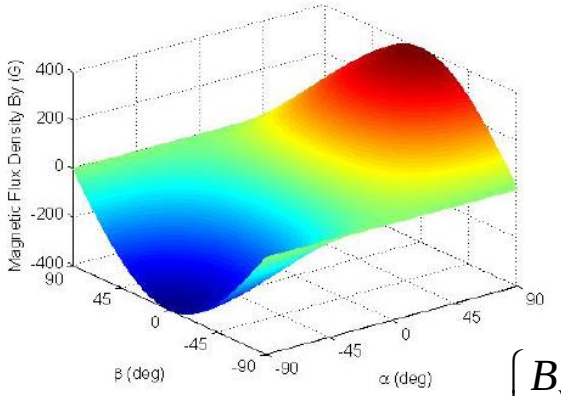




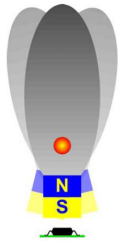
● 位置/角度传感器/Linear or Angular Position Sensor

◆三坐标磁场矢量芯片/3D Flux Strength Sensor MLX90333









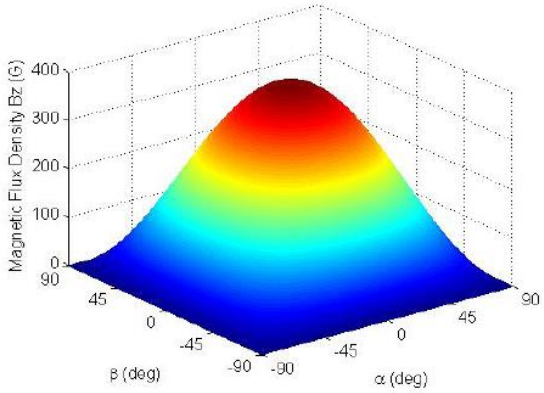


Figure 5 – Magnetic Flux Density – B_X, B_Y, B_Z

$$\begin{cases} B_x = B\sin(\alpha)\cos(\beta) \\ B_y = B\cos(\alpha)\sin(\beta) \\ B_z = B\cos(\alpha)\cos(\beta) \end{cases}$$

$$\alpha = ATAN\left(\frac{\sqrt{(k_Z V_Z)^2 + (k_t V_Y)^2}}{V_X}\right)$$

$$\beta = ATAN\left(\frac{\sqrt{(k_Z V_Z)^2 + (k_t V_X)^2}}{V_Y}\right)$$



§1 传感器工作原理/The Principle of Sensors in AECs

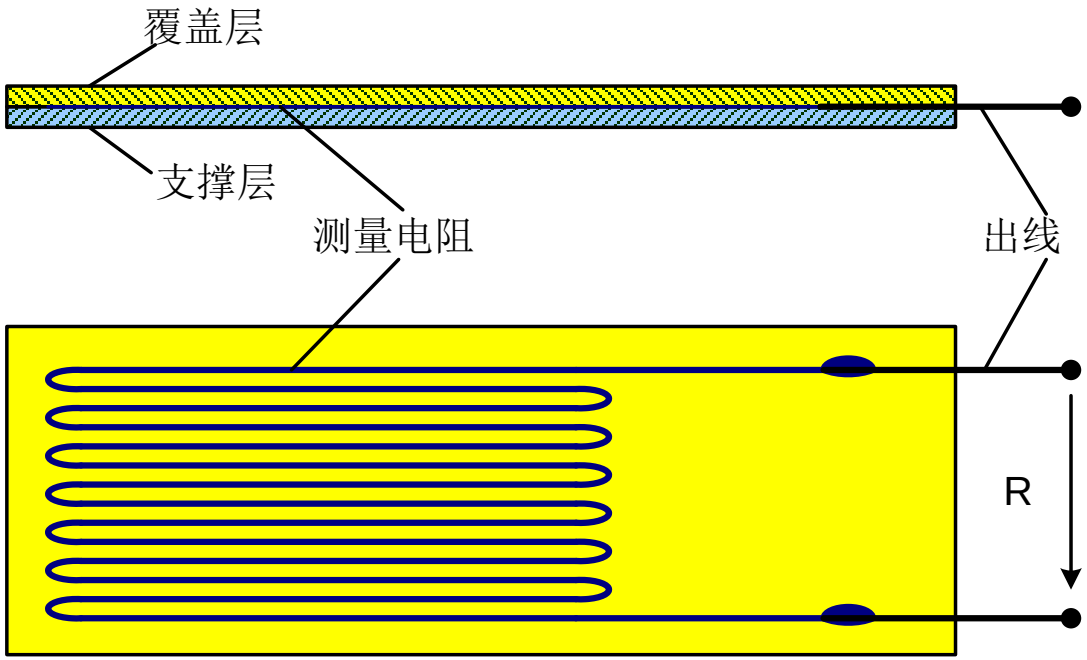
● 位置/角度传感器/Linear or Angular Position Sensor

◆ 霍尔传感器在汽车中的应用/Hall sensor in vehicle

序号	用途和特点/ Applications	序号	用途和特点/ Applications
1	后视镜位置反馈/ Mirror position	14	车前灯位置反馈/ Front Main Lamp position
2	雨刮极限位置检测/ Wiper position	15	车顶窗位置反馈/ Window position
3	制动踏板位置反馈/ Brake position	16	各种简单的按钮开关/ Simple switch
4	方向盘位置反馈/ Steering angle	17	安全带锁止状态检测开关/ Safe belt
5	悬架高度反馈/ Suspension height	18	无刷直流电机位置反馈/ BLDC position
6	燃油液面高度反馈/ Fuel level	19	座椅位置/重量反馈/Seat position, Weight
7	EGR阀的位置反馈/ EGR valve position	20	流量测量/ Flow sensor
8	VVT控制阀位置反馈/ VVT valve position	21	电流传感器/ Current sensor
9	凸轮轴位置传感器/ Cam shaft sensor	22	磁场测量/ Magnet flux measurement
10	曲轴位置传感器/ Crank shaft sensor	23	手机打开/合上的感应开关/ mobile open
11	电子节气门位置反馈/ ETC position	24	机械操纵手柄传感器/ 3d manual stick
12	轮速传感器/ Wheel speed	25	游戏杆传感器/ 3d Joystick
13	变速器轴速传感器/ Gear box speed	26	针阀升程传感器/ valve needle lift sensor

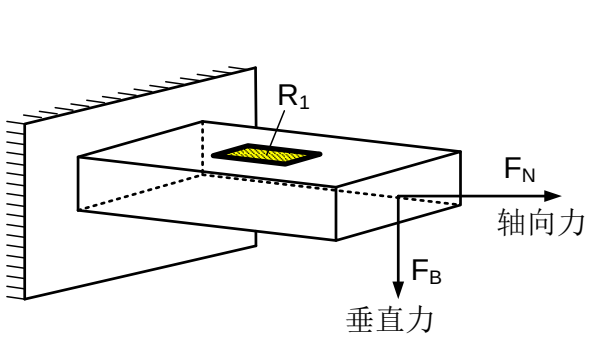


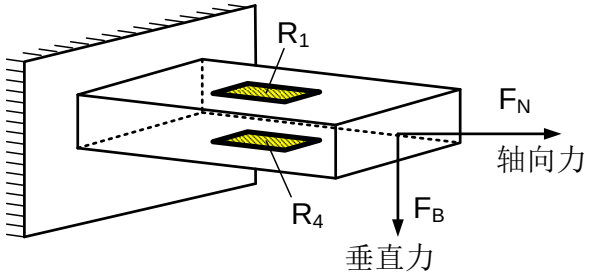
- 力/压力测量/Force / Pressure Measurement
  - ◆ 电阻

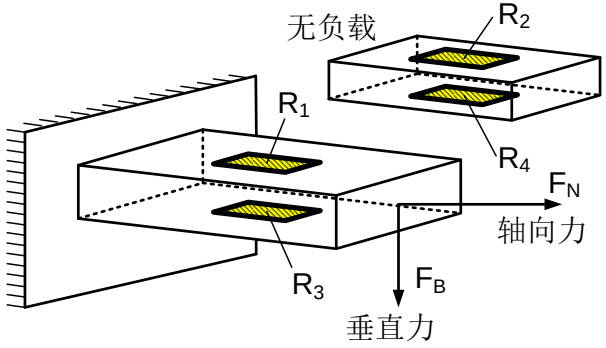


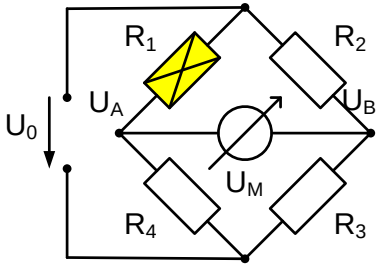
● 力/压力测量/Force / Pressure Measurement

◆ 电阻



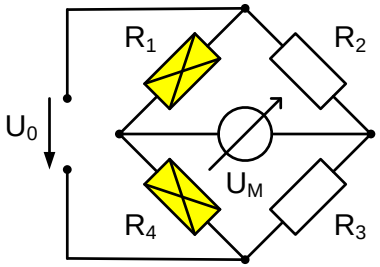






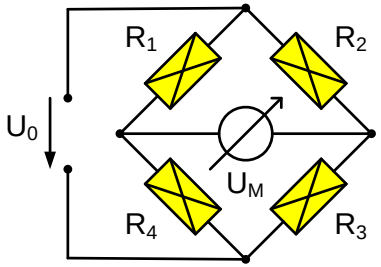
¼ 惠斯通桥

（a）



½ 惠斯通桥

（b）



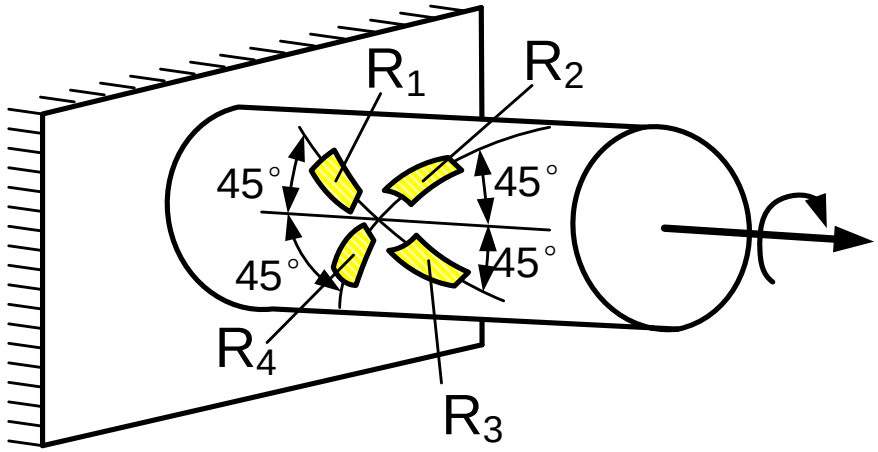
惠斯通全桥

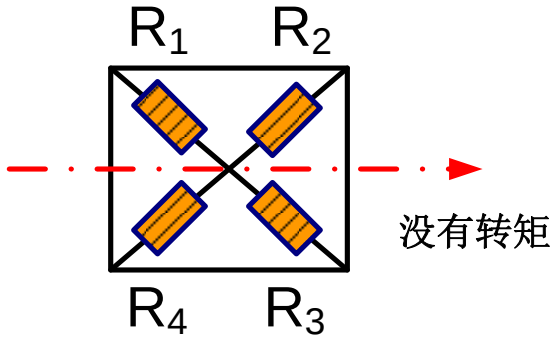
（c）

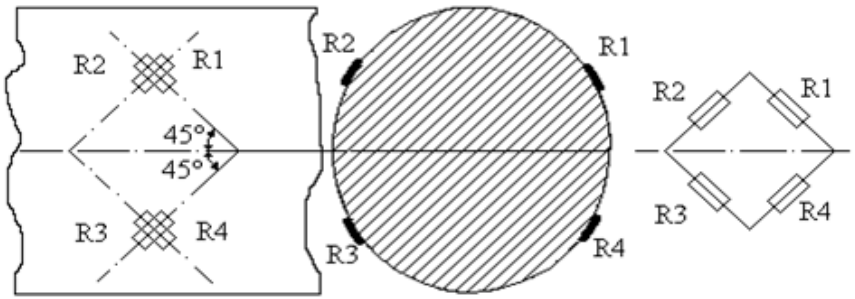
三种惠斯通桥(Wheatstone bridge)

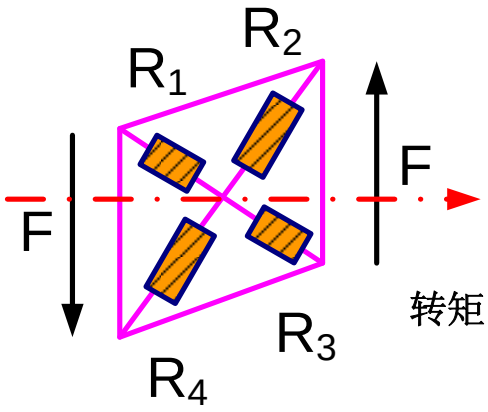
注意应变片的结构安装位置和惠斯通桥电路位置的对应关系

- 力/压力测量/Force / Pressure Measurement
  - 电阻









惠斯通桥全桥

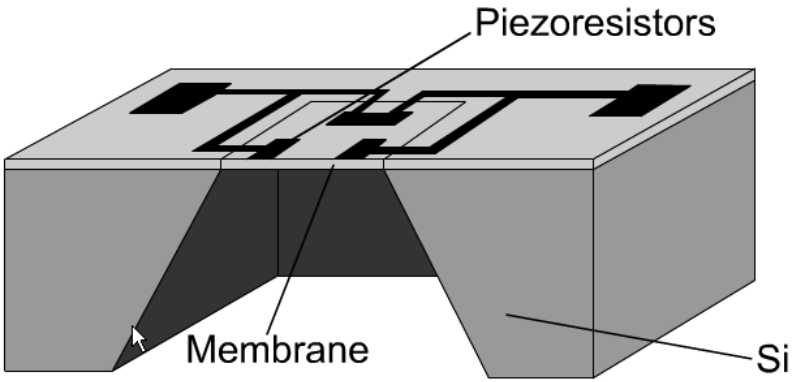
注意应变片的结构安装位置和惠斯通桥电路位置的对应关系

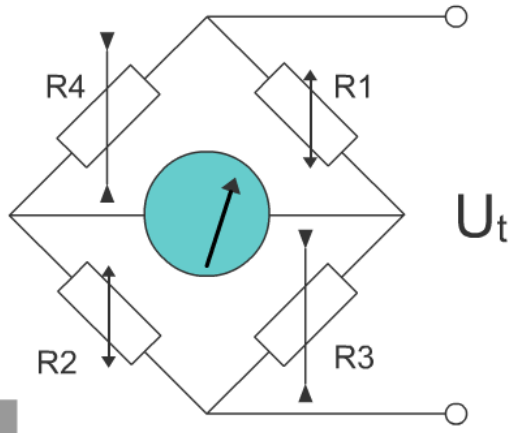
● 力/压力测量/Force / Pressure Measurement

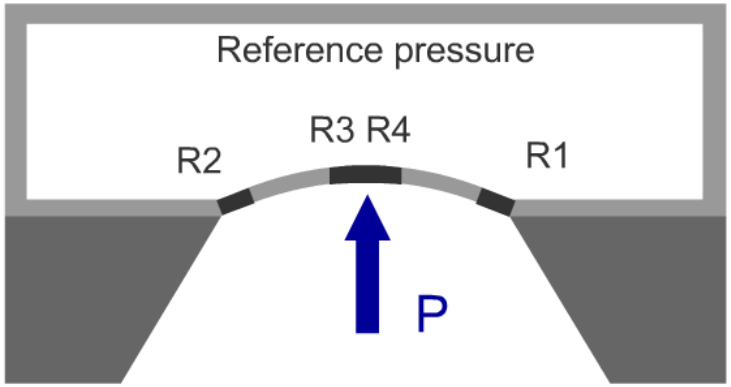
◆ 电阻

Piezoresistive Si Pressure Sensor

Related topics







Show :

Piezoresistive

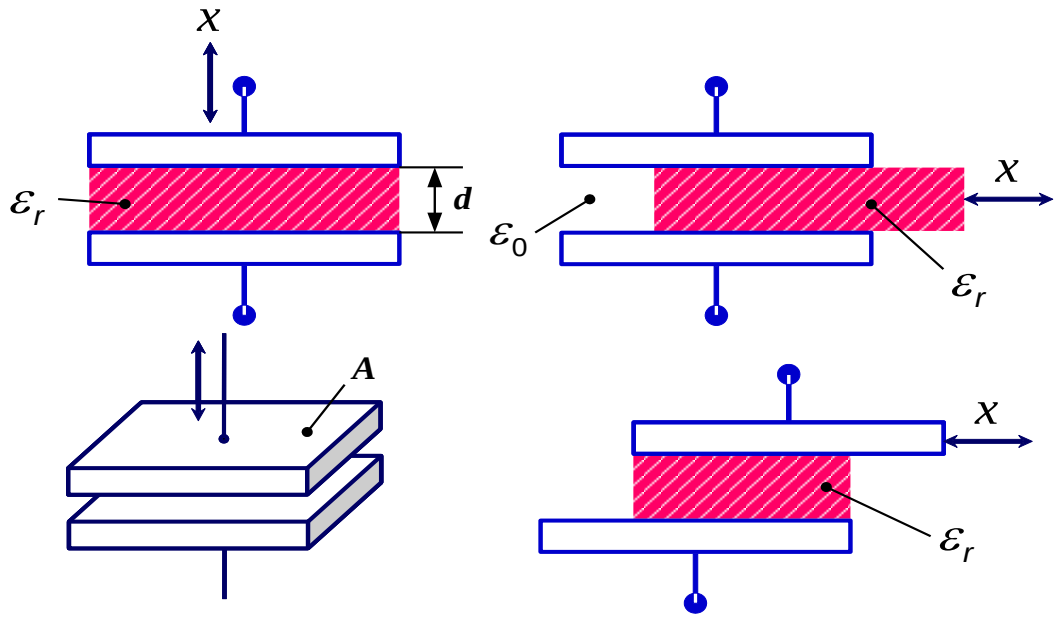
Capacitive

压电电阻Piezo-resistance

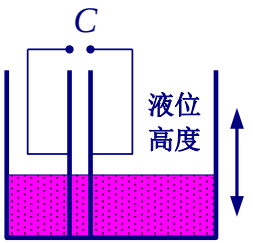
- **力/压力测量/Force / Pressure Measurement**
  - **电容**

电容测量原理/Capacitive Principle Sensor–MEMS
改变平板距离/ Change distance of plates–d
改变电介质区域/ Change dielectric area–ε_r
改变平板相对面积/ Change overlap area–A

Source:
www.ett.bme.hu/memsedu/
On line lessons about MEMS, very good!



$$C = \varepsilon_0 \cdot \varepsilon_r \cdot \frac{A}{d}$$

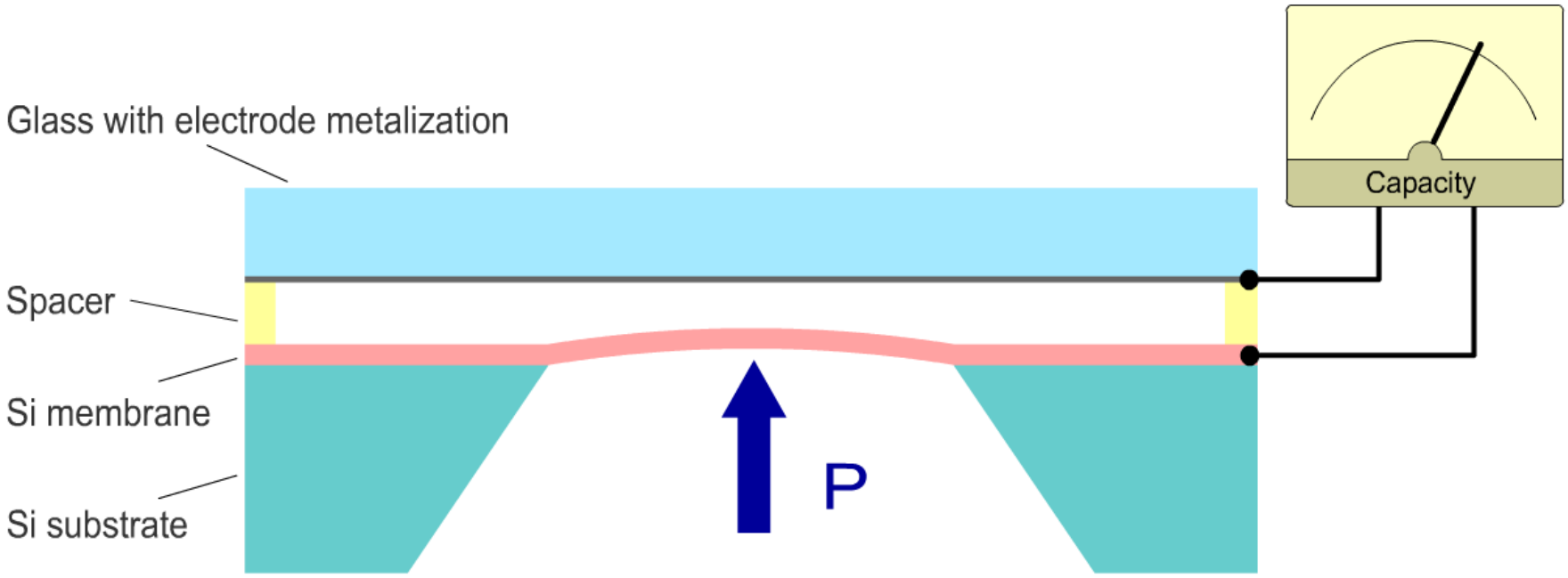


电容测量基本原理：充电过程特征曲线

- 力/压力测量/Force / Pressure Measurement
  - ◆ 电容

Capacitive Pressure Sensor

Related topics



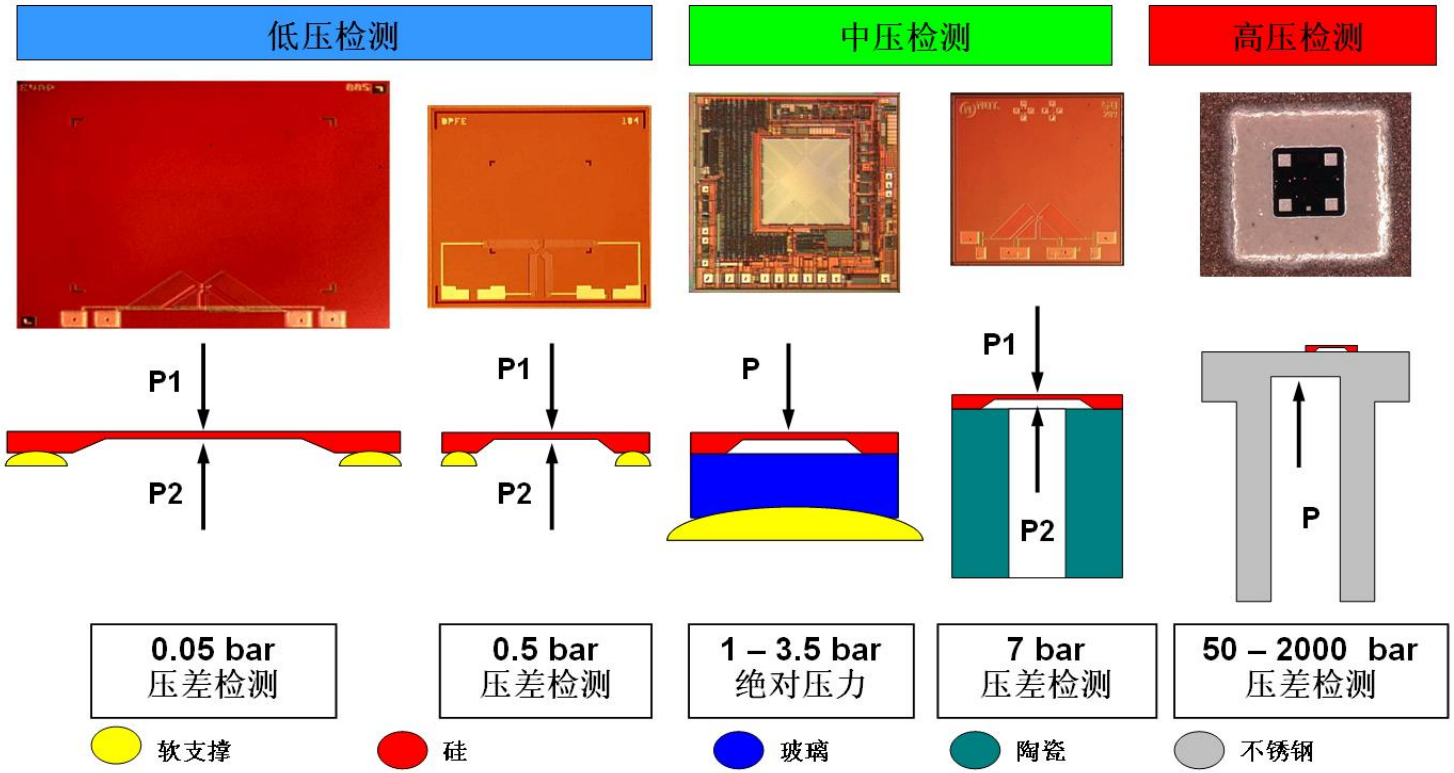
Show :

Piezoresistive

Capacitive

电容性的 Capacitive

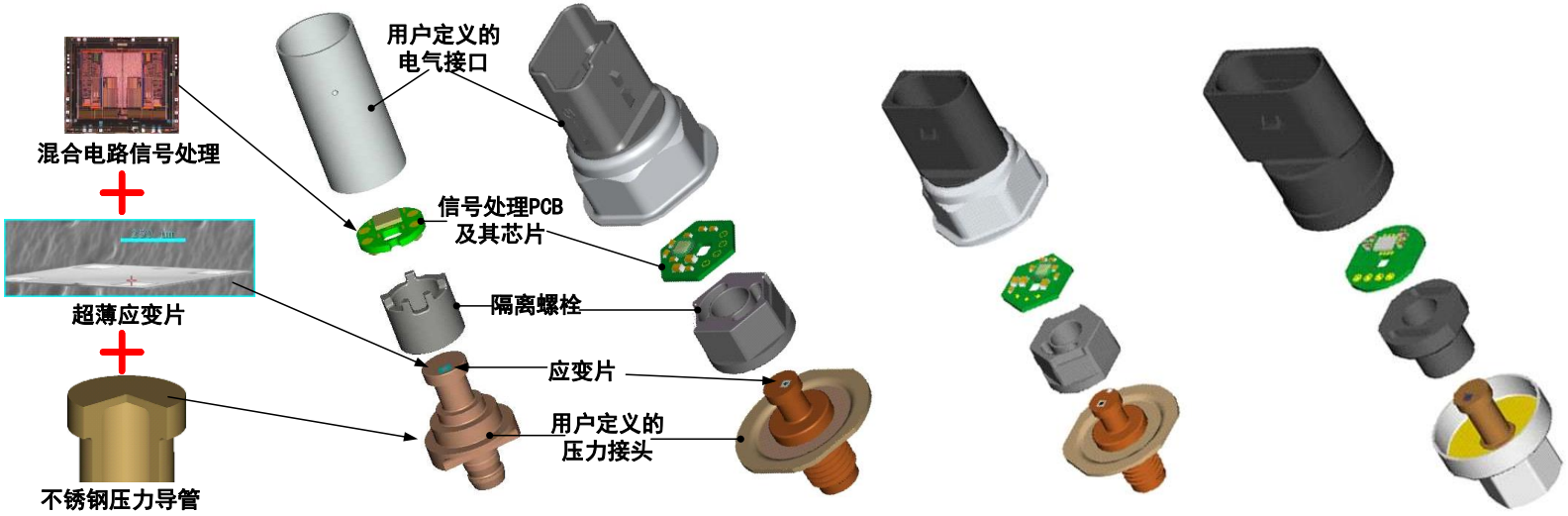
- 力/压力/微机电传感器测量/Force / Pressure Measurement MEMS (Micro–Electro–Mechanical System)





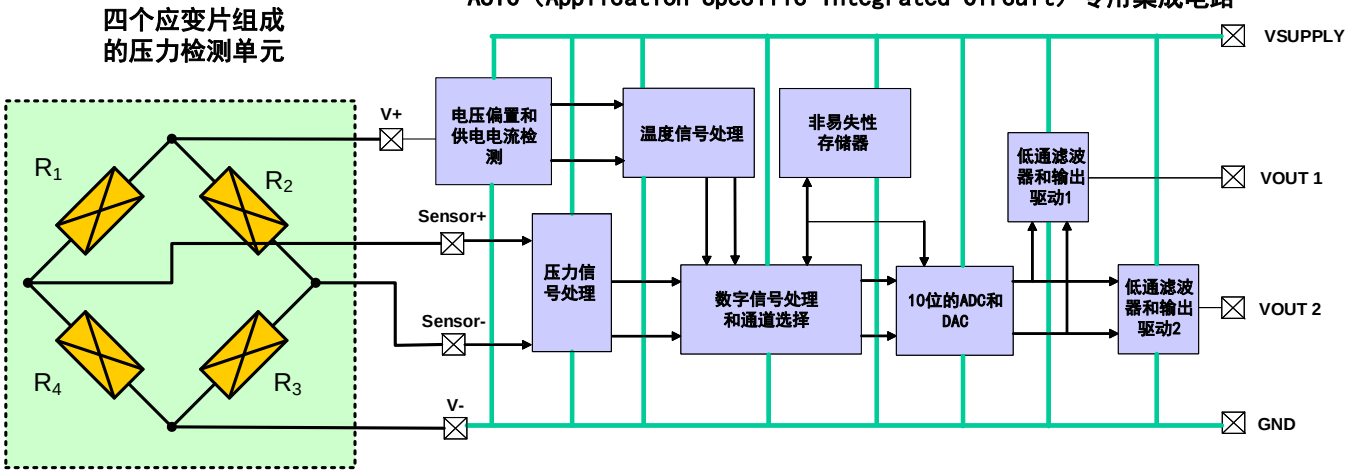
§1 传感器工作原理/The Principle of Sensors in AECs

● 力/压力测量/Force / Pressure Measurement MEMS

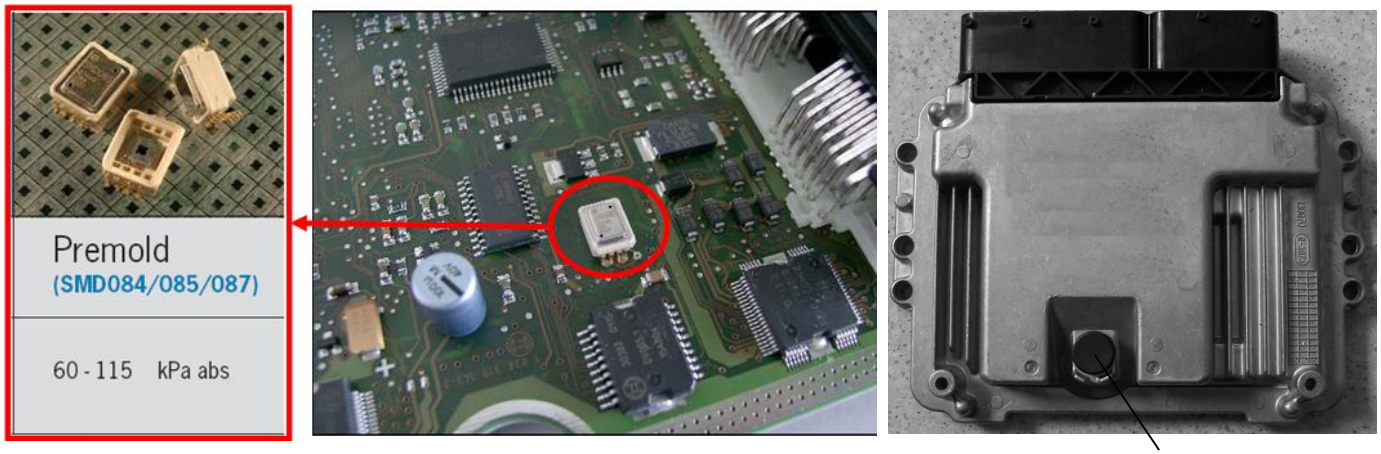


传感器信号处理芯片

ASIC（Application Specific Integrated Circuit）专用集成电路

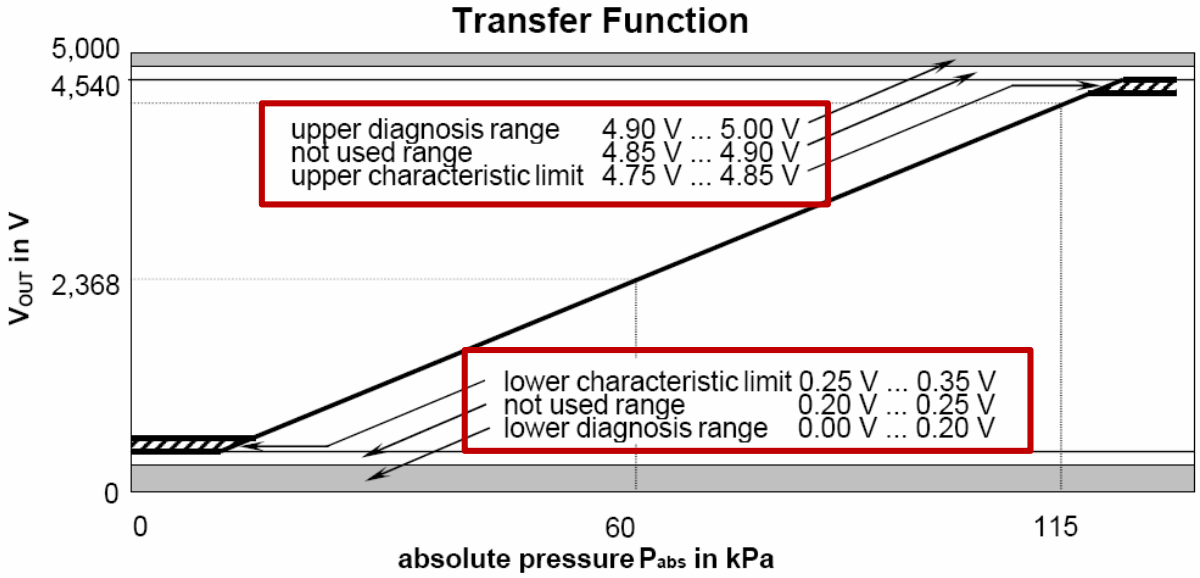


● 力/压力测量/Force / Pressure Measurement MEMS



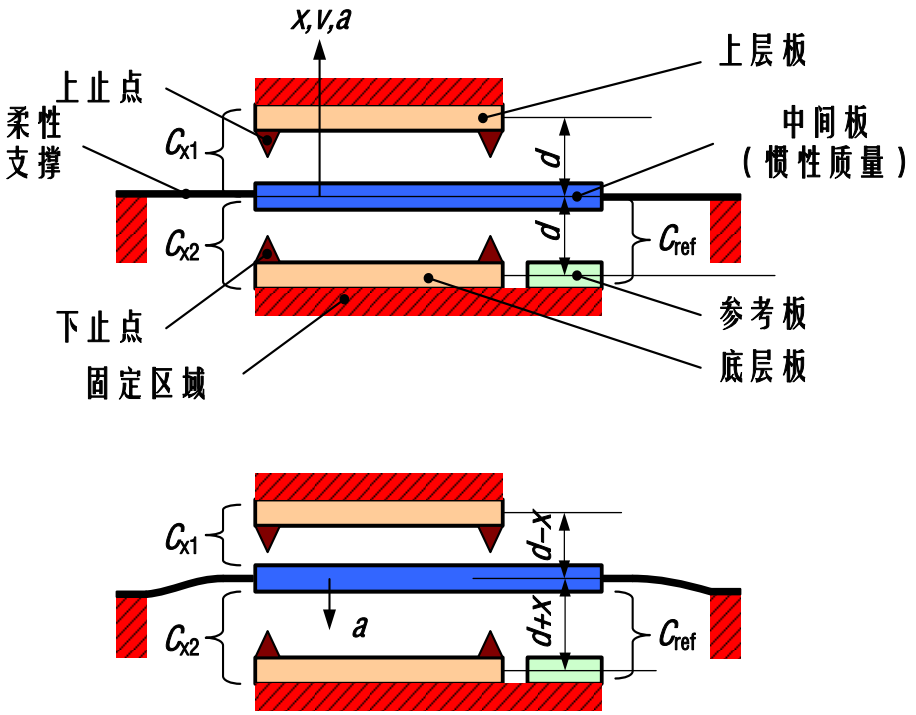
大气压力传感器的外形和在PCB板上的安装形式

ECU机壳上的通气塞

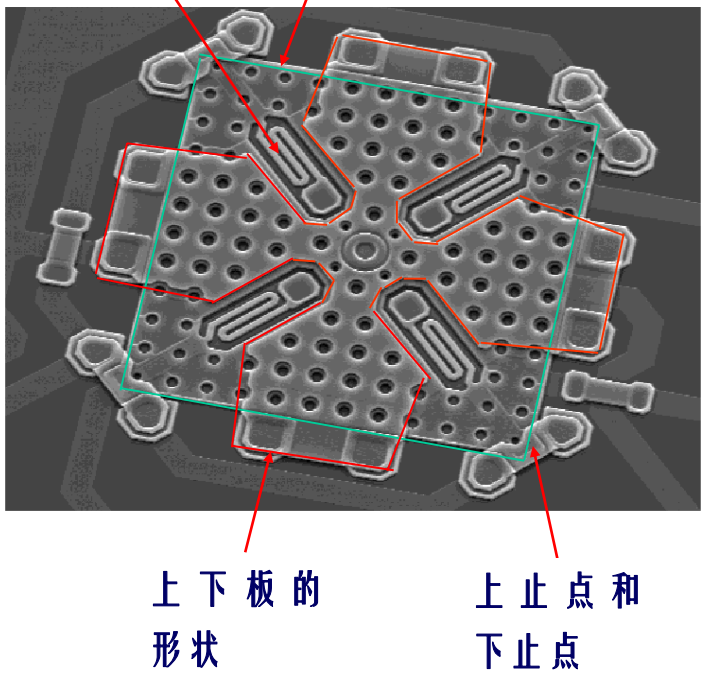


● 加速度传感器/Acceleration Sensor–MEMS

◆电容式加速度计/ Capacitive Acc Meter:



$$C = \varepsilon_0 \cdot \varepsilon_r \cdot \frac{A}{d}$$



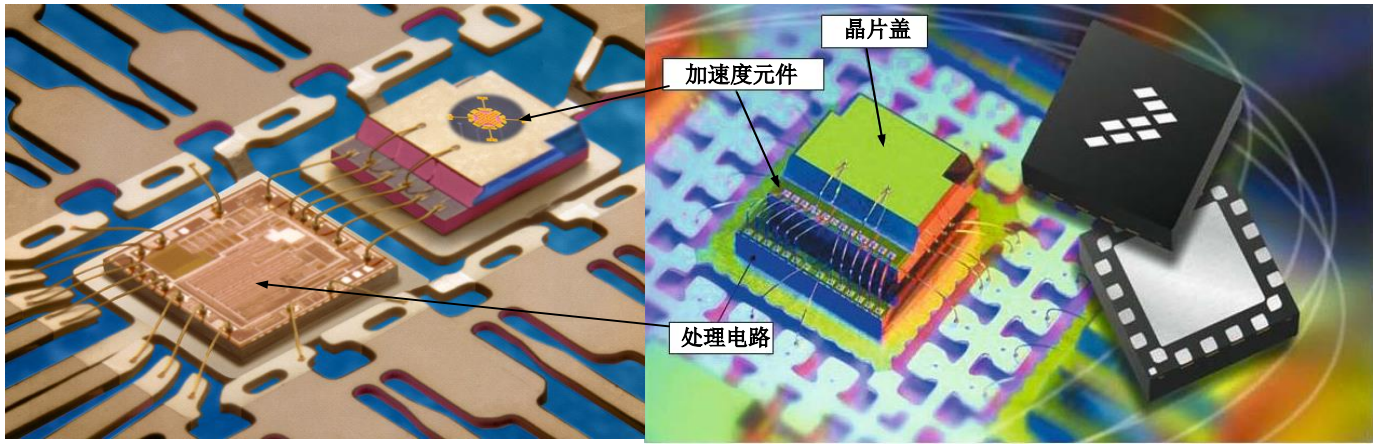


● 加速度传感器/Acceleration Sensor–MEMS

◆电容式加速度计/ Capacitive acc. meter:

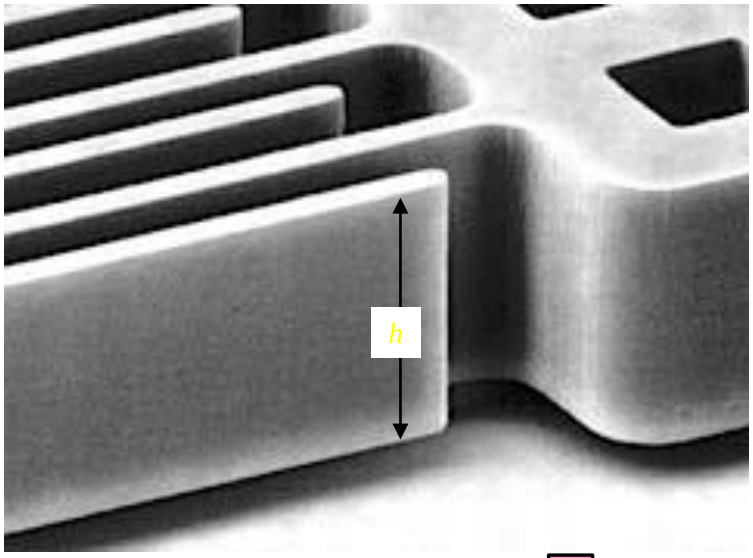
$$x=0,\quad C_{x1}=C_{x2}=\frac{\varepsilon_0\varepsilon_r A}{d} \qquad x>0,\quad C_{x1}=\frac{\varepsilon_0\varepsilon_r A}{d-x},\quad C_{x2}=\frac{\varepsilon_0\varepsilon_r A}{d+x}$$

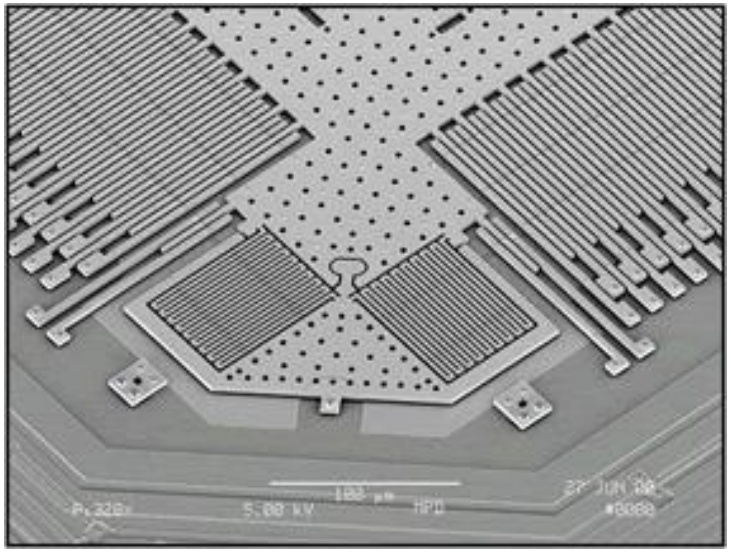
$$\Rightarrow x=\frac{1}{2}\varepsilon_0\varepsilon_r A\left(\frac{1}{C_{x2}}-\frac{1}{C_{x1}}\right) \Rightarrow a=\frac{F}{m}=\frac{kx}{m}=\frac{1}{2m}\varepsilon_0\varepsilon_r Ak\left(\frac{1}{C_{x2}}-\frac{1}{C_{x1}}\right)$$

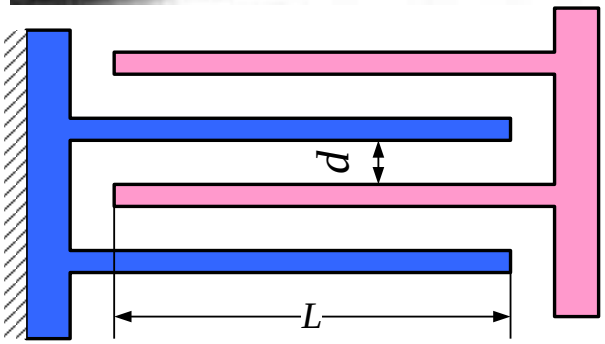


● 加速度传感器/Acceleration Sensor–MEMS

◆电容式加速度计/ Capacitive acc. meter:







v

d=1.4μm;

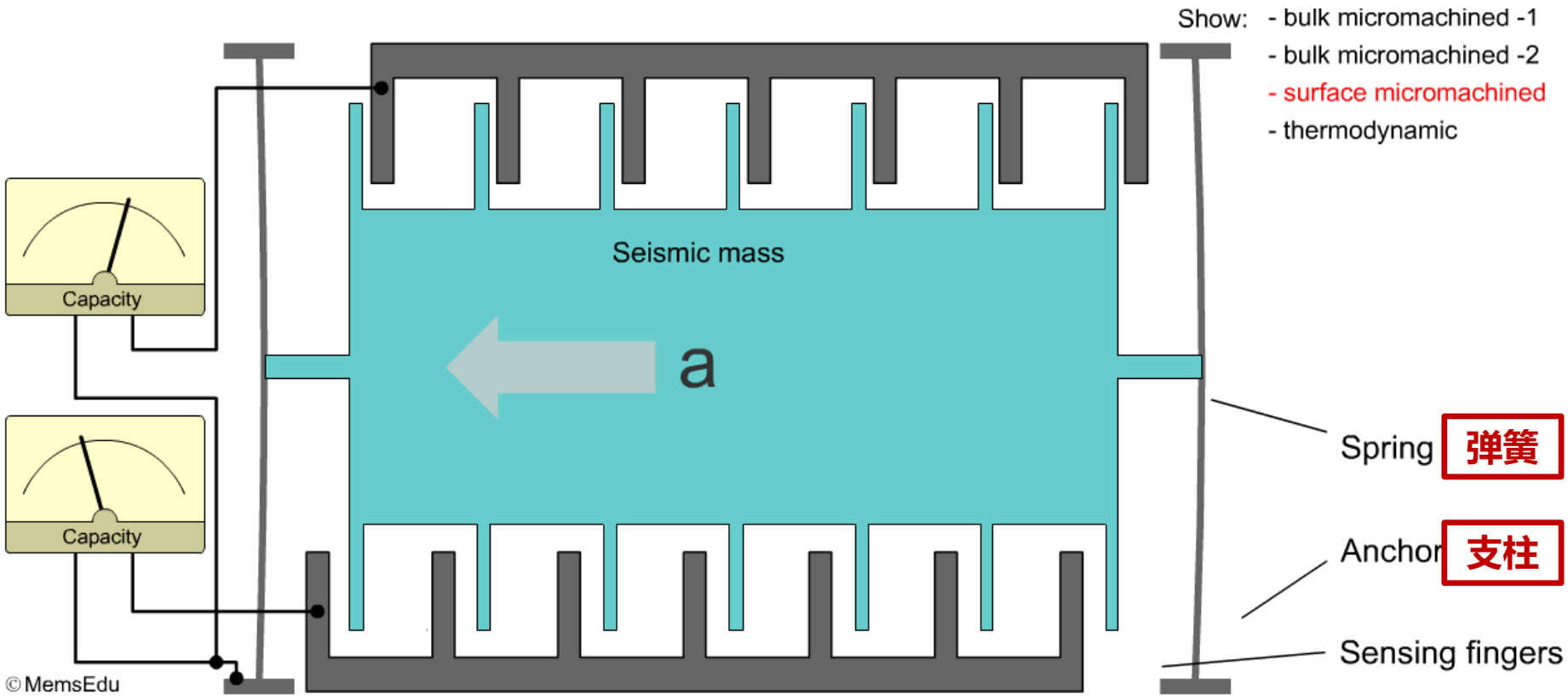
L=150μm

N=50对

$$C = N\varepsilon_0 \frac{Lh}{d} \Rightarrow \Delta C \approx N\varepsilon_0 \frac{Lh}{d^2}\Delta x$$

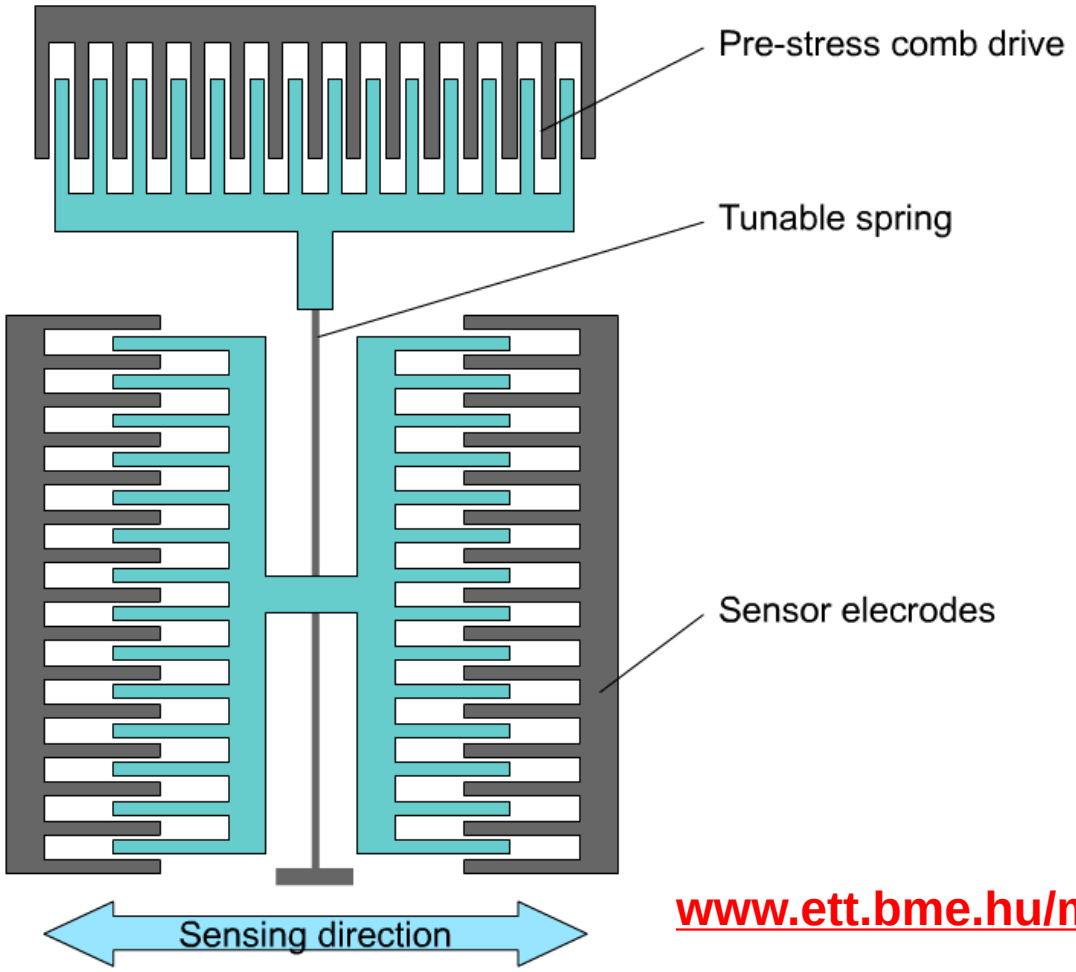
$$\Delta x = 0,01nm \Rightarrow \Delta C = 1aF = 10^{-18}F$$

- 加速度传感器/Acceleration Sensor–MEMS
  - ◆电容式加速度计/ Capacitive acc. meter:



www.ett.bme.hu/memsedu/

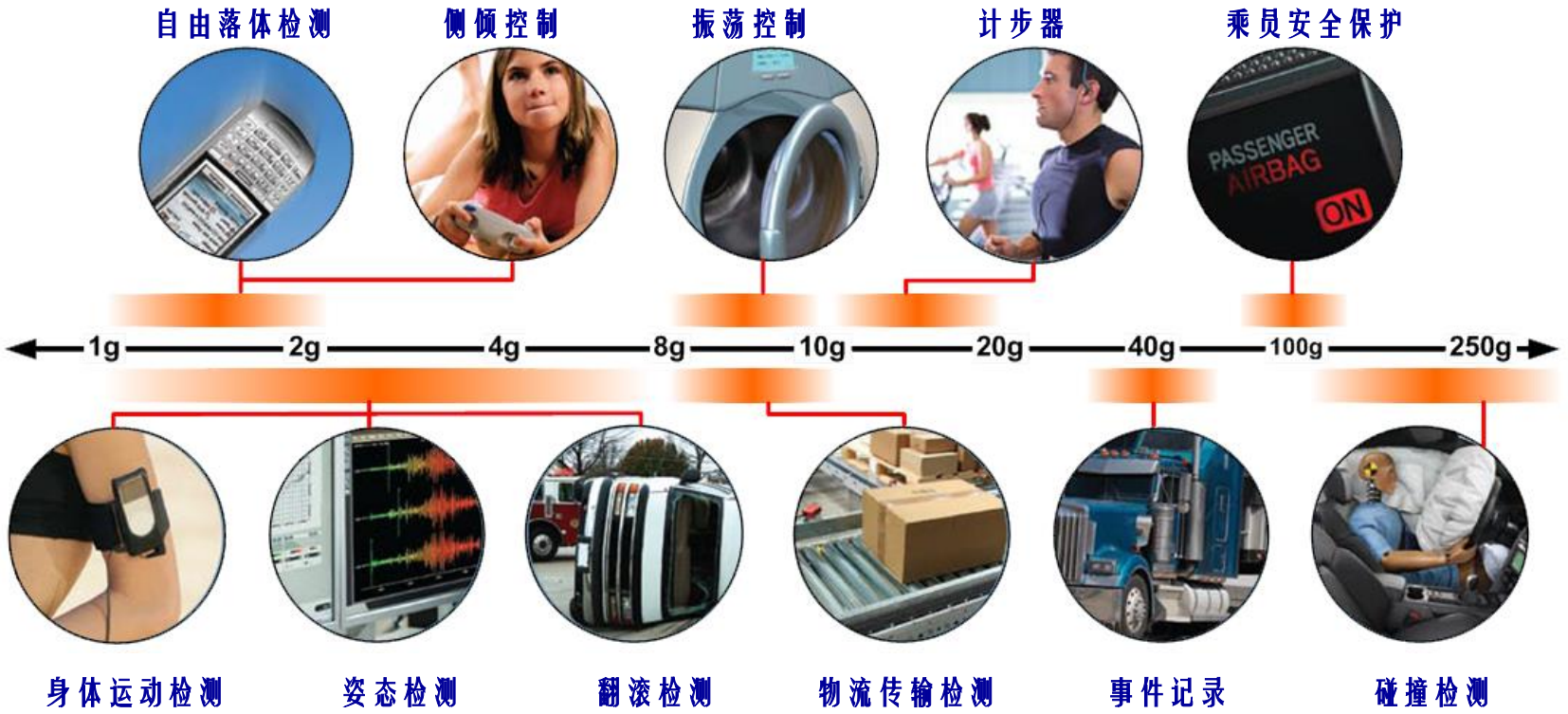
- 加速度传感器/Acceleration Sensor–MEMS
  - ◆电容式加速度计/ Capacitive acc. meter:



www.ett.bme.hu/memsedu/



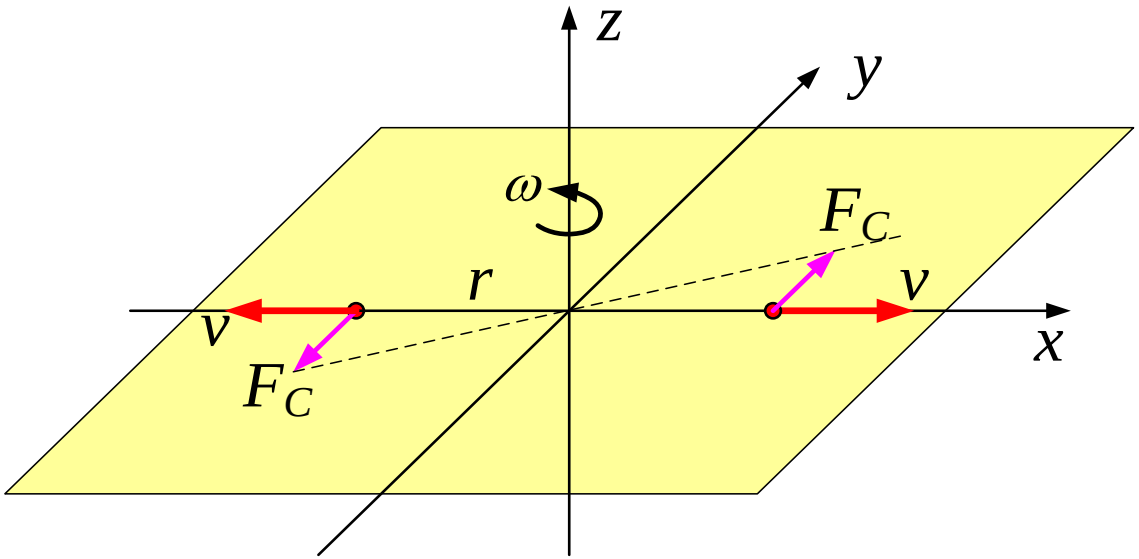
● 加速度传感器的应用/Application of acc. sensor



● 角速度–陀螺仪/Rotational Speed–Gyro–MEMS

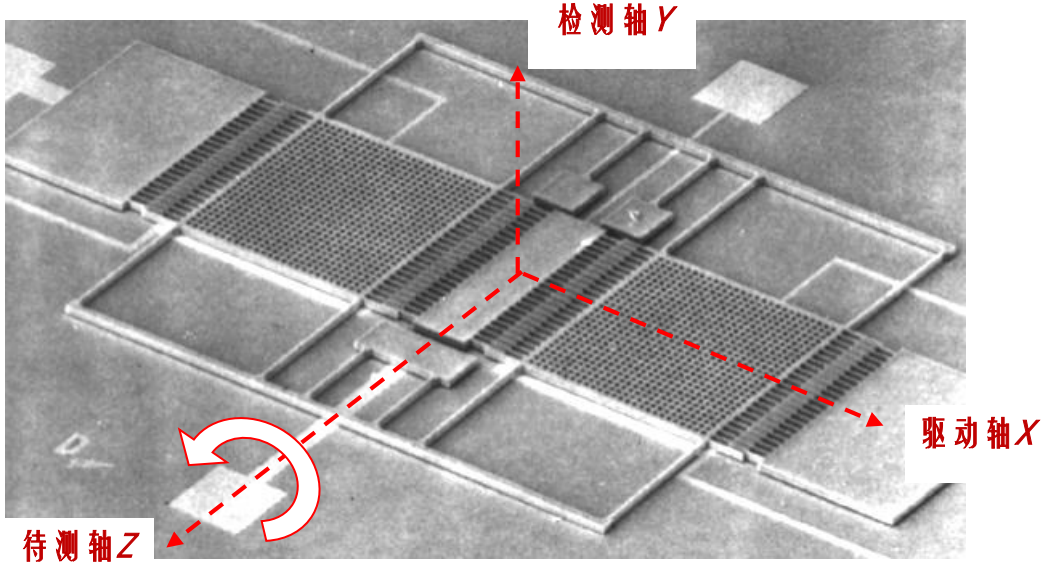
◆利用科氏加速度（科里奥利，Coriolis）的测量原理/Coriolis force.

$$\vec{F}_C = 2m\vec{\omega}\times\vec{v}$$

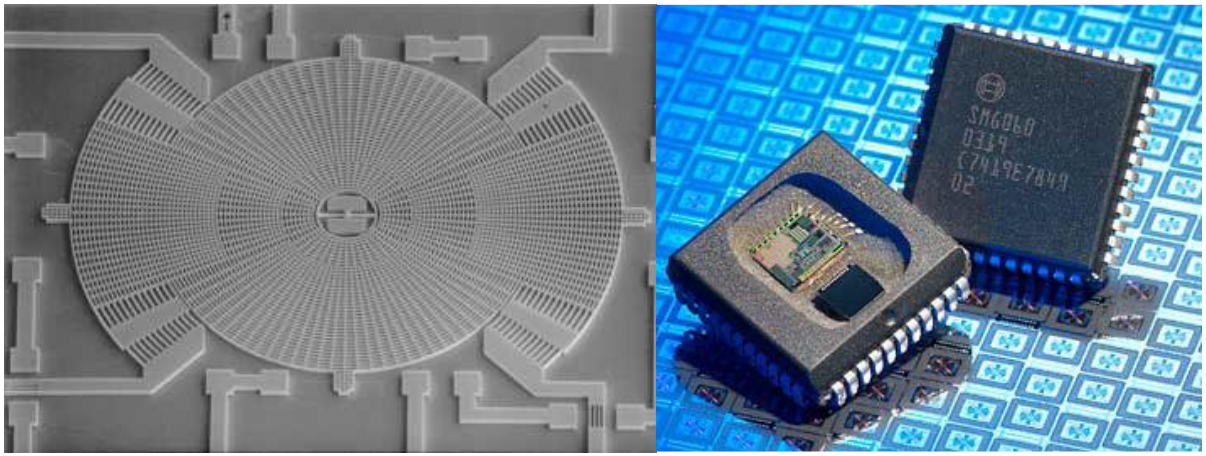


右手法则

● 角速度–陀螺仪/Rotational Speed–Gyro–MEMS

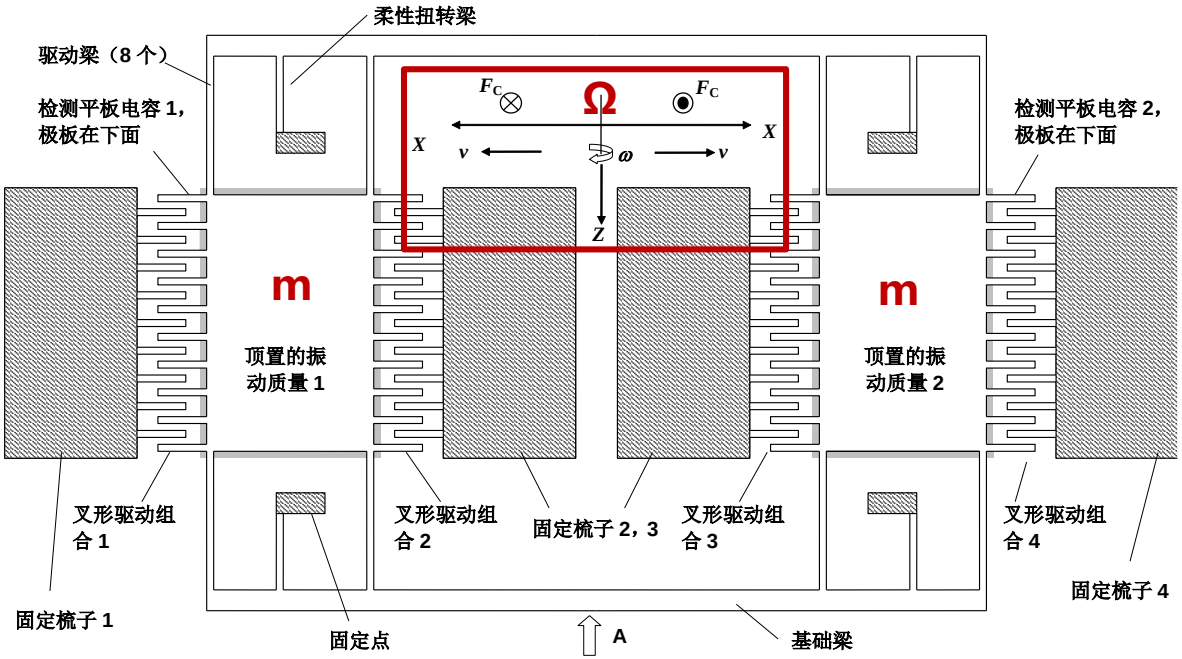


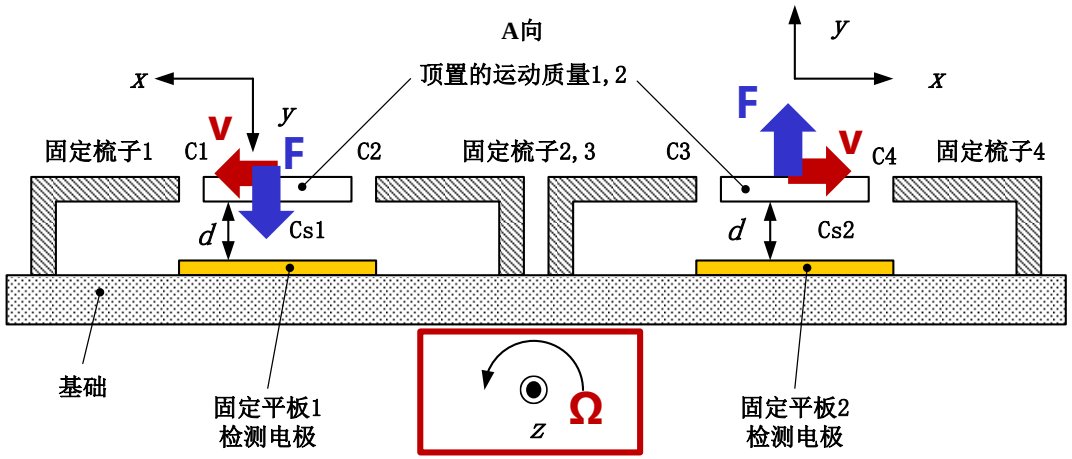
角速度传感器检测元件照片/Photo of rotational speed measurement element, Freescale



陀螺测试仪及其芯片照片/Photo of Gyrometer, Bosch

● 角速度-陀螺仪/Rotational Speed-Gyro-MEMS

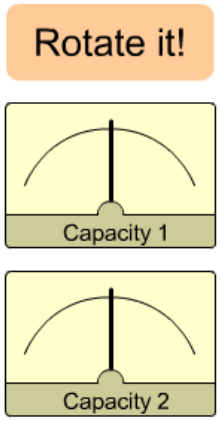




- 待测量：刚体转速
- 注入信号：两个质量块以一定的规律沿系统表面水平运动
- 直接测量：两个质量块与参考电极电容变化产生电压信号

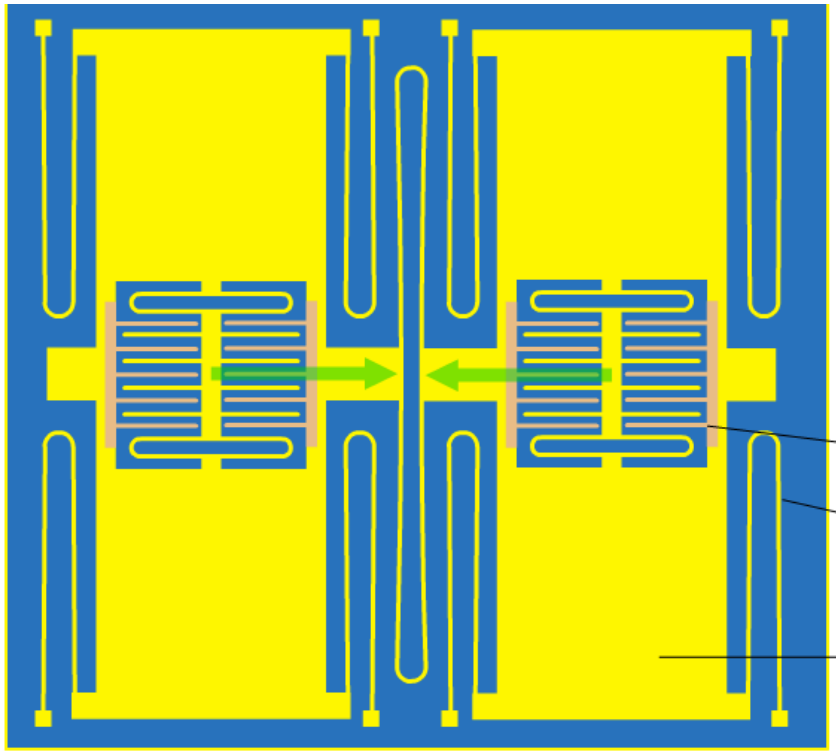
● 角速度-陀螺仪/Rotational Speed-Gyro-MEMS

Source:
www.ett.bme.hu/memsedu/



Select type:
- suspended mass
- rotating plate
- tuning fork

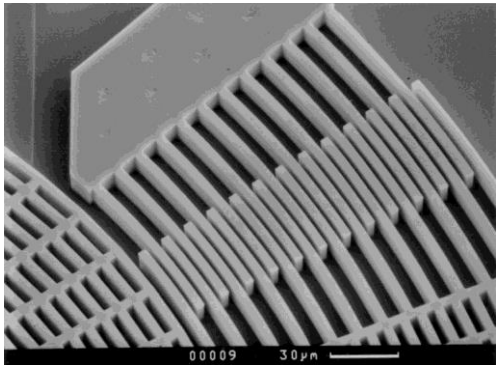
© MemsEdu

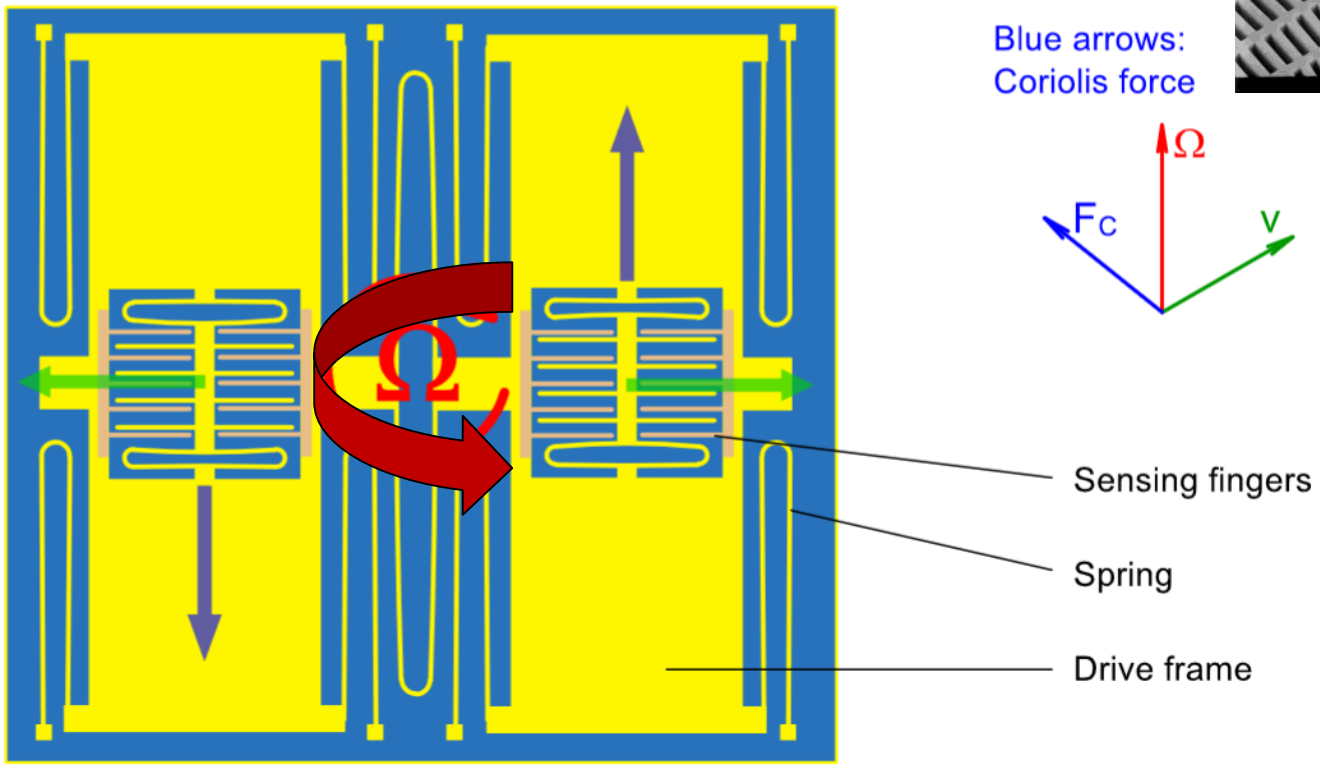


http://www.ett.bme.hu/memsedu/anim/index.html?anim=4/4114

● 角速度–陀螺仪/Rotational Speed–Gyro–MEMS

Source:
www.ett.bme.hu/memsedu/



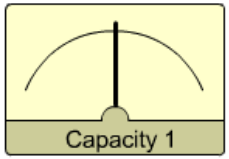


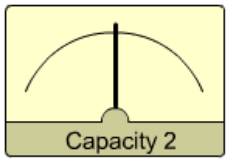
Blue arrows:
Coriolis force

● 角速度–陀螺仪/Rotational Speed–Gyro–MEMS

Source:
www.ett.bme.hu/memsedu/

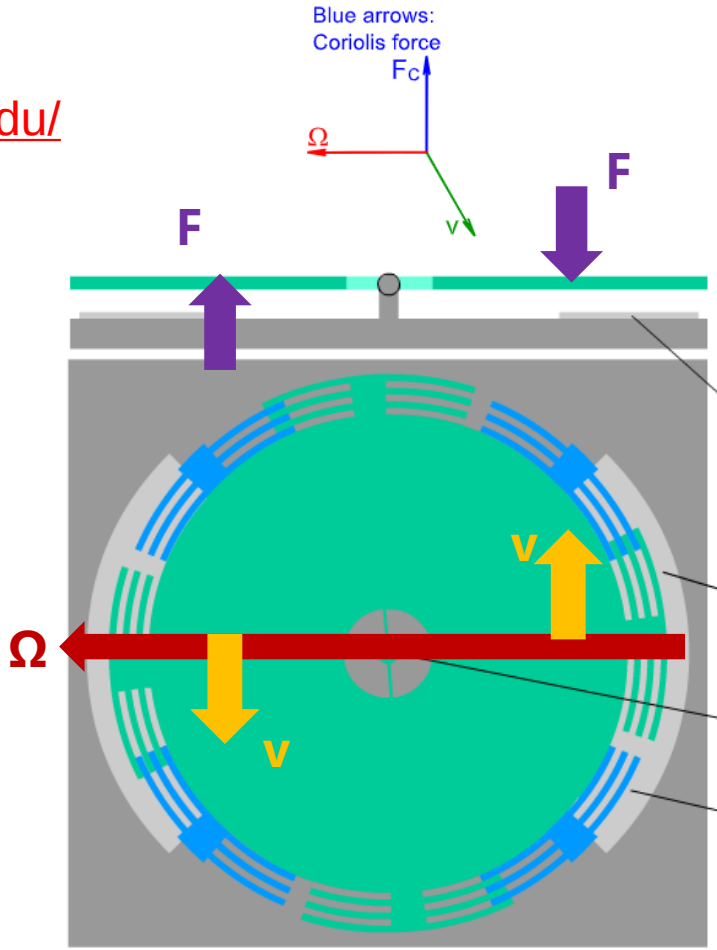
Rotate it!

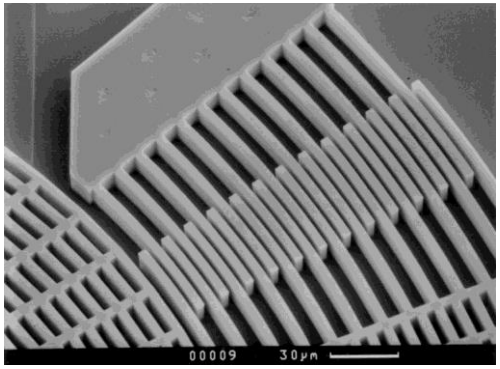




Select type:
- suspended mass
- rotating plate
- tuning fork

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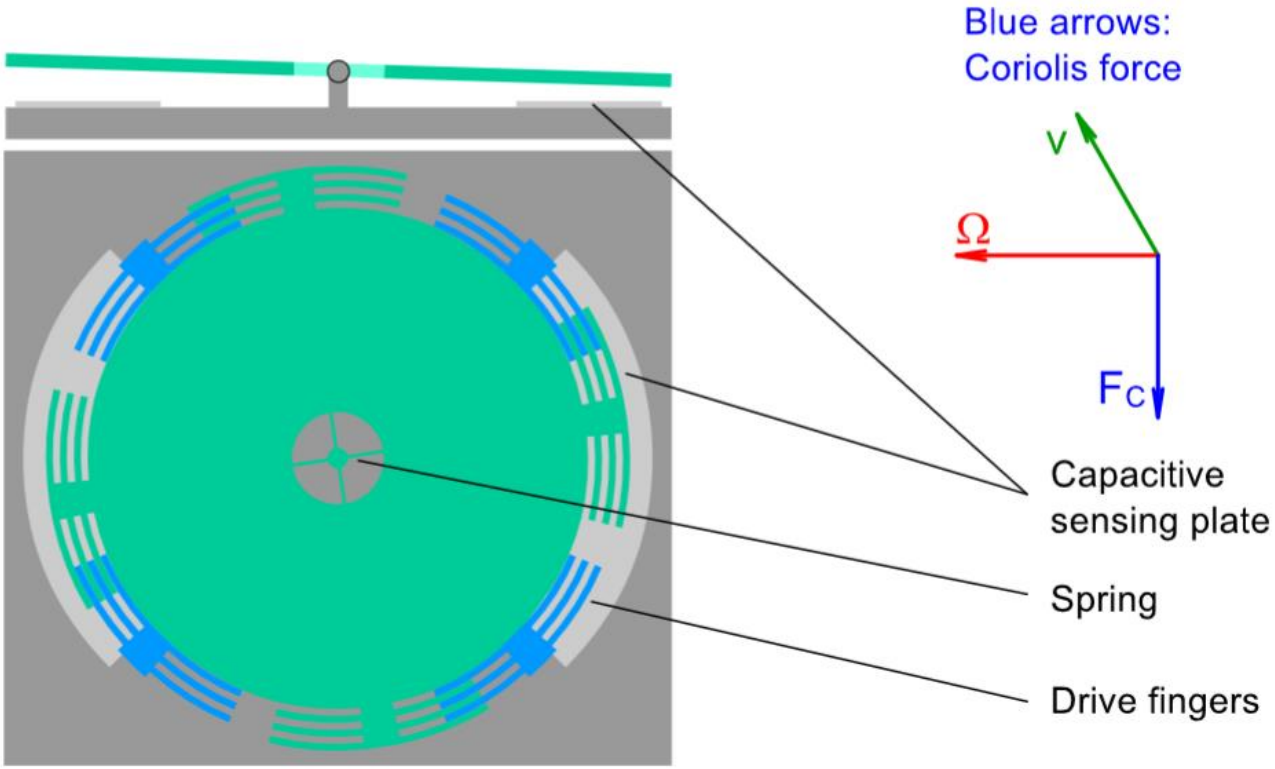




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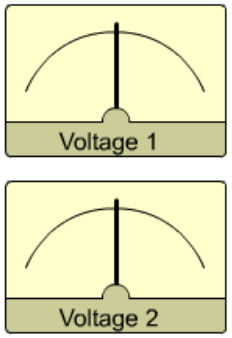
● 角速度-陀螺仪/Rotational Speed-Gyro-MEMS

Source:
www.ett.bme.hu/memsedu/



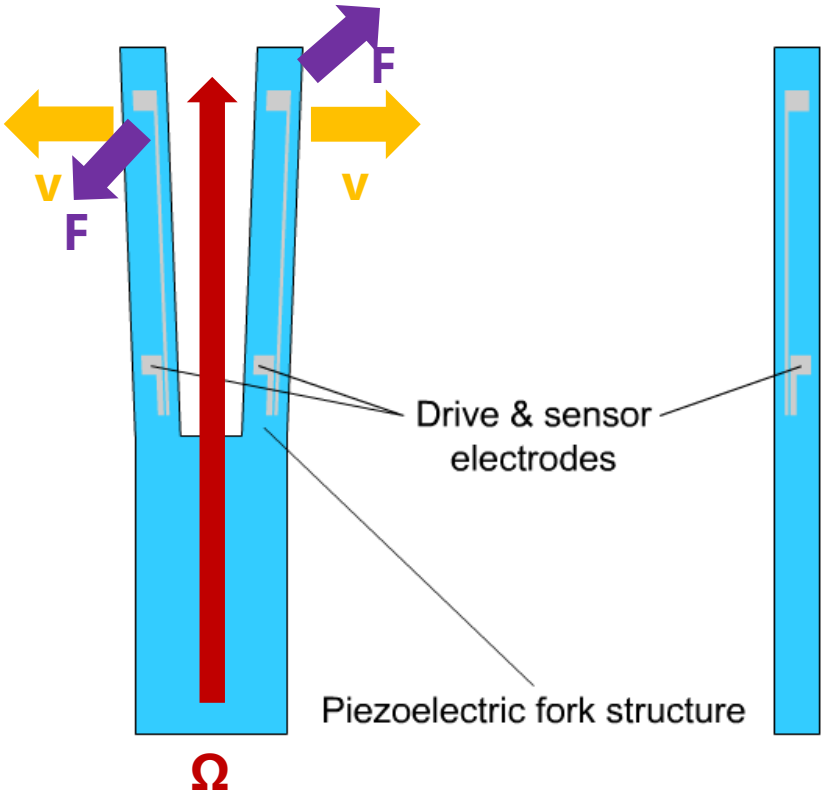
● 角速度–陀螺仪/Rotational Speed–Gyro–MEMS

Rotate it!



Select type:
- suspended mass
- rotating plate
- tuning fork

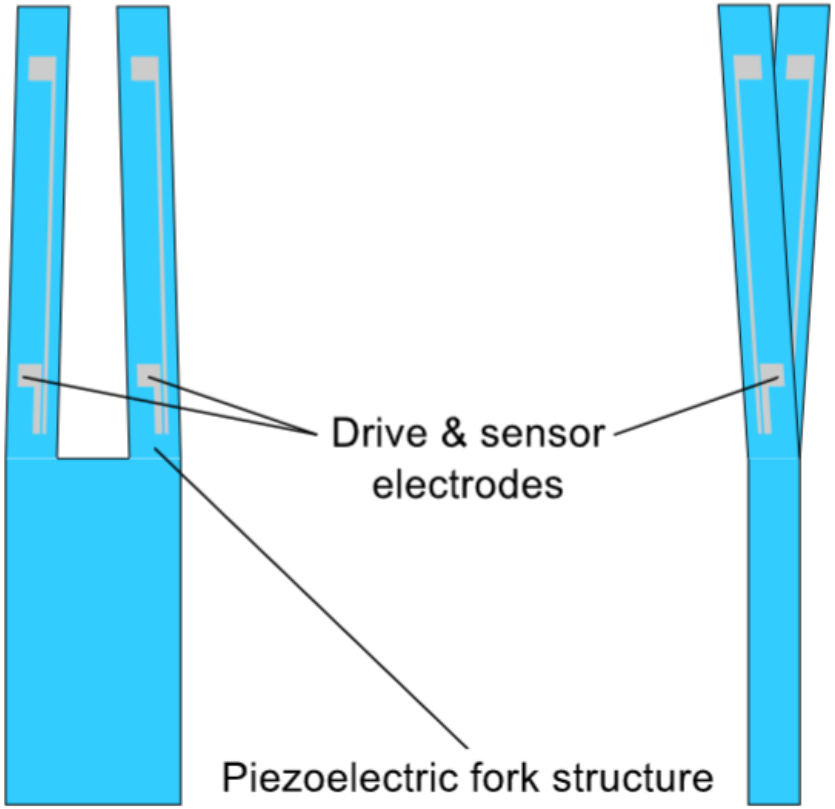
© MemsEdu

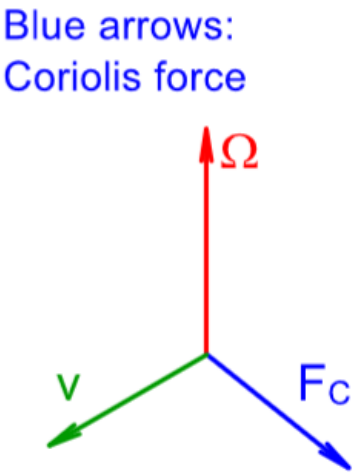


http://www.ett.bme.hu/memsedu/anim/index.html?anim=4/4114



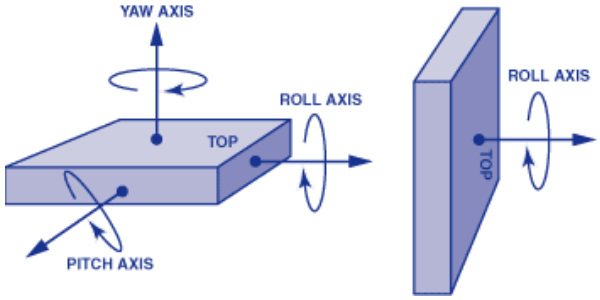
● 角速度-陀螺仪/Rotational Speed-Gyro-MEMS

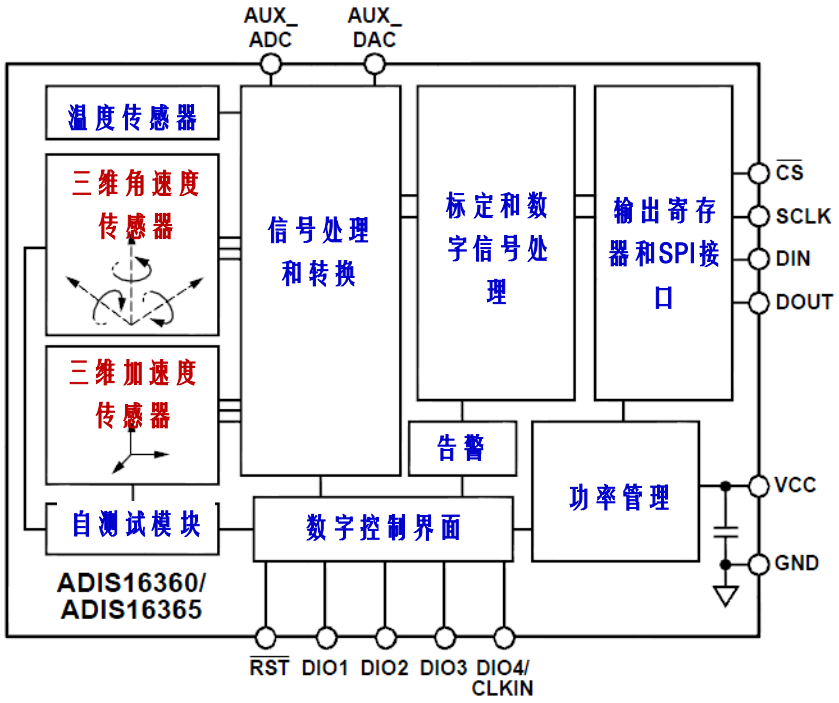


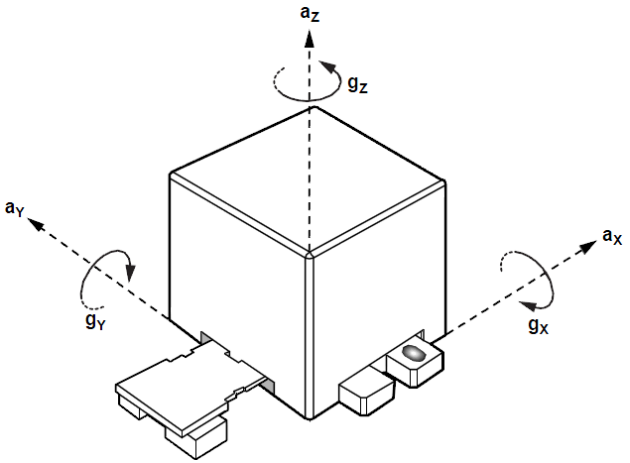




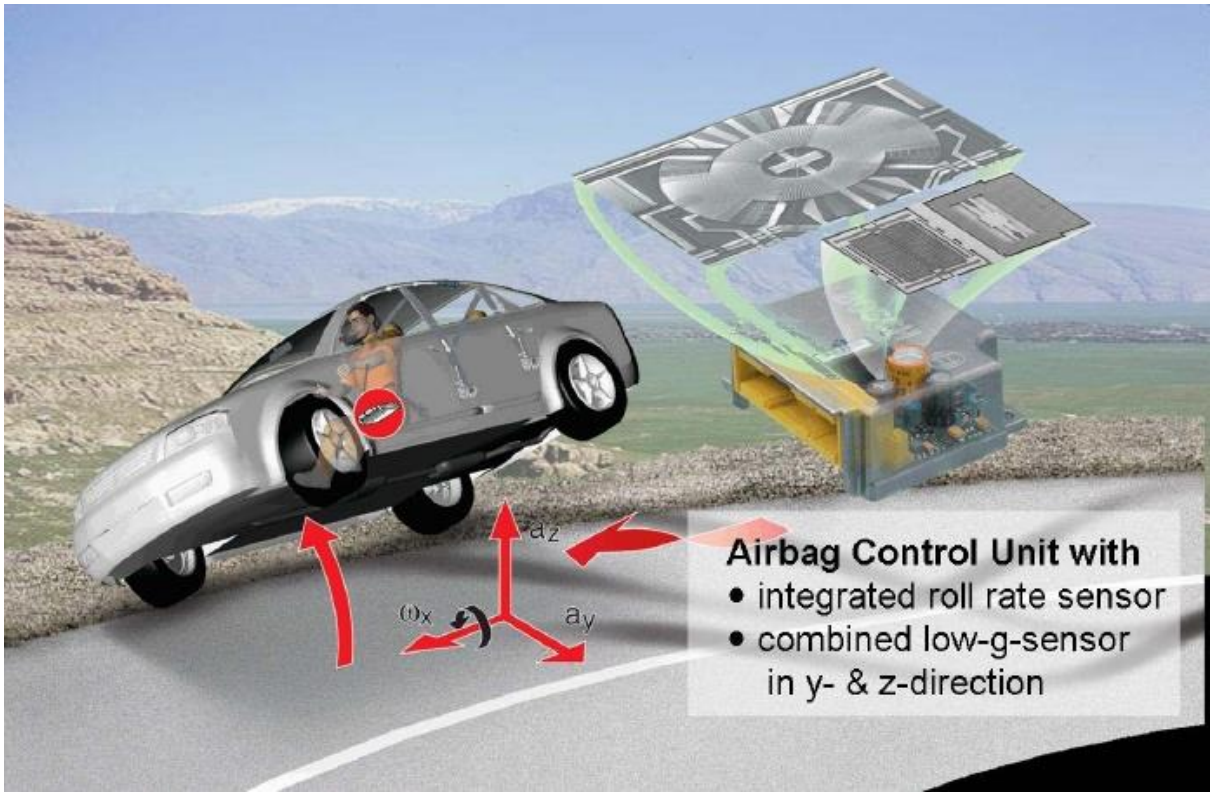
- 角速度–陀螺仪/Rotational Speed–Gyro–MEMS
  - ◆集成的六自由度的惯性传感器/ Integrated 6D Inertial Sensor
    ADIS16360, Analog Device





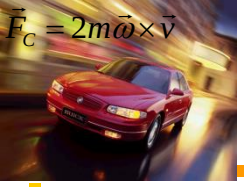


● 角速度-陀螺仪/Rotational Speed-Gyro-MEMS

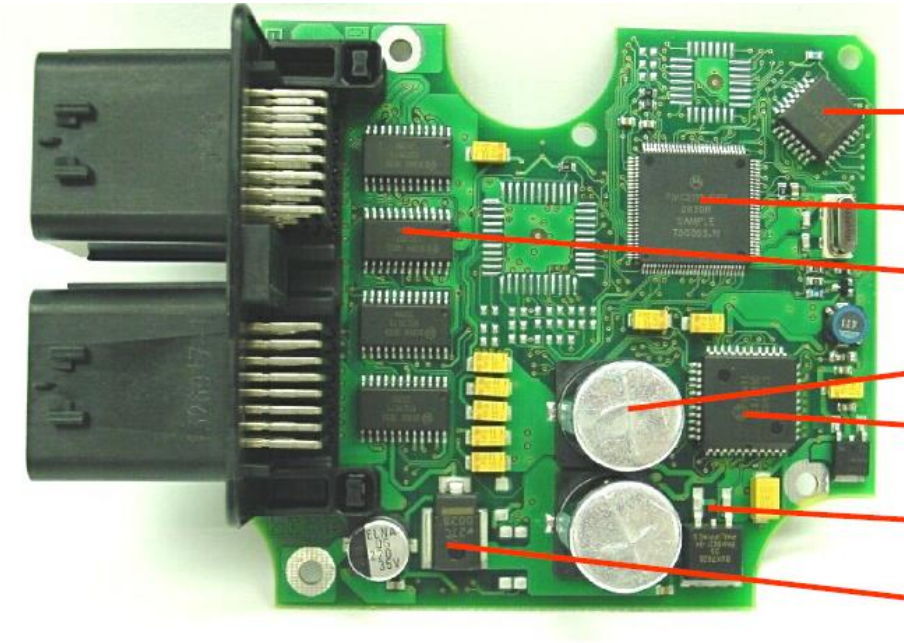


安全气囊控制单元

集成了翻滚角速度传感器和Y方向/Z方向的低g值加速度传感器，以及X方向的高g值加速度传感器；

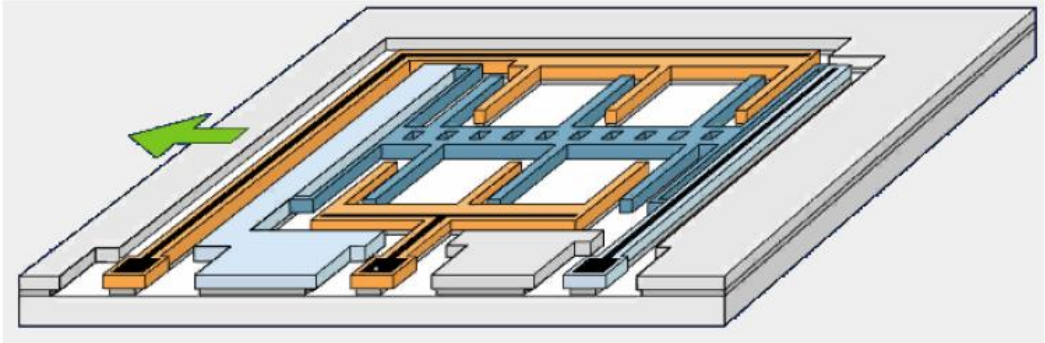


● 角速度-陀螺仪/Rotational Speed-Gyro-MEMS



Key Components: / 关键部件

2 accelerometers (saving sensor) / 2个加速度计
➔ 45° / 135° orientation /45度/135度布置
Microcontroller (32 bit) 32位MCU
Firing loop IC' s (each for 4 loops) 点火IC，每片4路
Energy Reserve capacitors 储能电容
➔ i.e. 200ms after battery loss 断电后可保持200ms
Peripheral IC' s w/ watchdog 带有看门狗的外围芯片
Inductor for up/down convert 升压/降压转换电感
i.e. 1 s after ignition on 用于点火后1s
Diode for load dump protection 电源过载保护二极管
➔ against voltage peaks 过压保护



Micro-machined 微机械加速度传感器 acceleration sensor:

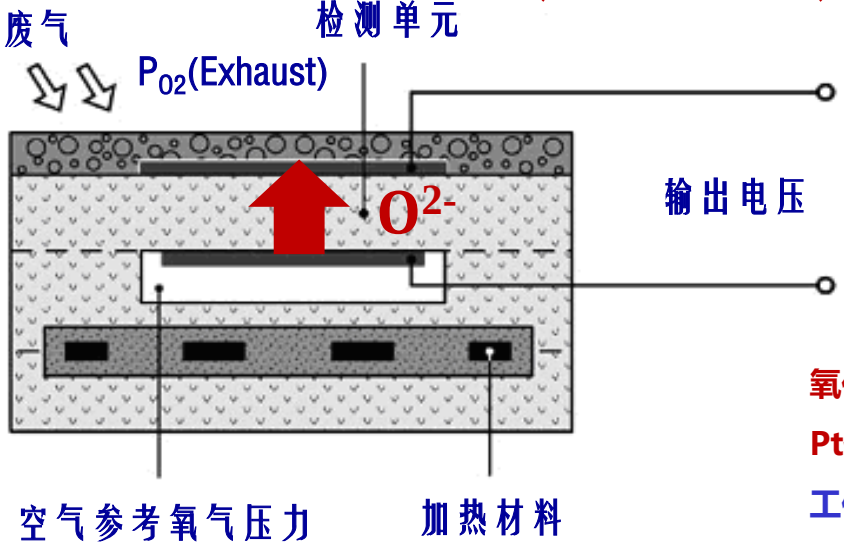
capacitive 电容式

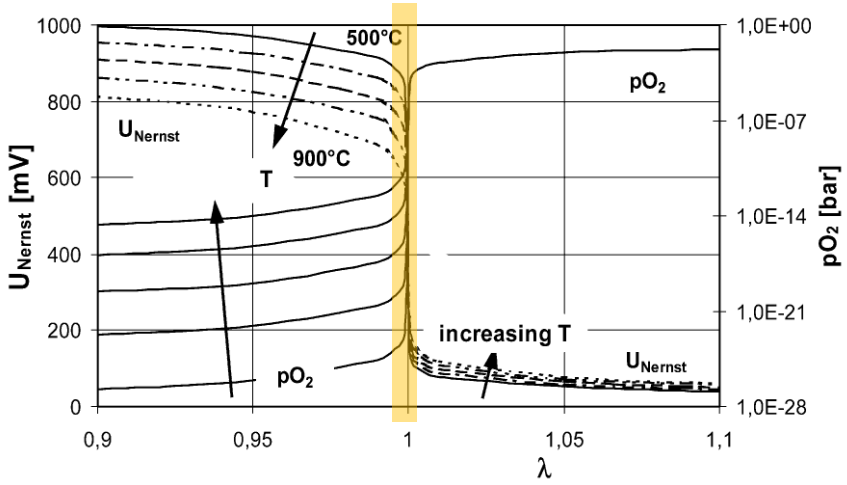


§1 传感器工作原理/The Principle of Sensors in AECs

● 窄范围氧传感器/Narrow Range Lambda Sensor

$$V = \frac{RT}{2F} \ln\left(\frac{\sqrt{P_{o2}(air)}}{P_{o2}(exhaust)}\right)$$



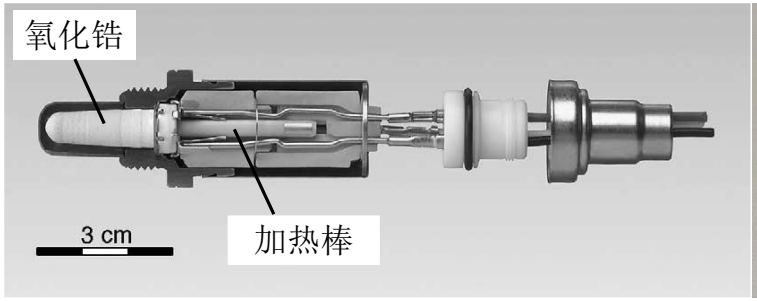


氧化锆（ZrO2, Zirconia）内外表面有Pt催化剂+多孔陶瓷层

Pt催化剂：将排气中的氧和CO结合变成CO2，增大两侧浓度差

工作原理：氧化锆对氧离子有扩散作用，氧离子浓度差->产生电压

工作范围：适用于理论空燃比14.7附近区域



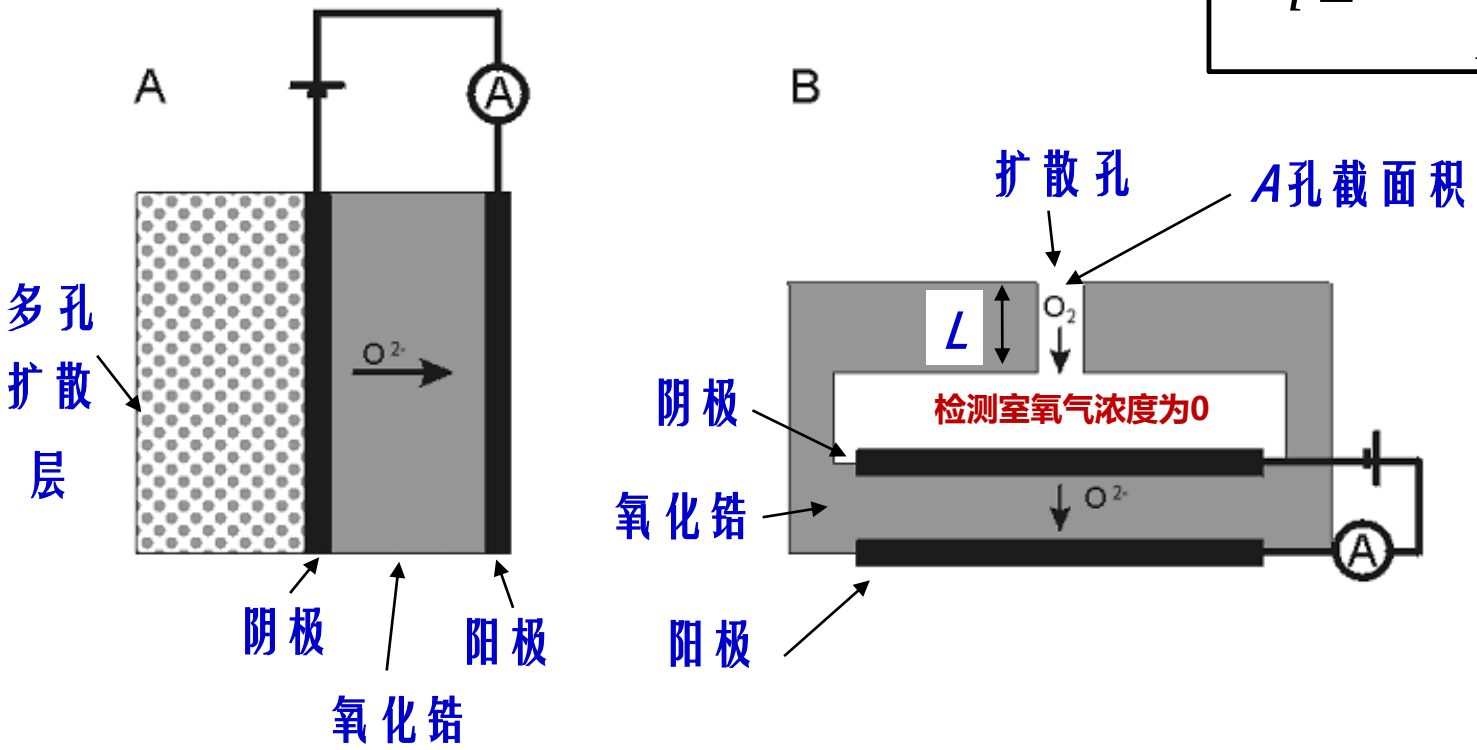


宽范围氧传感器/Wide Range Lambda Sensor

- C_0 为扩散孔外部氧的浓度 / Oxygen density in exhaust gas
- I 为扩散散电流 / diffusion current
- F 为法拉第常数/ Faraday constant
- Do 为扩散系数（取决于材料） / Coefficient of diffusion

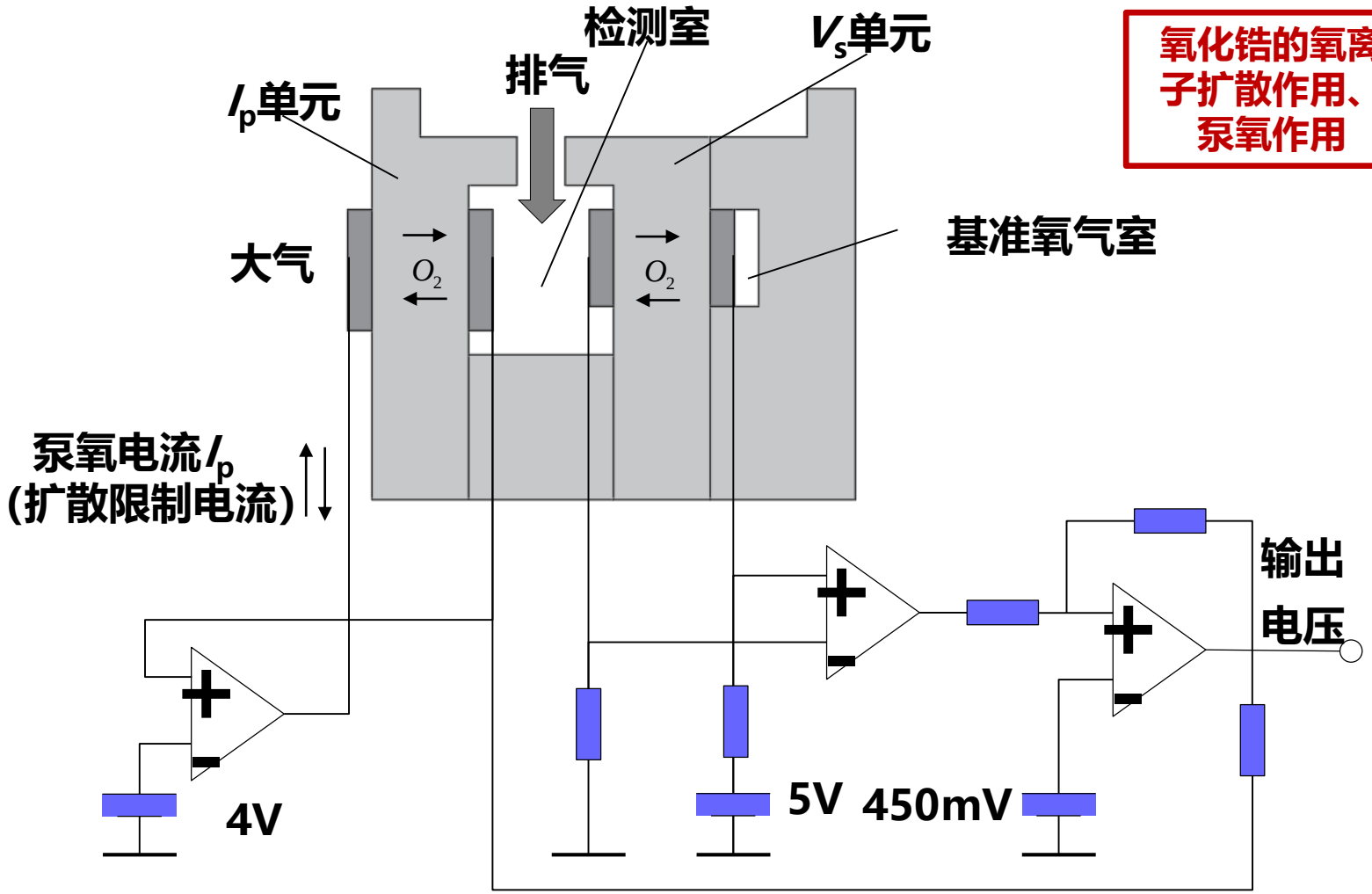
检测室氧气浓度接近0时

$$i = \frac{4FD_O Ac_O}{L}$$



泵氧原理：通过主动控制外部电流，可调控检测室氧气浓度

● 宽范围氧传感器/Wide Range Lambda Sensor





● 宽范围氧传感器/Wide Range Lambda Sensor

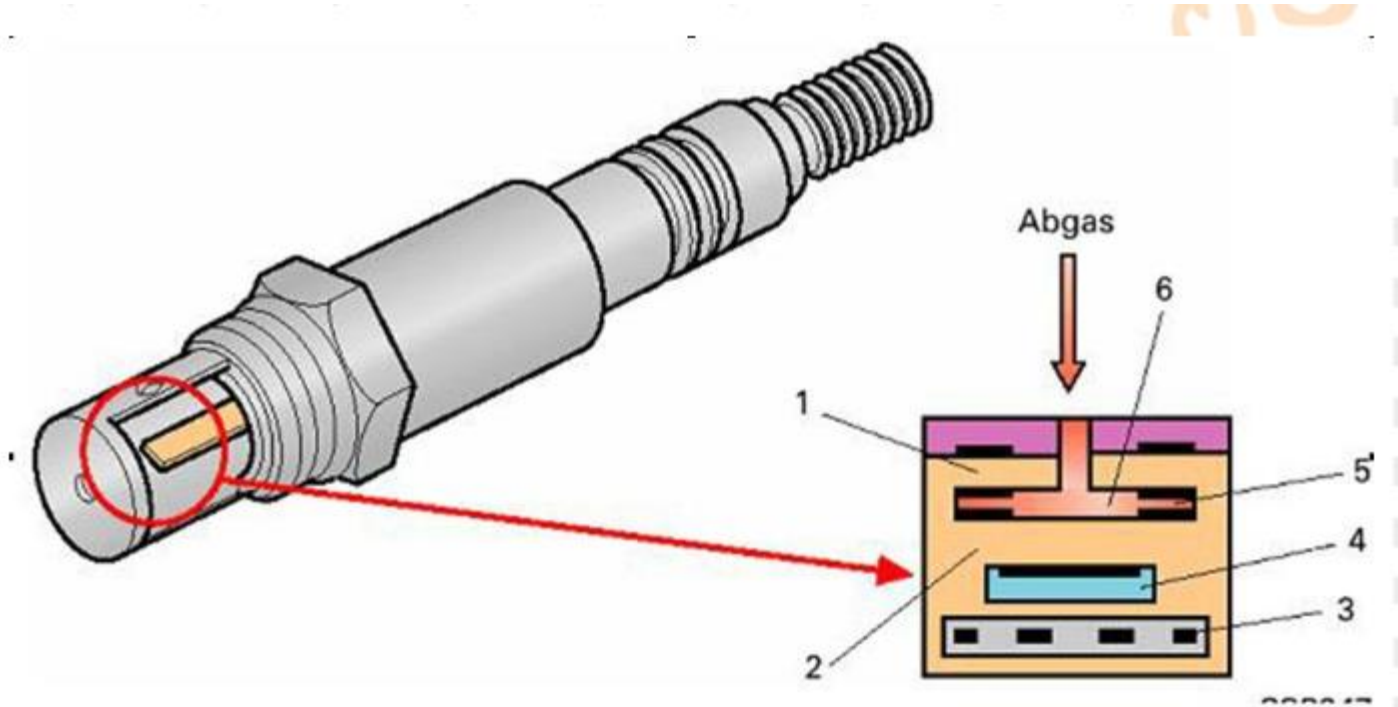
泵氧单元 (Ip)：控制泵氧电流 Ip （又称为扩散限制电流，diffusion limit current），可调节从检测室泵入或泵出的氧气量
检测单元 (Vs)：由小固定电流供电，将微量的氧气从检测室移到基准氧气室，“窄氧传感器”

工作原理：
排气进入检测室，在 Vs 检测单元中检测到电压；
通过控制泵氧电流 Ip ，调节检测室中气体浓度，将 Vs 控制在 0.45V；
此时，检测室中氧气浓度接近零（理论空燃比），此时 Ip 电流与排气空燃比之间存在对应关系

工作范围：适用于宽范围空燃比范围，比如10~20

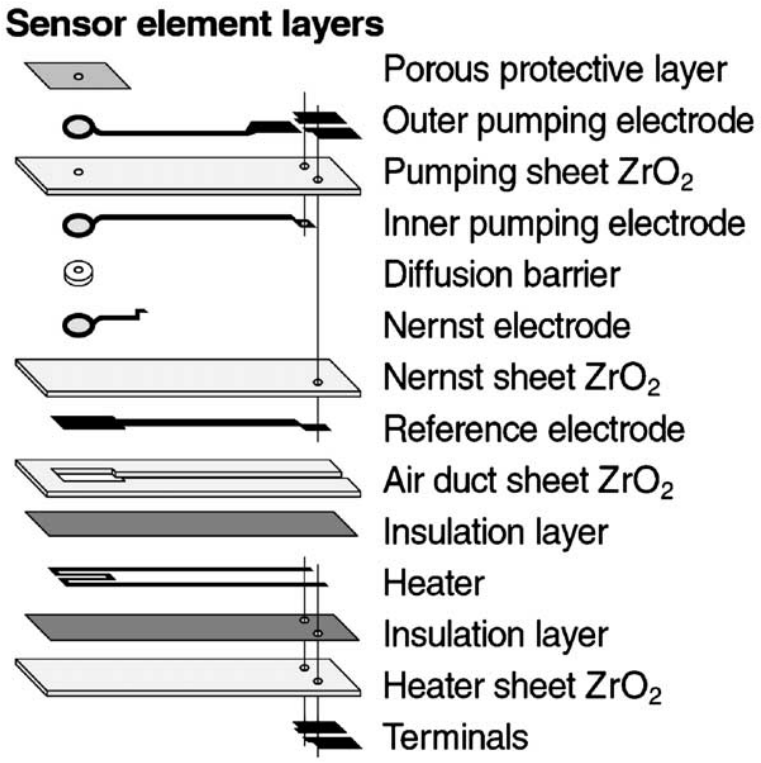
泵氧单元+检测单元（窄范围氧传感器）

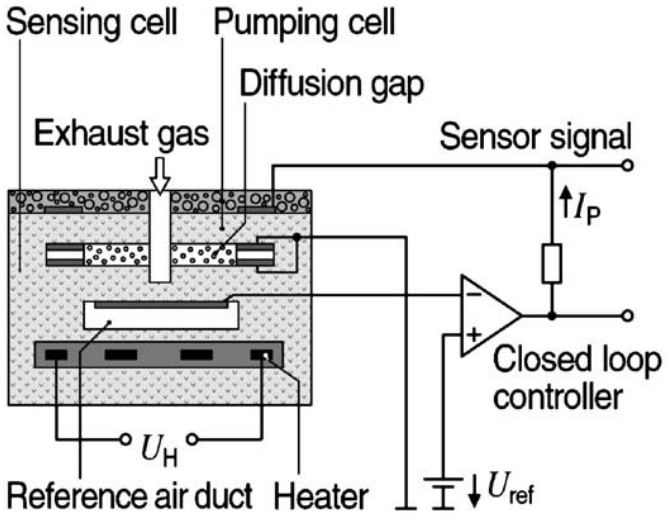
- 宽范围氧传感器/Wide Range Lambda Sensor

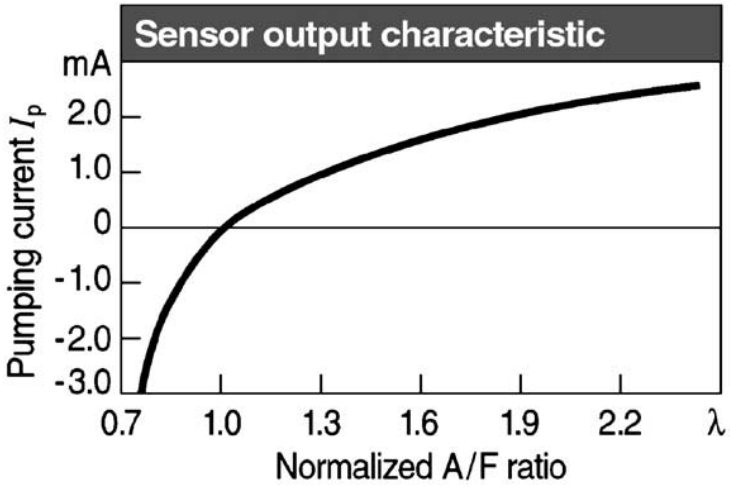


1 单元泵 2 能斯托单元3 λ传感器加热器4 外界空气通道 5测量室 6 放氧通道

● 宽范围氧传感器/Wide Range Lambda Sensor







● 氮氧化物传感器/NOx Sensor

主泵（宽氧传感器）+辅泵+NOx测量泵

当排气浓/稀时，主泵泵入/出氧气，确保室内的氧气浓度接近零；

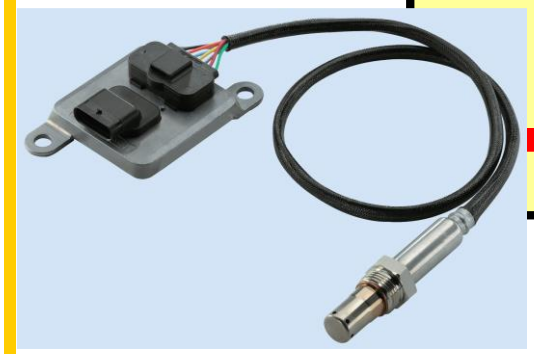
空燃比（Lambda）输出信号

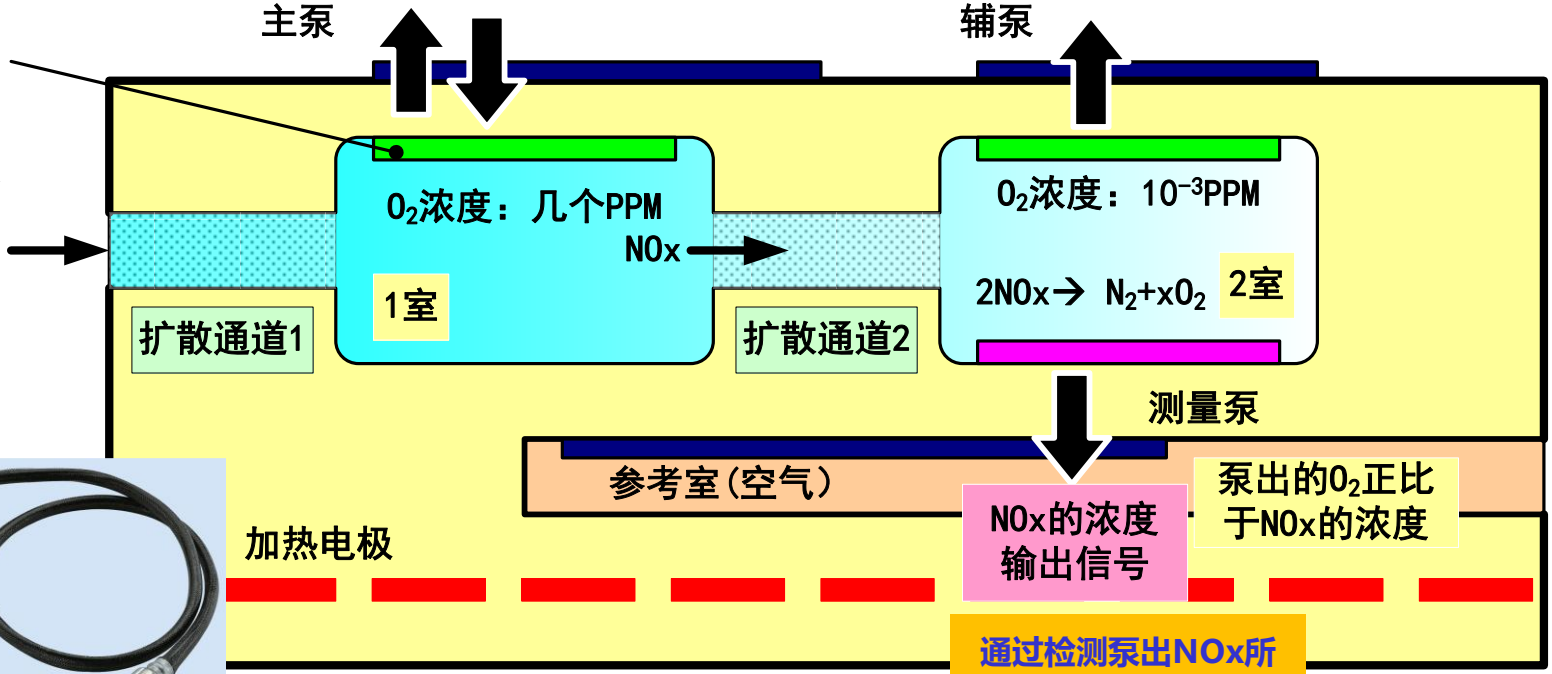
主泵工作原理与宽氧传感器完全一致

将剩余的氧全部泵出，为检测NOx创造条件

主泵中的铂催化电极将使得HC、CO和H_2氧化；

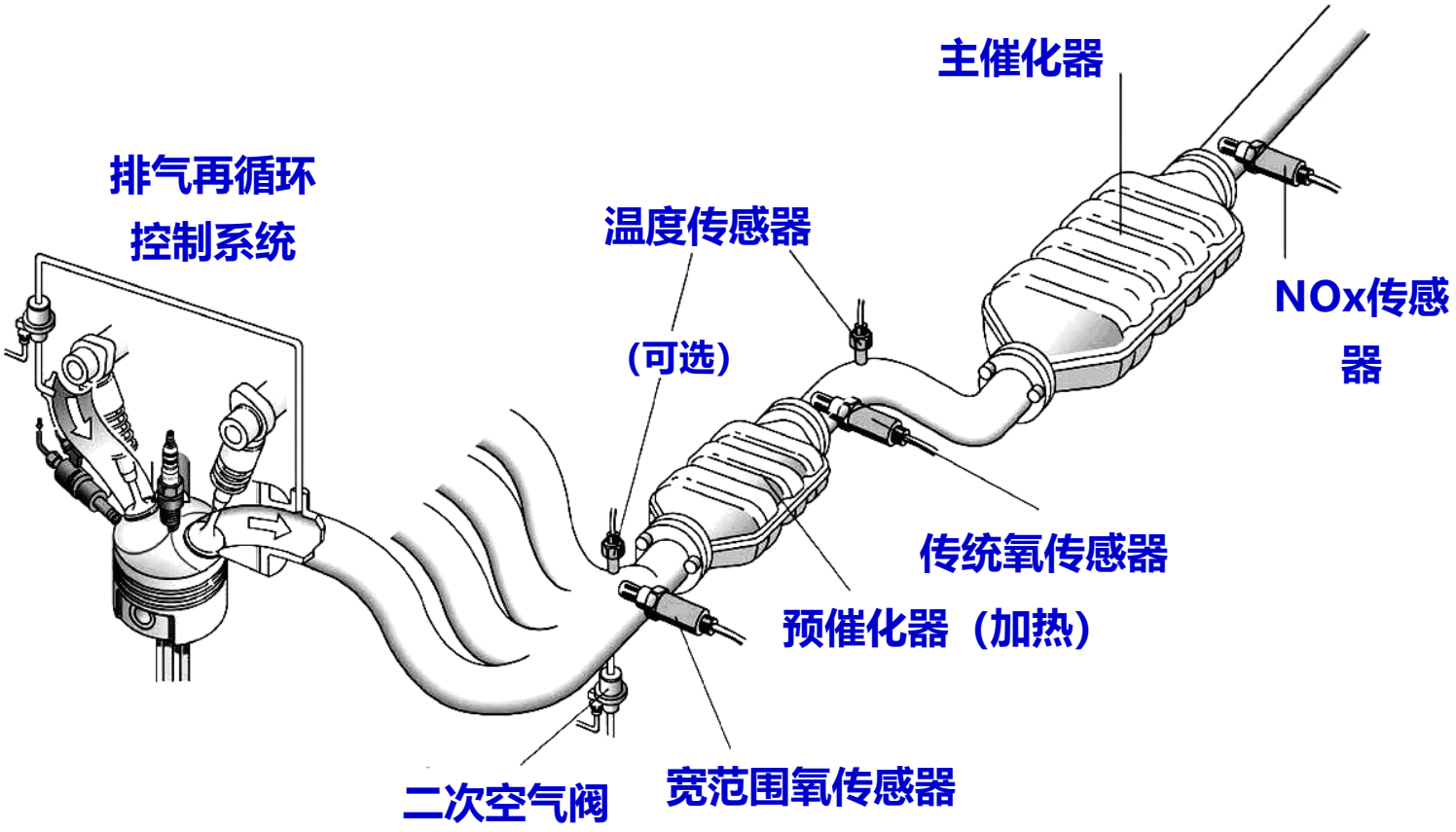
排气，包含多种成分：O_2、NOx、HC、CO、H_2，...





通过检测泵出NOx所产生的O2的所需电流大小，得到NOx浓度

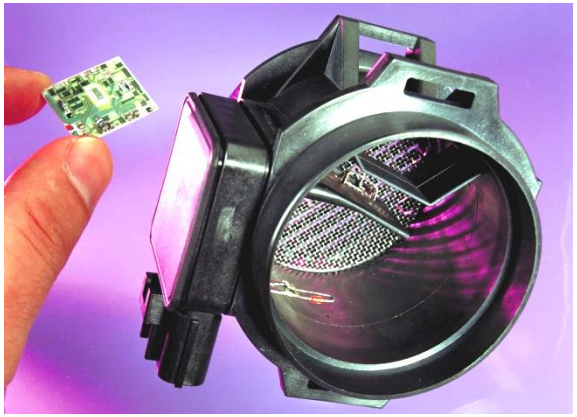
- 排放后处理系统/After treatment system





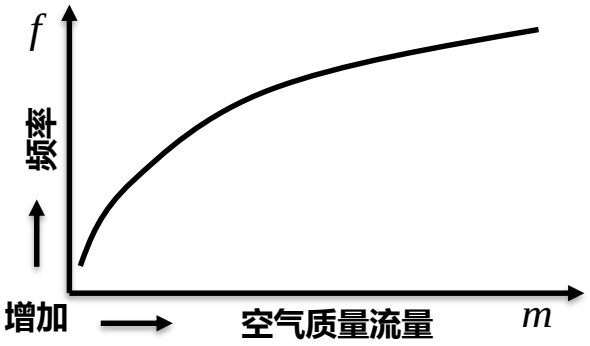
§1 传感器工作原理/The Principle of Sensors in AECs

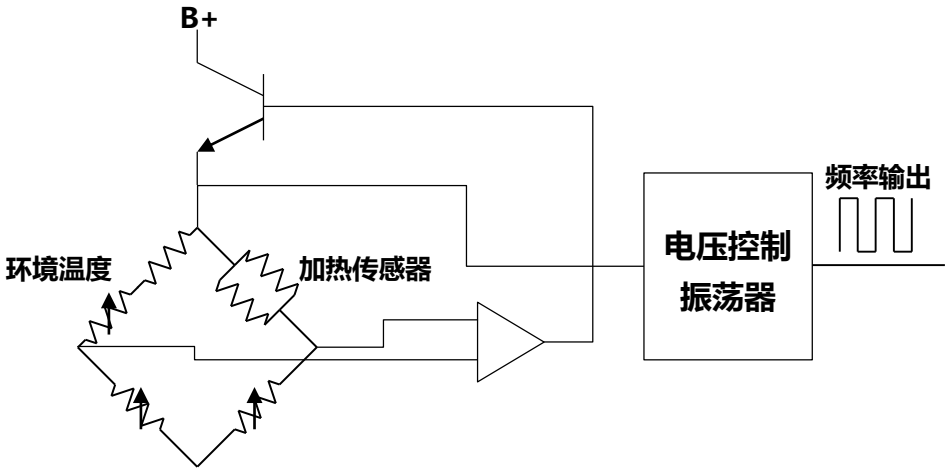
● 热线式空气流量计/Hot Wire Air Mass Flow Sensor



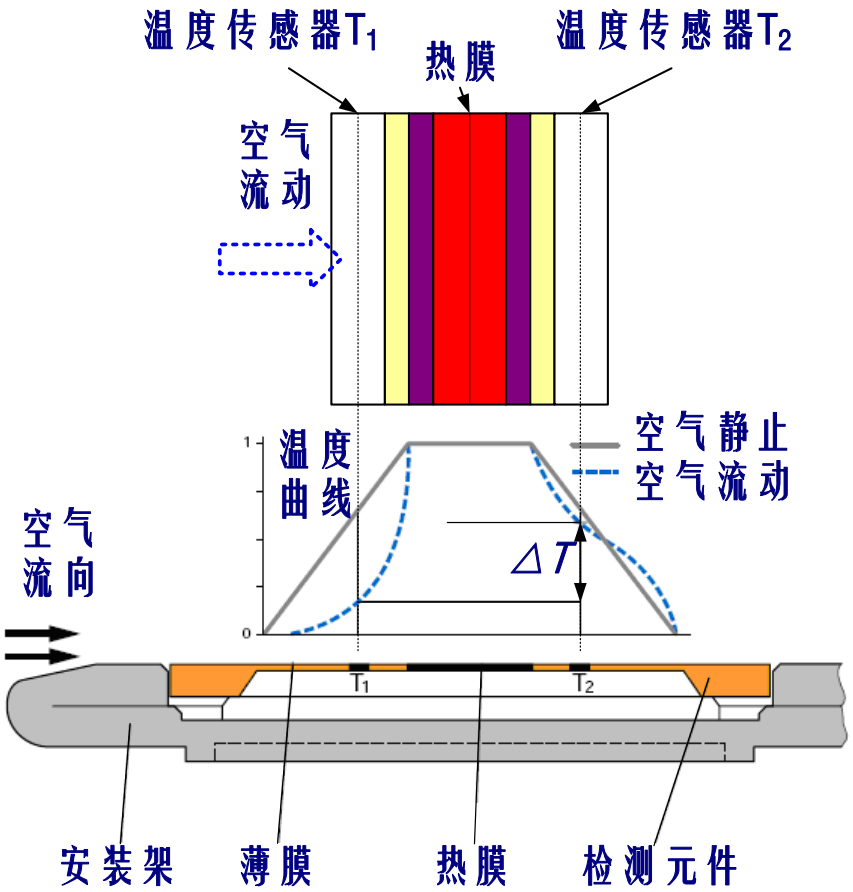
原理：

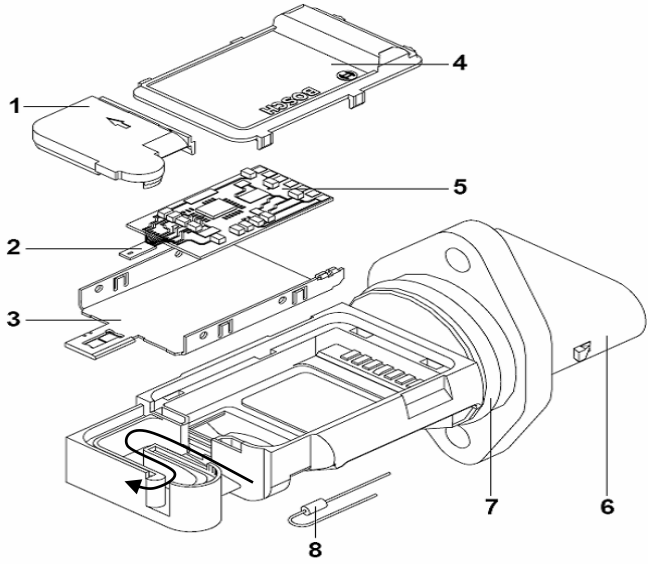
- 传感器（热线）的温度受流量影响
- 当热线温度比吸入空气高100℃时，惠斯通桥保持平衡
- 流量越大，温度越低，电阻越高，惠斯通桥电压失衡
- 提高电压控制器频率->提高供电电压->加大供电电流->提升热线温度->回复惠斯通桥平衡
- 电压频率与空气流量之间有对应关系





● 热膜式空气流量计/Hot Film Air Mass Flow Sensor



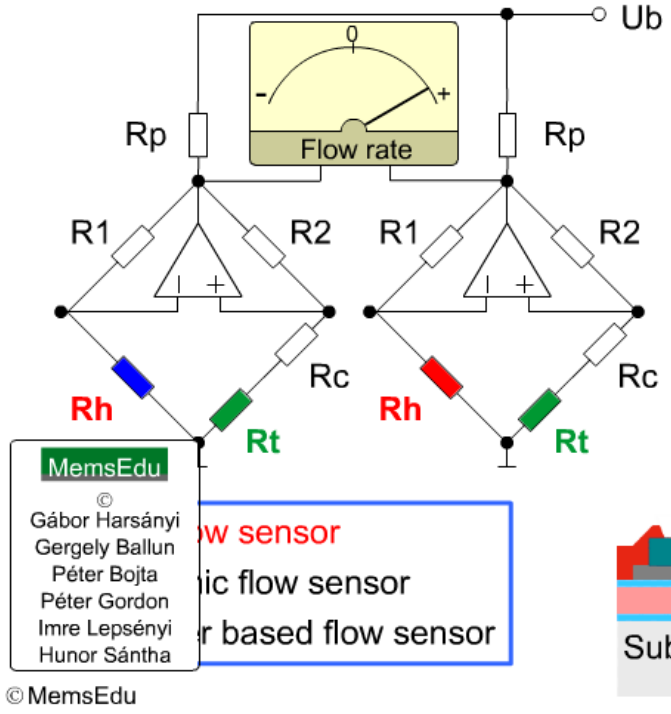


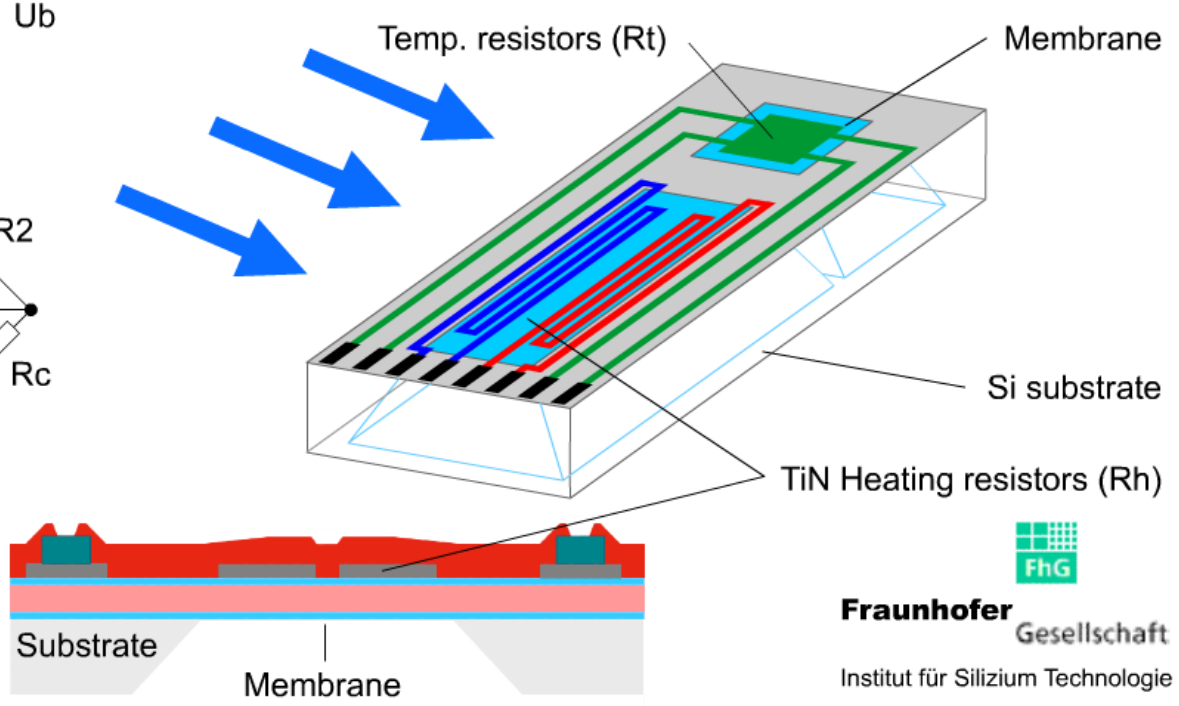
1. 测量通道上盖
2. 传感器
3. 安装板
4. 混合电路上盖
5. 混合信号处理电路
6. 接插件
7. O型圈
8. 附加的空气温度传感器

两个温度传感器的温差与流量有单调对应关系



● 热膜式空气流量计/Hot Film Air Mass Flow Sensor

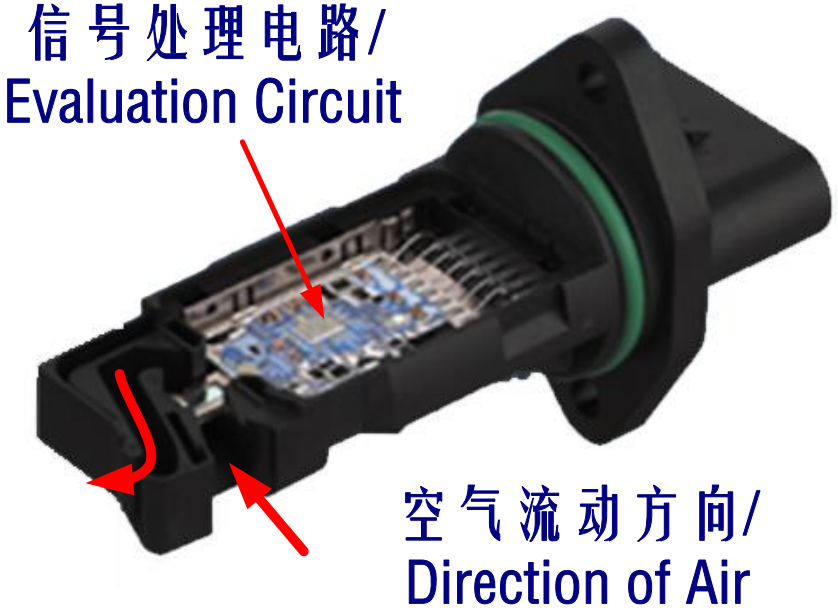




通过惠斯通桥测量两个温度传感器的值

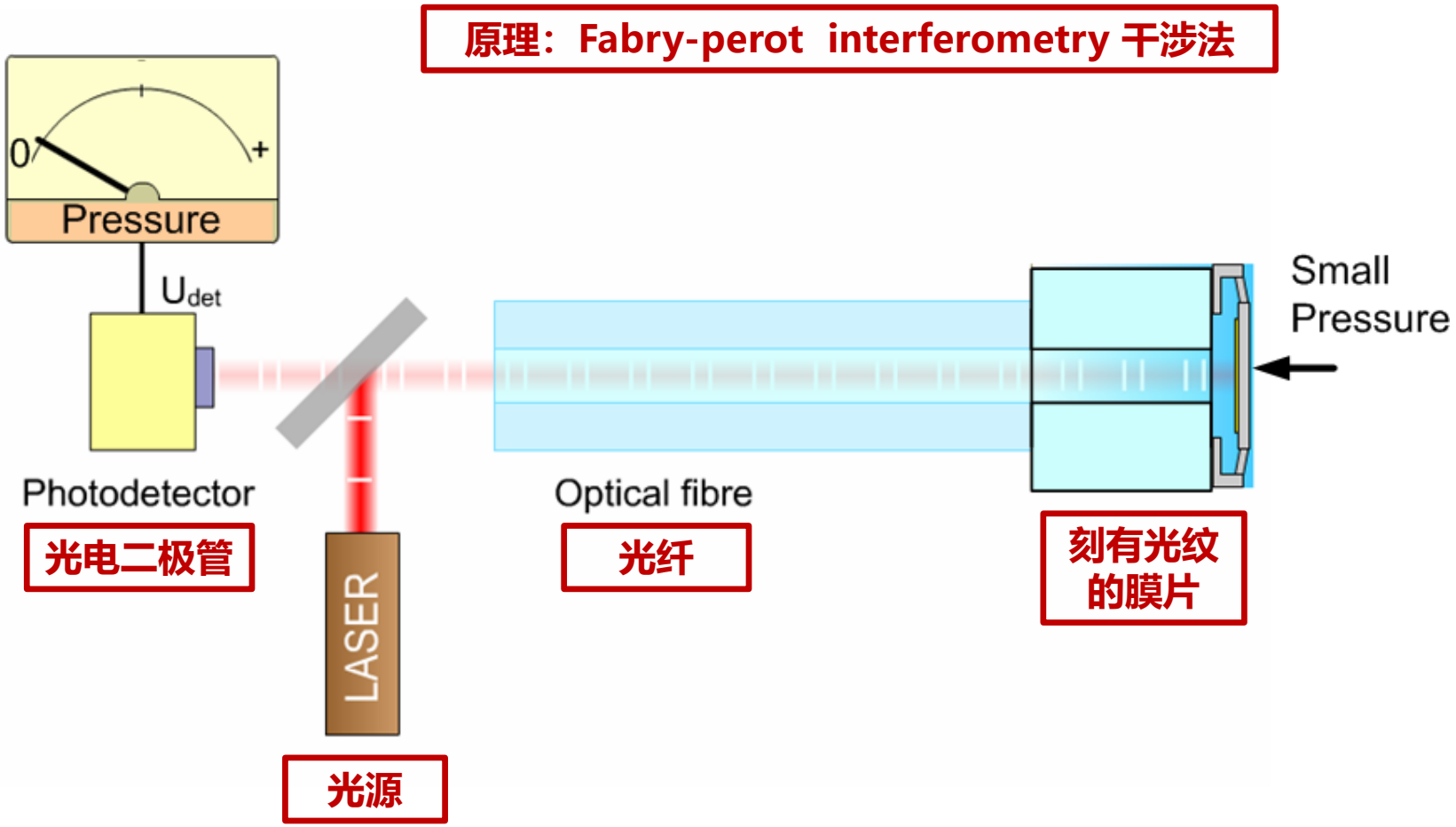


- 热膜式空气流量计/Hot Film Air Mass Flow Sensor

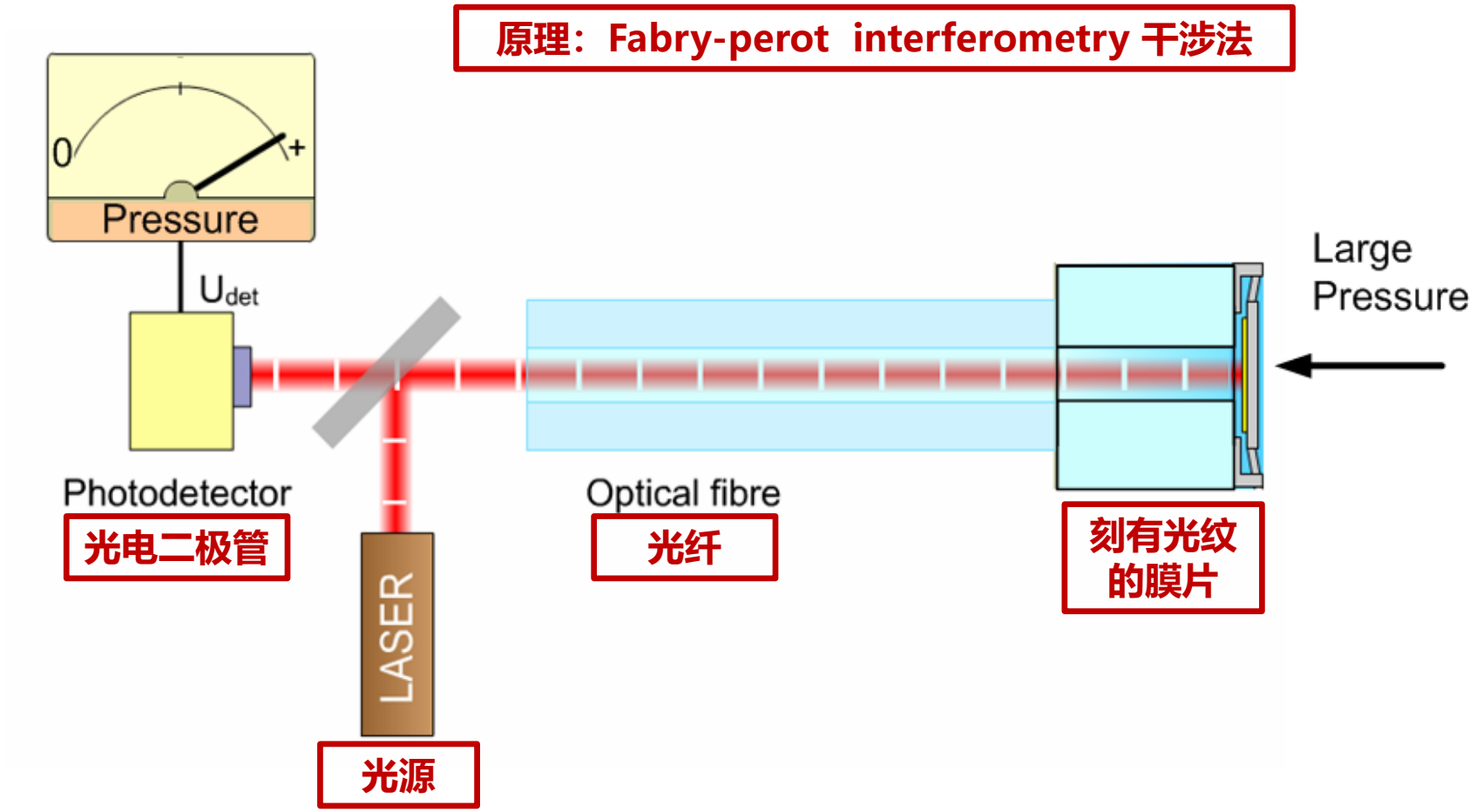




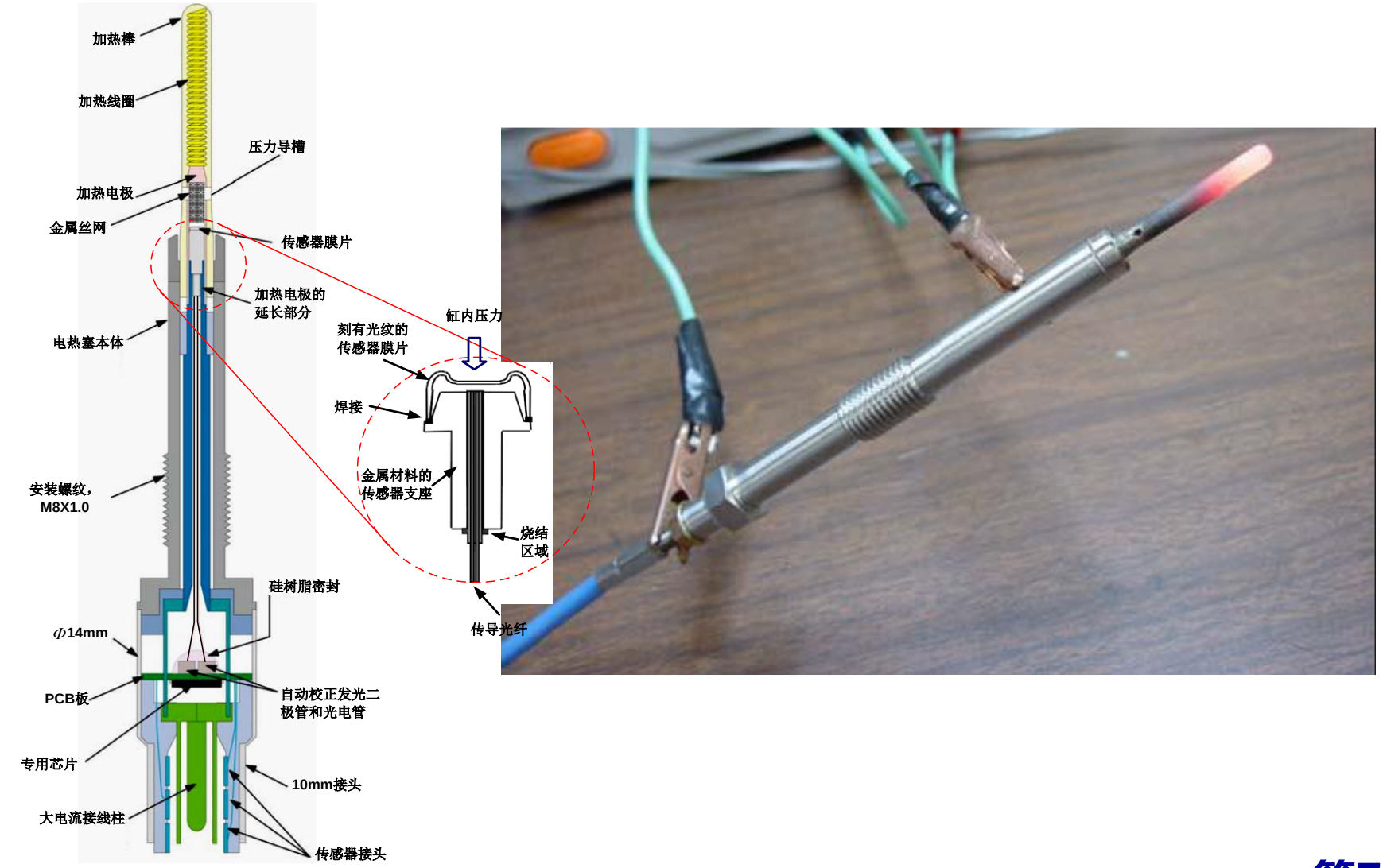
- 缸内压力传感器/In-Cylinder Pressure Sensor



- 缸内压力传感器/In-Cylinder Pressure Sensor



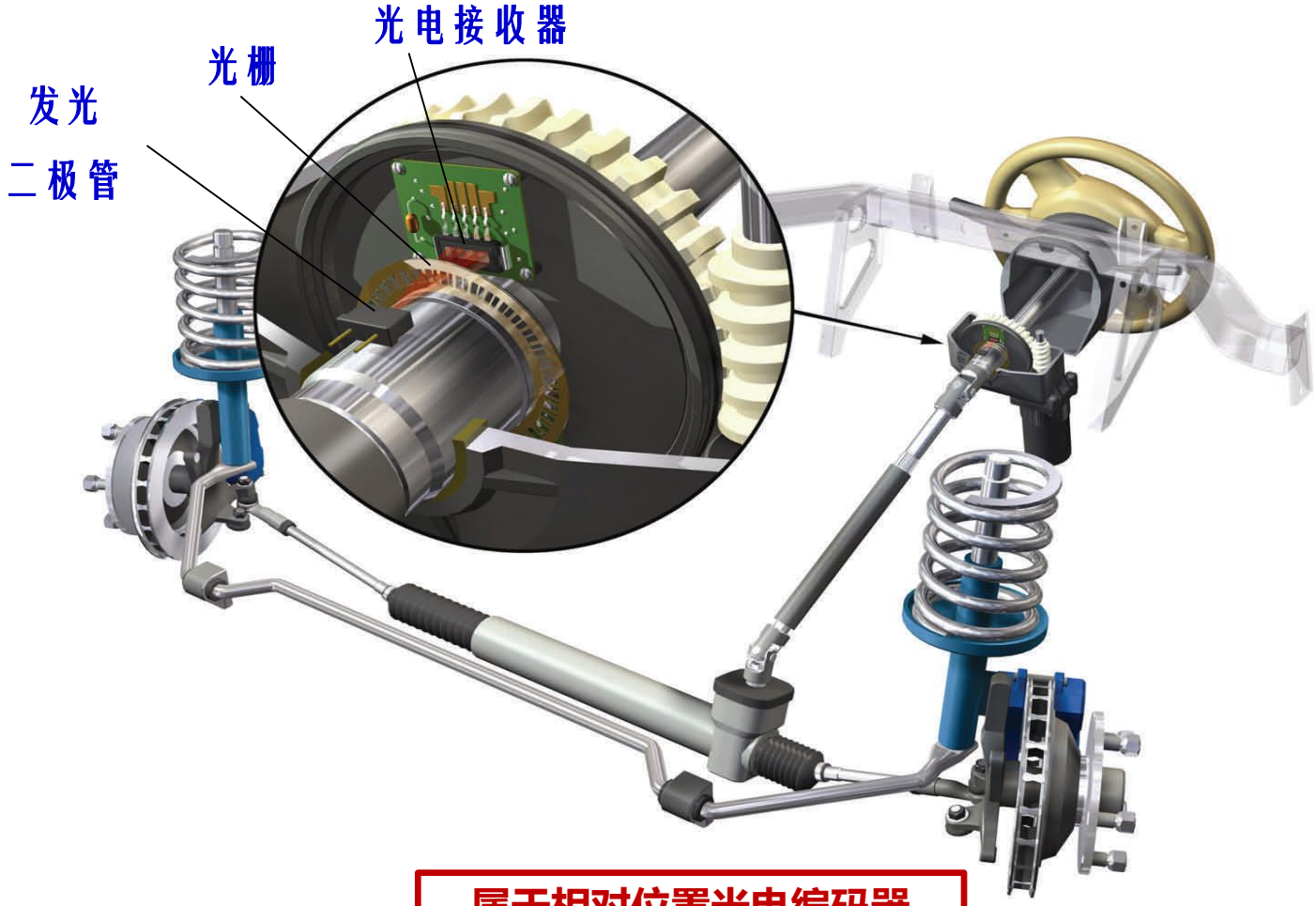
● 缸内压力传感器/In-Cylinder Pressure Sensor





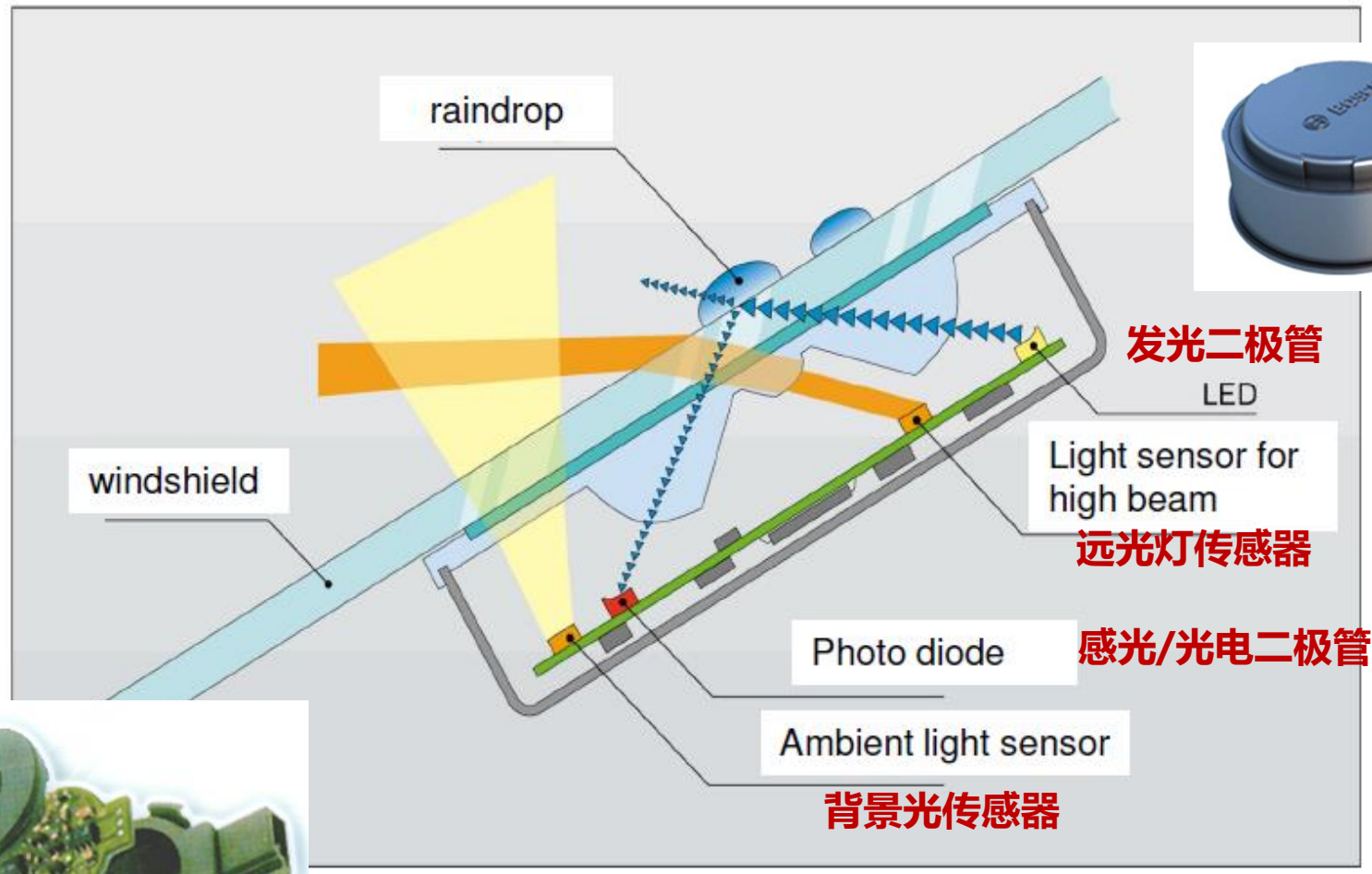
§1 传感器工作原理/The Principle of Sensors in AECs

- 光电传感器：绝对/相对位置光电编码
- 方向盘转角传感器/Steering Angle Sensor



属于相对位置光电编码器

- 光电传感器：雨水/光强传感器/Rain Light Sensor









§目录 / Index

1. 传感器、信号接口电路及运算放大器
   - ① 传感器工作原理 （Principle of Sensors）
   - ② 传感信号接口电路 (Interface circuits of sensor signals)
   - ③ 典型运算放大器 (Typical Operational Amplifier)
2. 执行器、开关器件及驱动接口电路
   - ① 执行器及四种功率开关器件
   - ② 驱动信号接口电路：四拓扑结构
   - ③ 驱动电路的续流回路
   - ④ 驱动电路的故障诊断和容错控制
   - ⑤ 汽车专用功率驱动集成芯片介绍

重点：运算放大器(Operational Amplifier)和MOSFET的使用



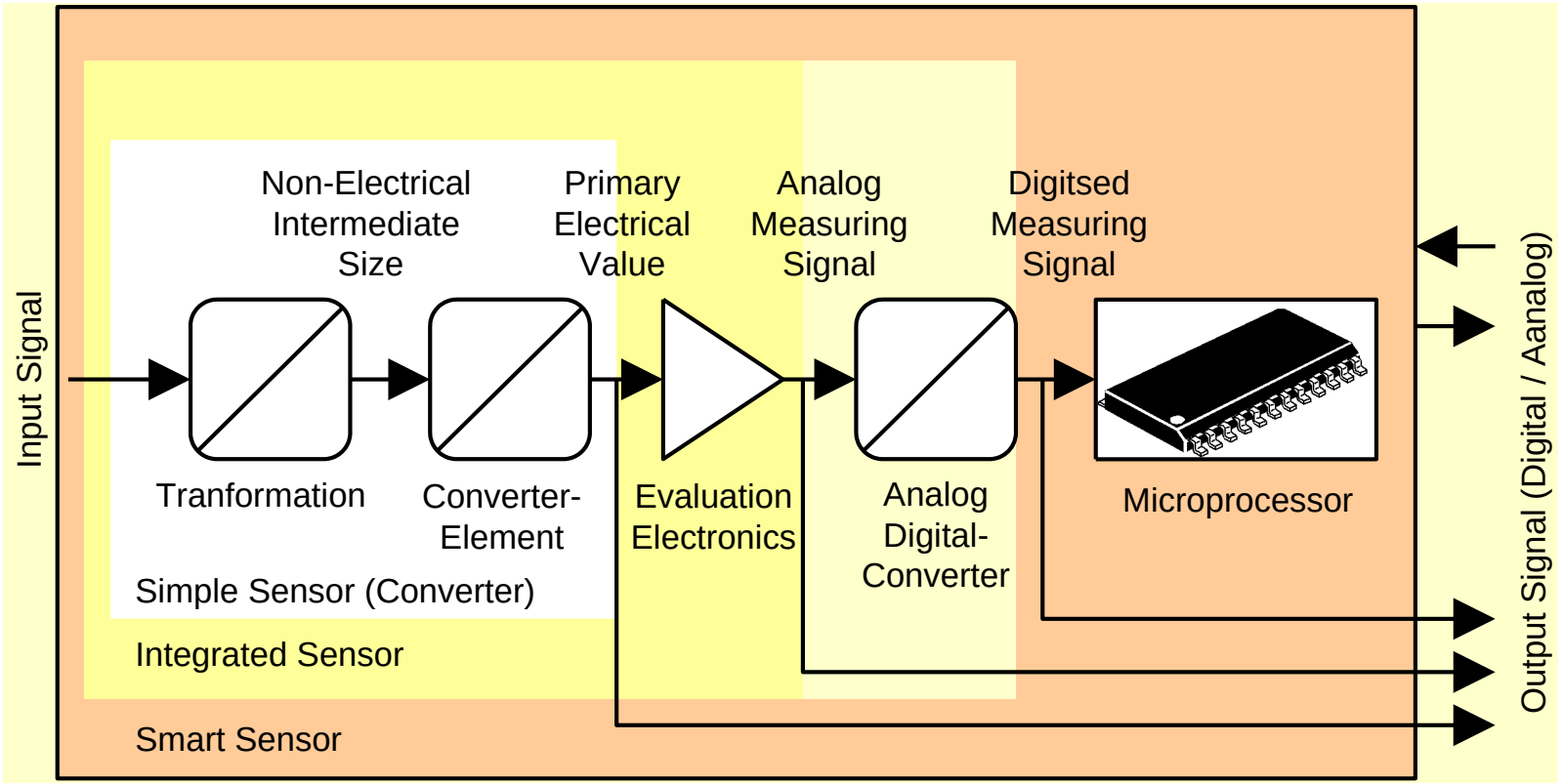
§2 传感信号接口电路
/ Interface circuits of sensor signals

- 传感器输出信号可以分为模拟量和数字量，其处理方法都是以运算放大器（简称运放）为基础/The output signal of sensor belongs analogue or digital, the Operational Amplifier is the basic element of construction the evaluation circuit.
- 以运放为基础的放大、滤波、积分和微分等电路单元，是ECU传感器接口电路的主要组成部分/ Typical signal process includes amplifying, filtering, integration, differentiation an so on.
- 掌握运放的基本性质和合理地选择和使用运放，是设计ECU传感器接口电路的基础/The basic properties of OA is the fundamental of designing ECU input signal conditioning circuit.



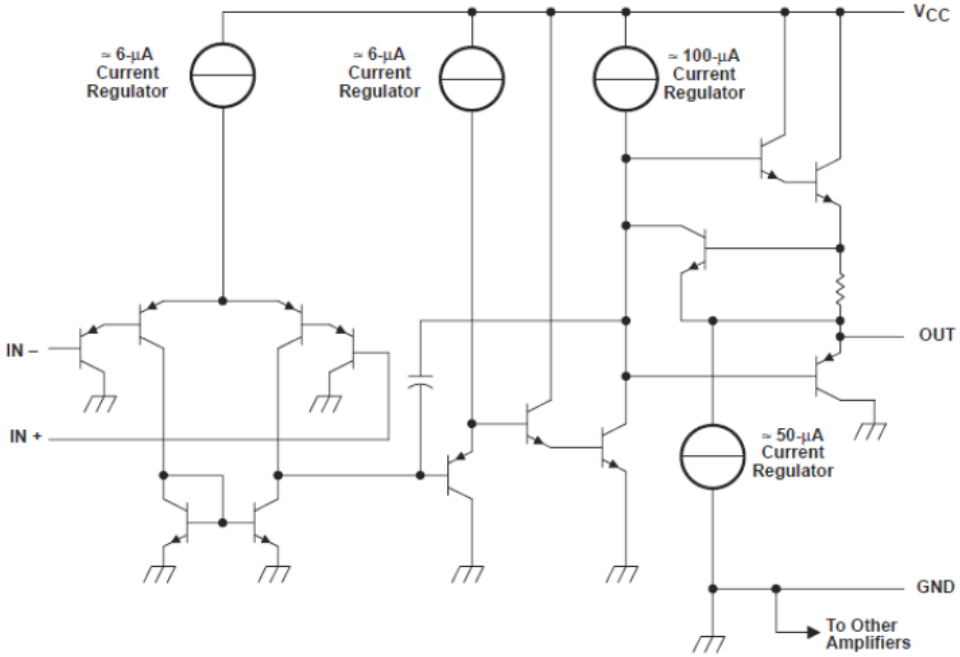
§2 传感信号接口电路
/ Interface circuits of sensor signals

● 车载传感器的集成度划分/ Classification of integration of sensors

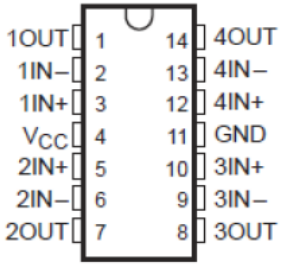


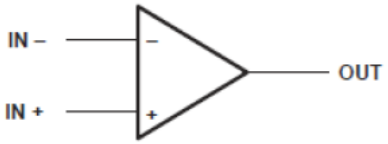
感知——传送——处理——采集

LM324:集成运算放大器示例



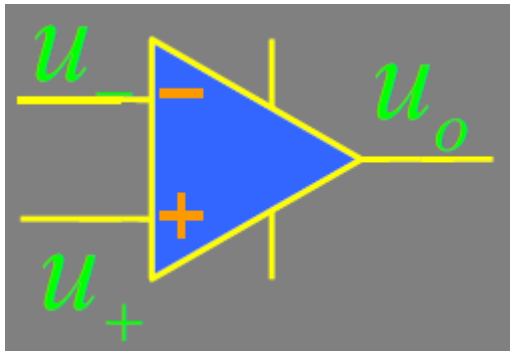
COMPONENT COUNT (total device)	
Epi-FET	1
Transistors	95
Diodes	4
Resistors	11
Capacitors	4

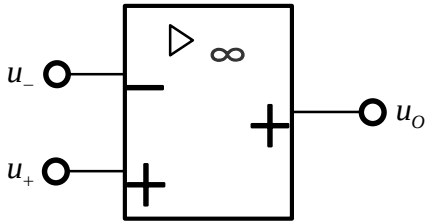






● 集成运算放大器基本知识：同相反相

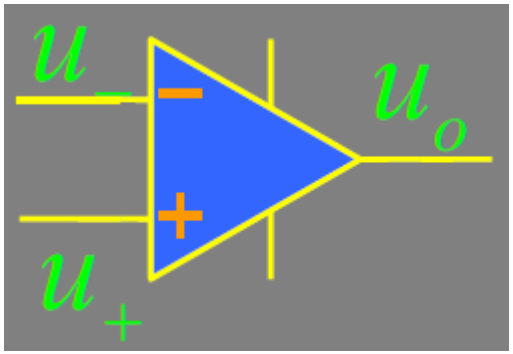


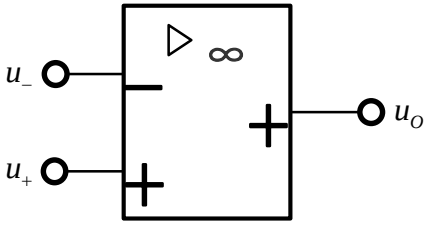


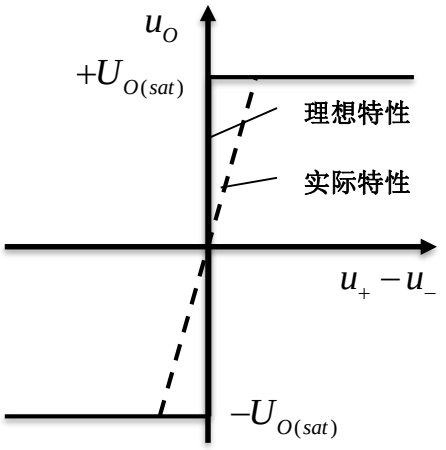
+：同相端，输出电压的极性与输入口相同

-：反相端，输出电压的极性与输入口相反

● 集成运算放大器基本知识：线性/饱和







线性区间 $u_o = A_{uo}\left(u_+ - u_-\right), \left|u_+ - u_-\right| \le \Delta u$

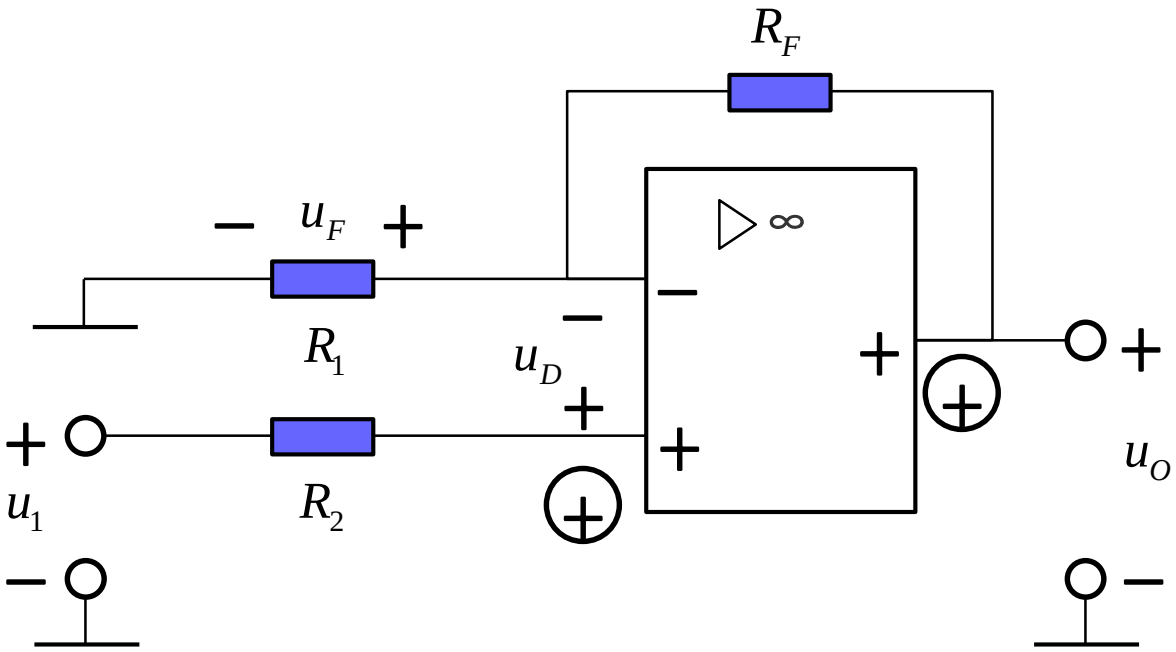
饱和区间 $u_o = \begin{cases} +u_{o(sat)}, u_+ - u_- > \Delta u \\ -u_{o(sat)}, u_+ - u_- < -\Delta u \end{cases}$

虚短、虚断

虚断：输入端的输入电流近似0

开环增益：A_{uo}

● 集成运算放大器基本知识：负反馈电路



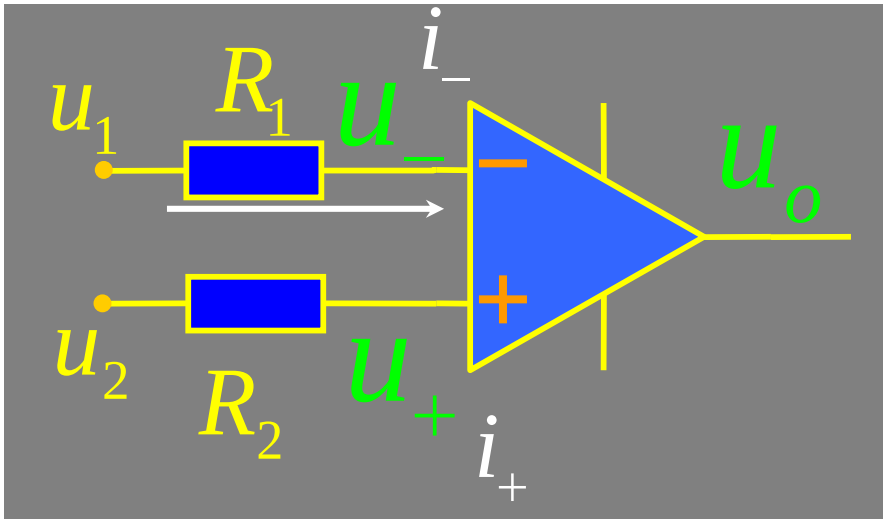
- 负反馈电路：输出端通过R/L/C器件与反相端(-)相连
- 负反馈电路运放单元一定工作于线性区→虚断虚短
- 闭环增益G：负反馈运放电路中，输出电压相对于输入电压的增益



- **集成运算放大器基本知识：线性区虚短虚断**
  - “虚短” / Virtual short circuit
  - “虚断” / Virtual open circuit

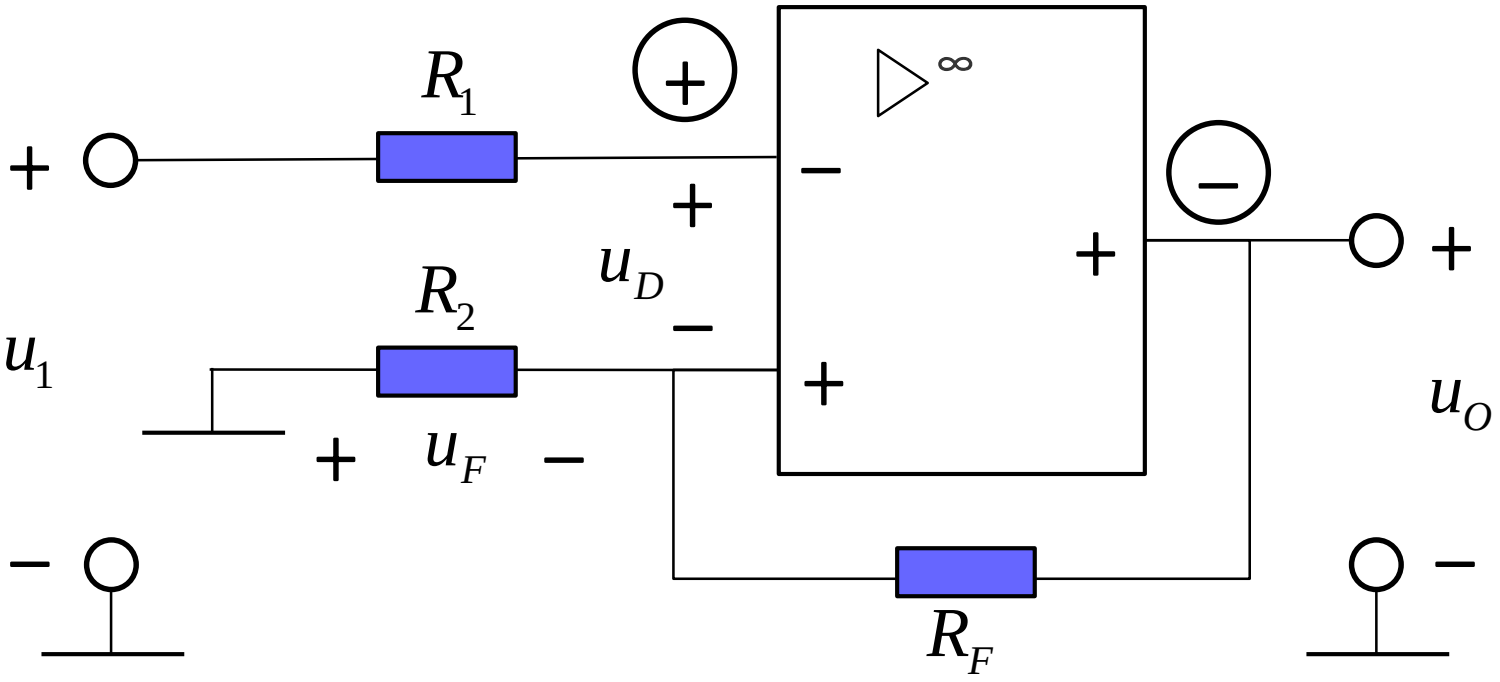
$$u_- = u_+$$

$$i_- = i_+ = 0$$





● 集成运算放大器基本知识：正反馈电路



- 正反馈电路：输出端通过R/L/C器件与同相端(+)相连
- 正反馈电路运放单元一定工作于饱和区→虚断

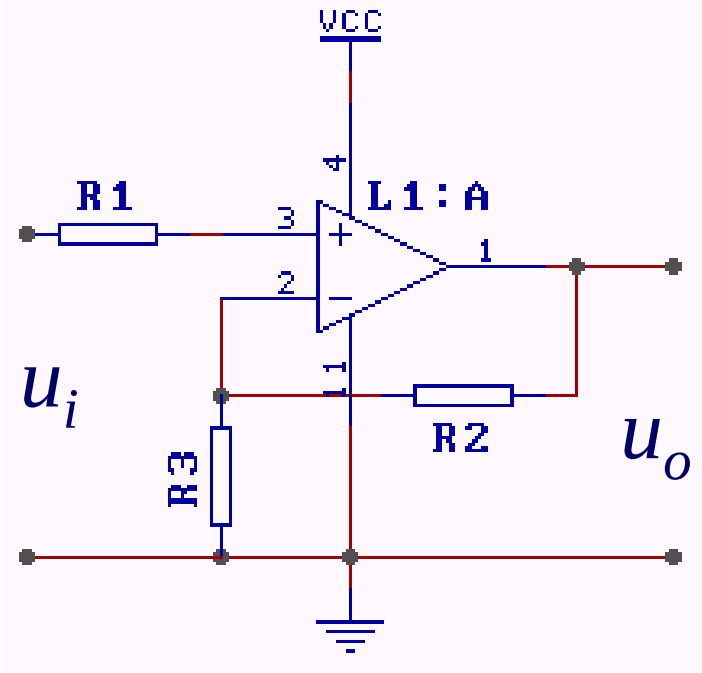


- **典型负反馈运放电路**
  - ◆同相比例电路/ Non-inverting amplifier
  - ◆同相端输入信号

思考：运放电路是否多此一举？意义何在?

$$u_o(t)=(1+R_2/R_3)\,u_i(t)$$

$$U_o(s)=(1+R_2/R_3)\,U_i(s)$$



$R_1=R_2//R_3$　　平衡电阻：消除静态基极电流对输出电压的影响

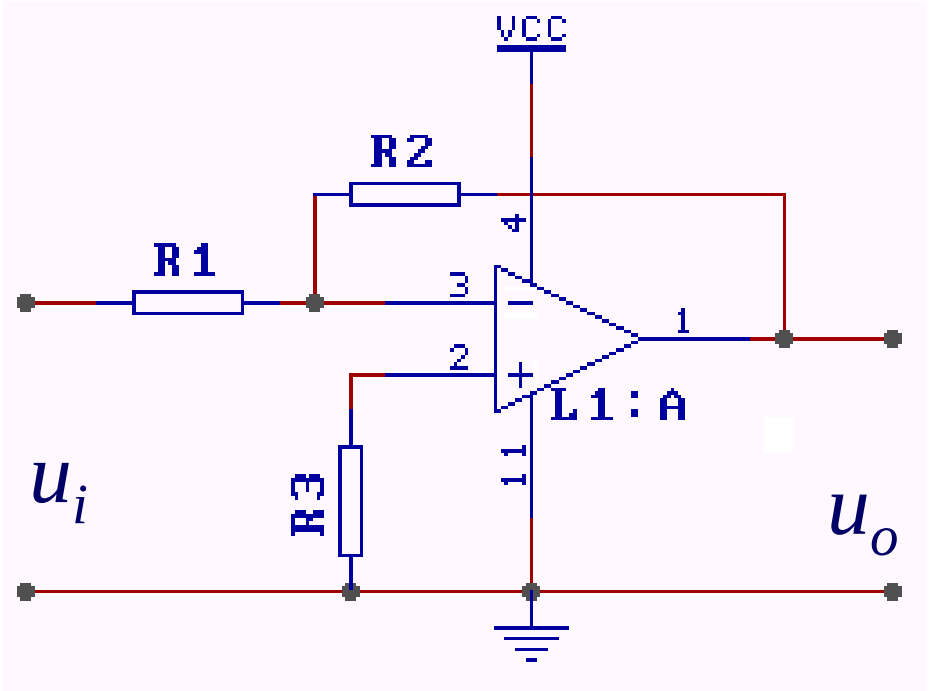
闭环增益：$G=1+R_2/R_3$



§2 传感信号接口电路
/ Interface circuits of sensor signals

◆反相比例电路/ Inverting amplifier

◆反相端输入信号



$$U_o(s) = (-R_1/R_2)\, U_i(s)$$

$R_3 = R_1 // R_2$　　平衡电阻：消除静态基极电流对输出电压的影响

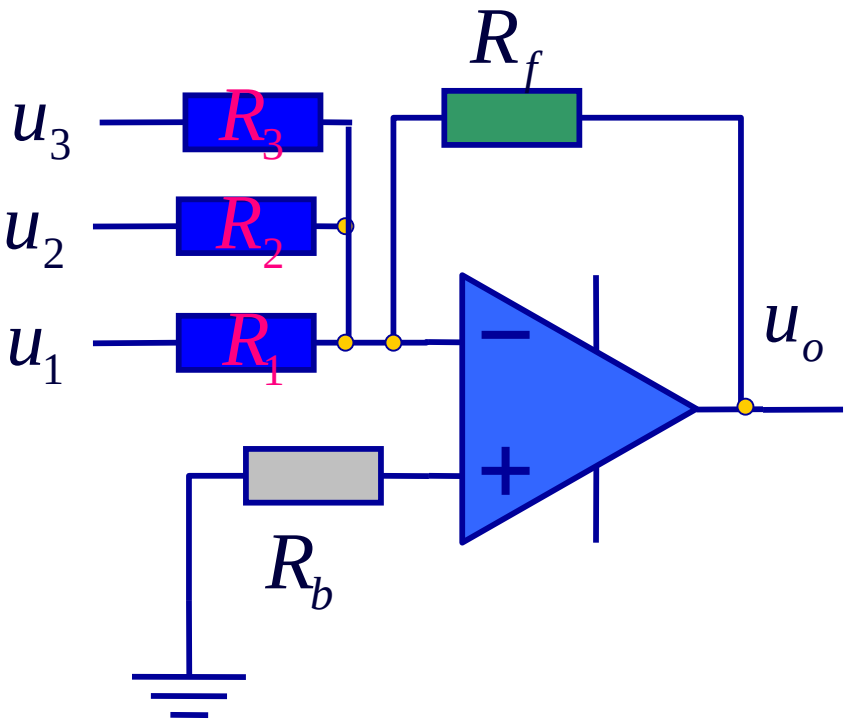
闭环增益：$G = -R_1/R_2$



● 加法器/ Summer

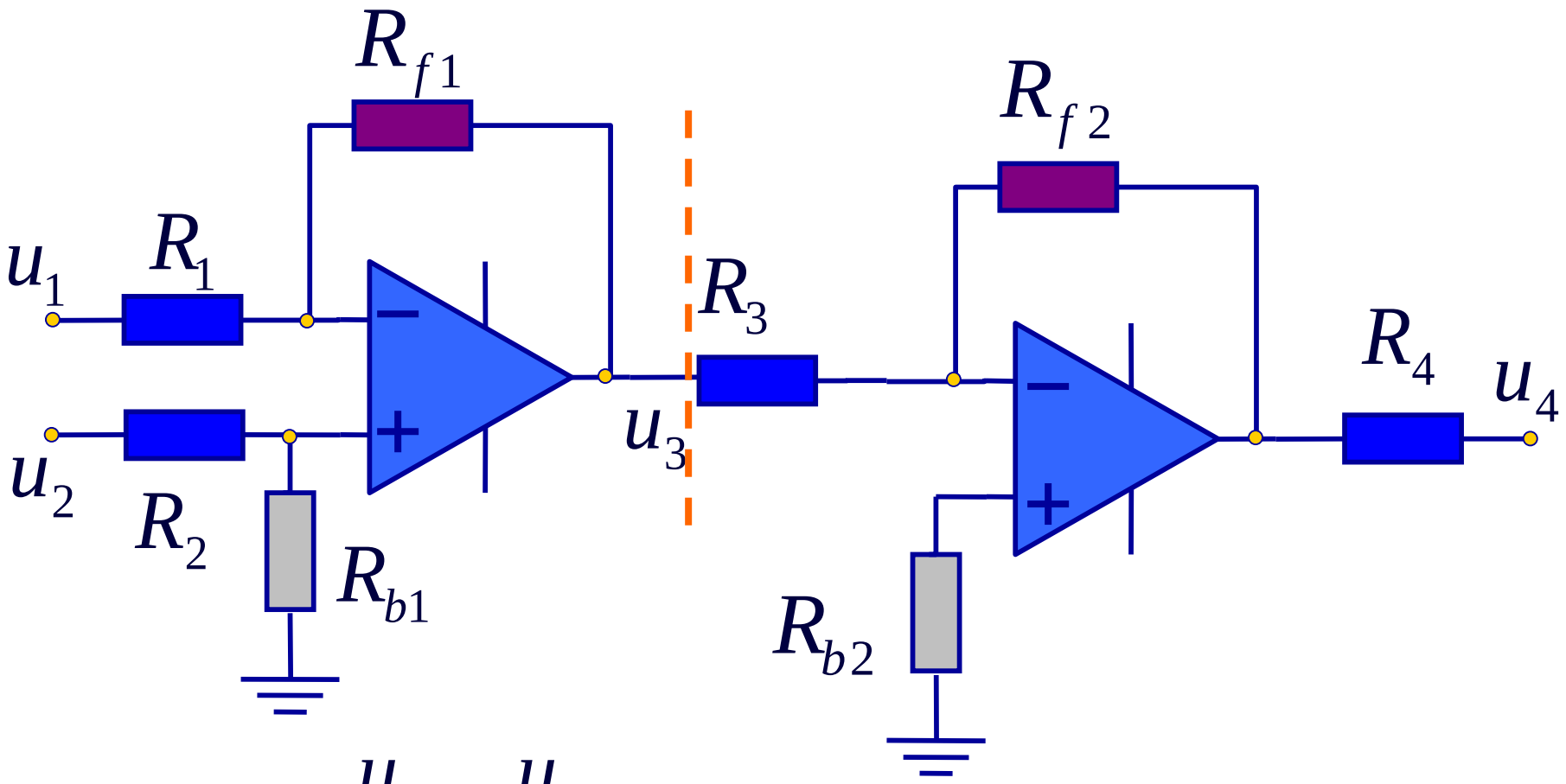
$$u_o = -R_f\left(\frac{u_1}{R_1} + \frac{u_2}{R_2} + \frac{u_3}{R_3}\right)$$

$$R_b = R_f \,//\, R_1 \,//\, R_2 \,//\, R_3$$





● 叠加原理/ Decoupling between OAs

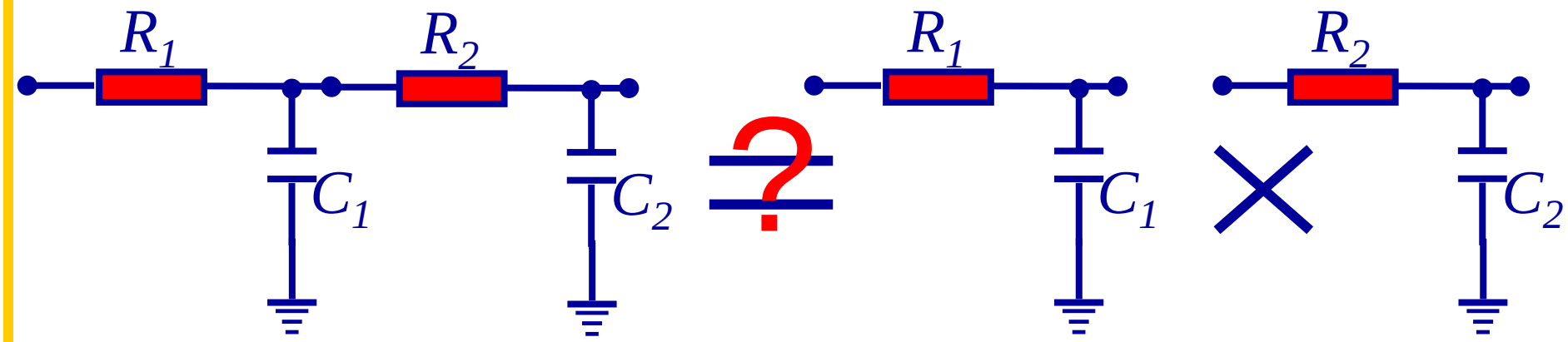


$$u_3 = R_{f1}(\frac{u_2}{R_2} - \frac{u_1}{R_1})$$

$$R_1 // R_{f1} = R_2 // R_{b1}$$



- 无源滤波器/ Passive filters

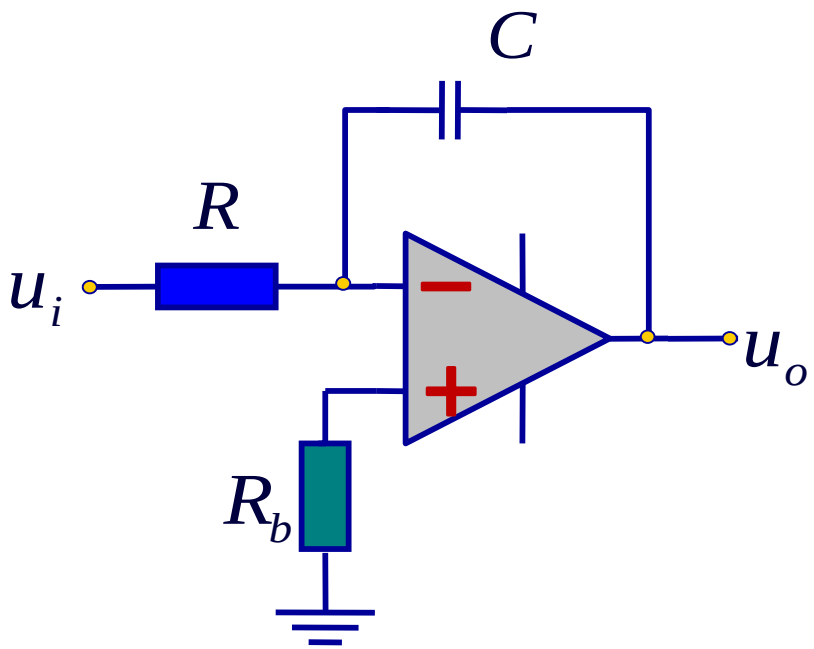


是否满足叠加原理/ Does the circuit fulfill decoupling principle?



◆ 积分电路/Integration

物理积分电路容易很饱和，所以目前一般采用数字积分，即微处理器将信号采集后再积分，优点是具有灵活的处理方法和手段。/ Become of saturation of analogue integration, it is recommended to use digital integration. By using MCU, the signal is sampled by AD first, then can be processed flexibly.



$$G(s)=\frac{U_o(s)}{U_i(s)}=-\frac{1}{sRC}$$

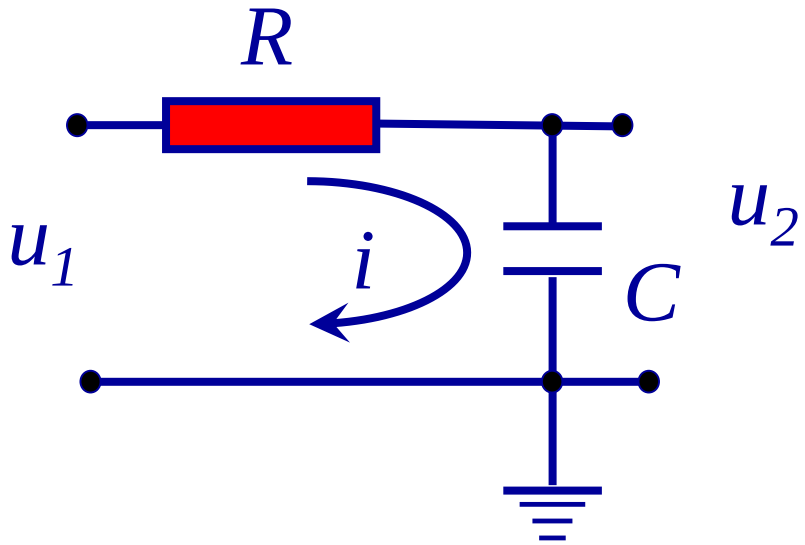


◆无源低通滤波器/ Low pass filter

$$Z_c = 1/(jwC)$$

$$i = C(du_2/dt)$$

$$i = (u_1 - u_2)/R$$



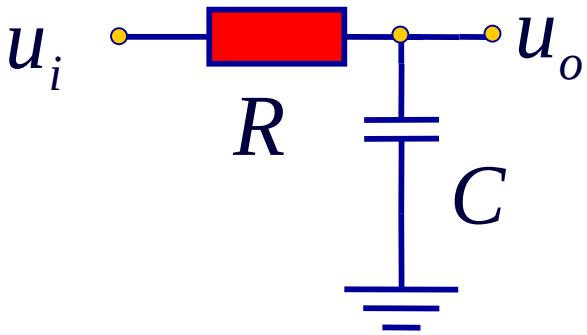
$$G(s) = \frac{U_2(s)}{U_1(s)} = \frac{1}{sRC + 1}$$

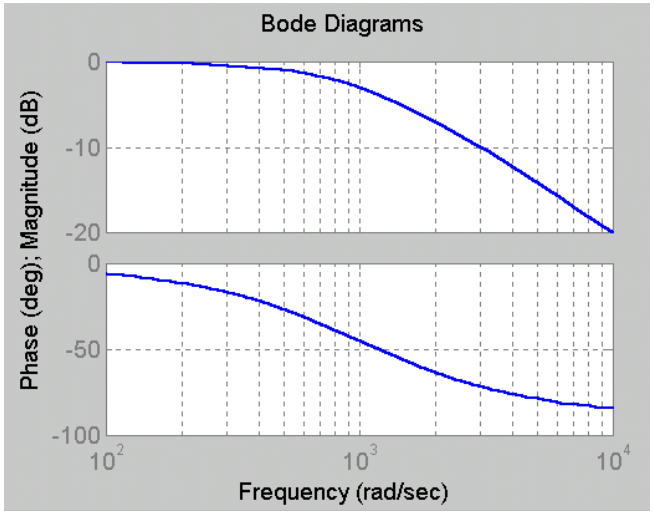


§2 传感信号接口电路
/ Interface circuits of sensor signals

理想接口电路：输入阻抗大、输出阻抗小（输出能力强、带载能力强）

- 大多数器件均涉及电压信号的处理
- 电压信号一般是指开路状况下的电压值，等效电路为开路电压+内阻
- 接口电路输入端：输入电阻较大时，将外部电压信号与接口电路相联，联结点的实际电压衰减不大
- 接口电路输出端：输出电阻较大时，接口电路产生的电压信号与其他接口电路相联，联结点的实际电压衰减不大
- 无源低通滤波电路缺点：输出电阻=R，R较大时输出电阻太大

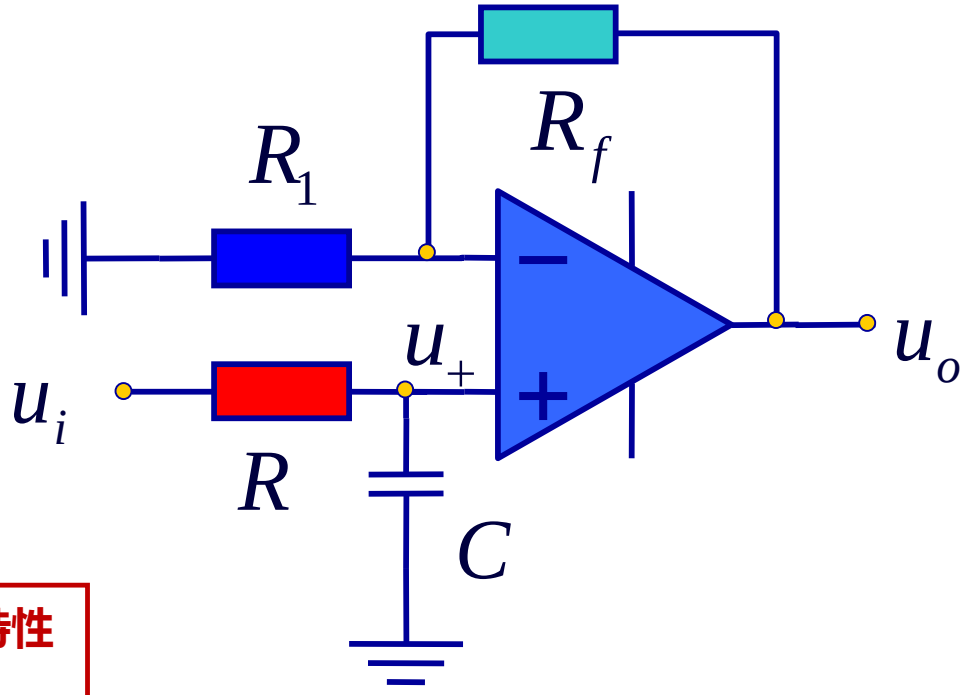






●有源RC滤波/ Active RC filter

- 输入电流小
- / Small input current;
- 可调幅/ Variant Gain
- 输出能力强/ Better Load Capability



与开路电路相比，输出信号不随负载特性发生明显衰退

带有数字接口的模拟滤波器

Configurable analogue filter with digital interface



§2 传感信号接口电路
/ Interface circuits of sensor signals

运放的保护/ Protection of OA

◎限制差模幅值（u_1-u_2）

/ Limits differential mode amplitude

◎限制共模幅值（u_1或u_2）

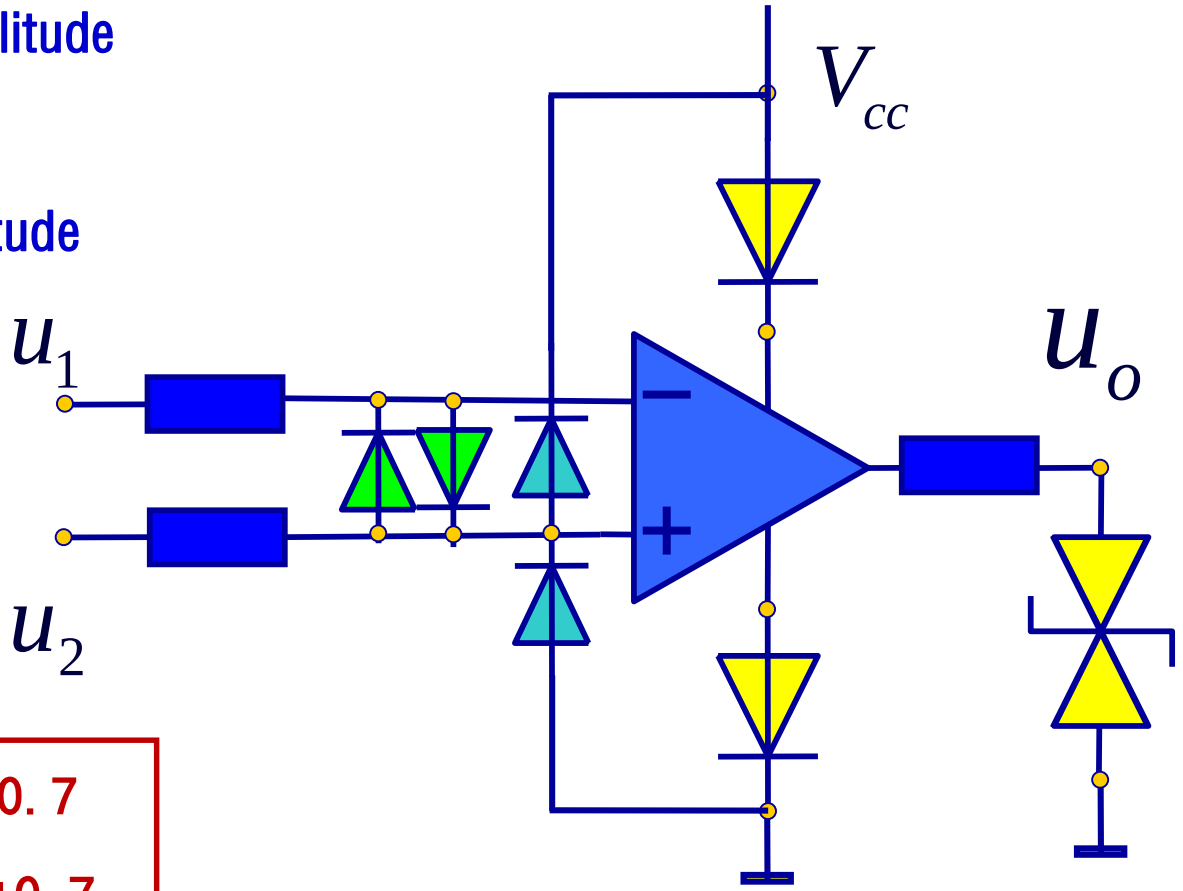
/ Limits common mode amplitude

◎限制输出幅值

/ Limits outputs

◎防止电源反接

/ Prevents inverse power

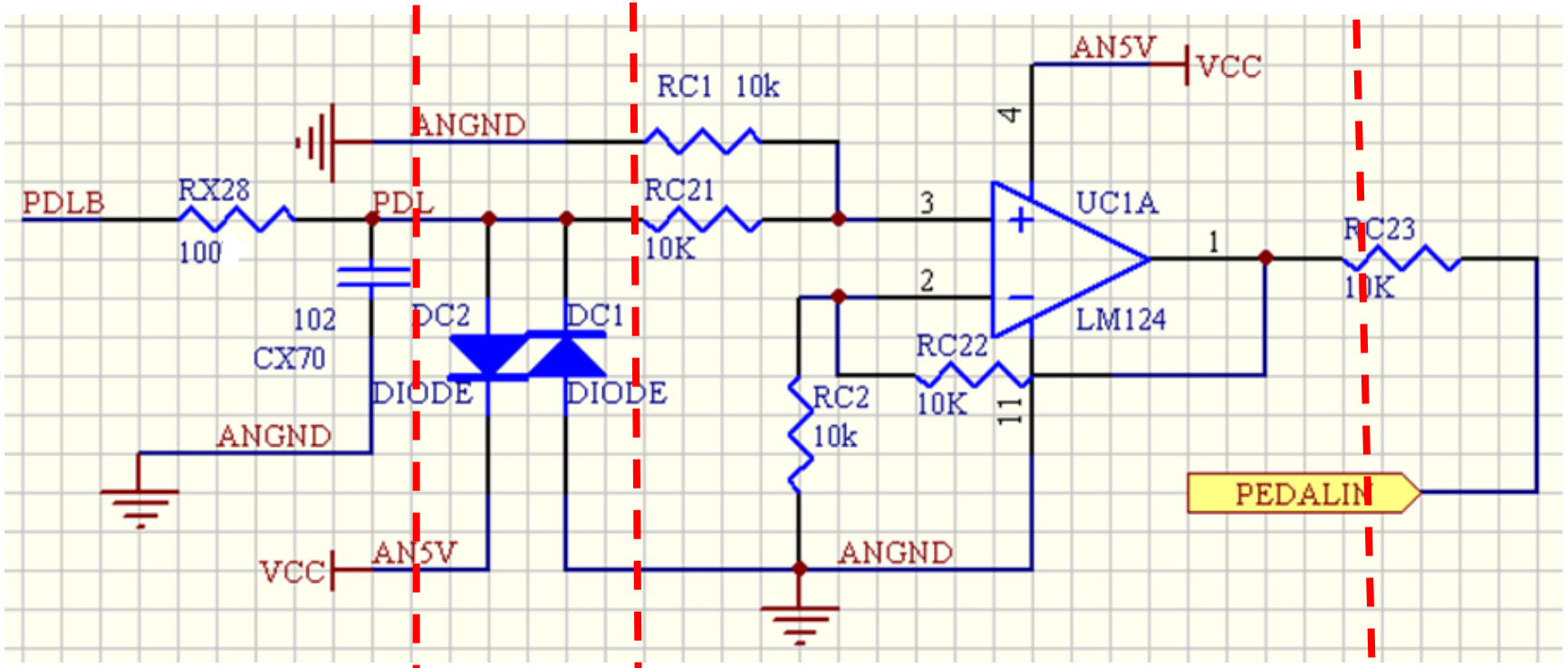


$-0.7 \leqslant u_1-u_2 \leqslant 0.7$

$-0.7 \leqslant u_2 \leqslant V_{cc}+0.7$



§2 传感信号接口电路
/ Interface circuits of sensor signals



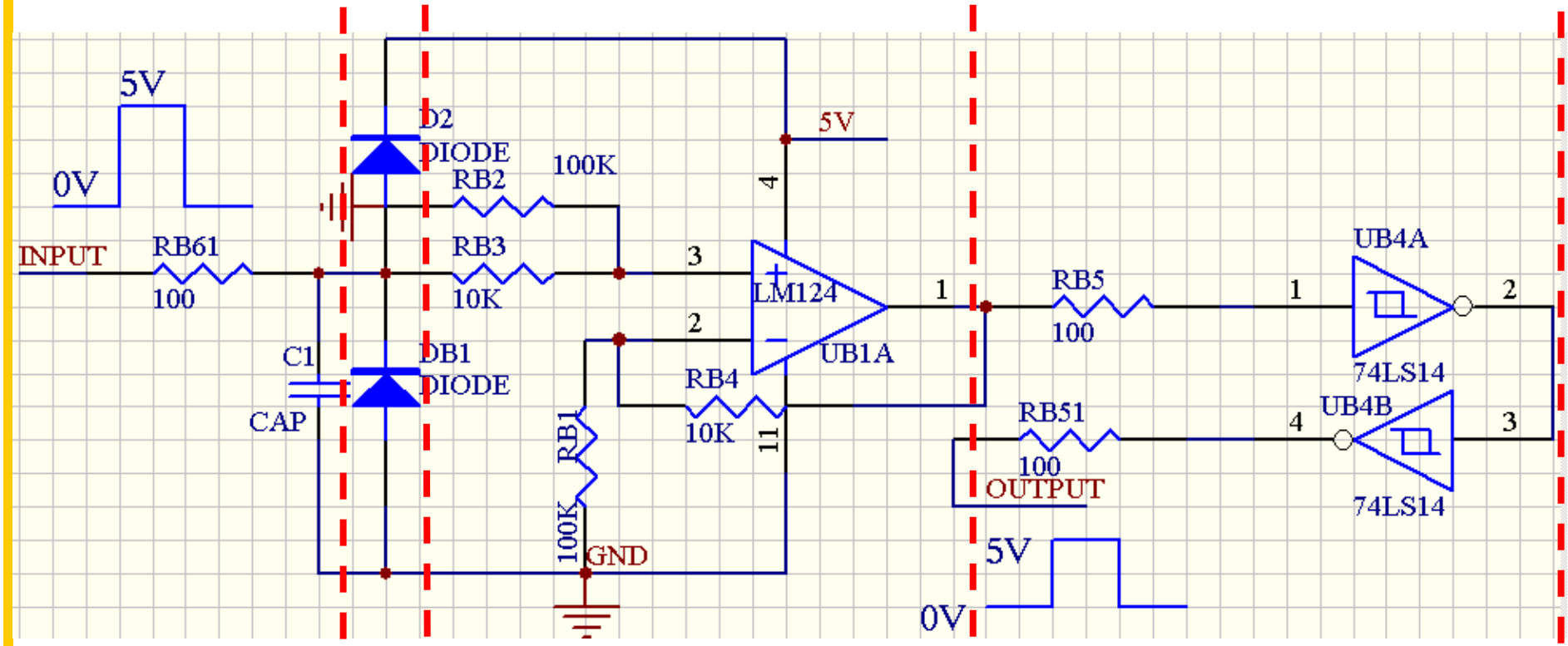
滤波/ Filter

限幅 Limits amplitude

幅值调整/ Tunable gain



§2 传感信号接口电路
/ Interface circuits of sensor signals



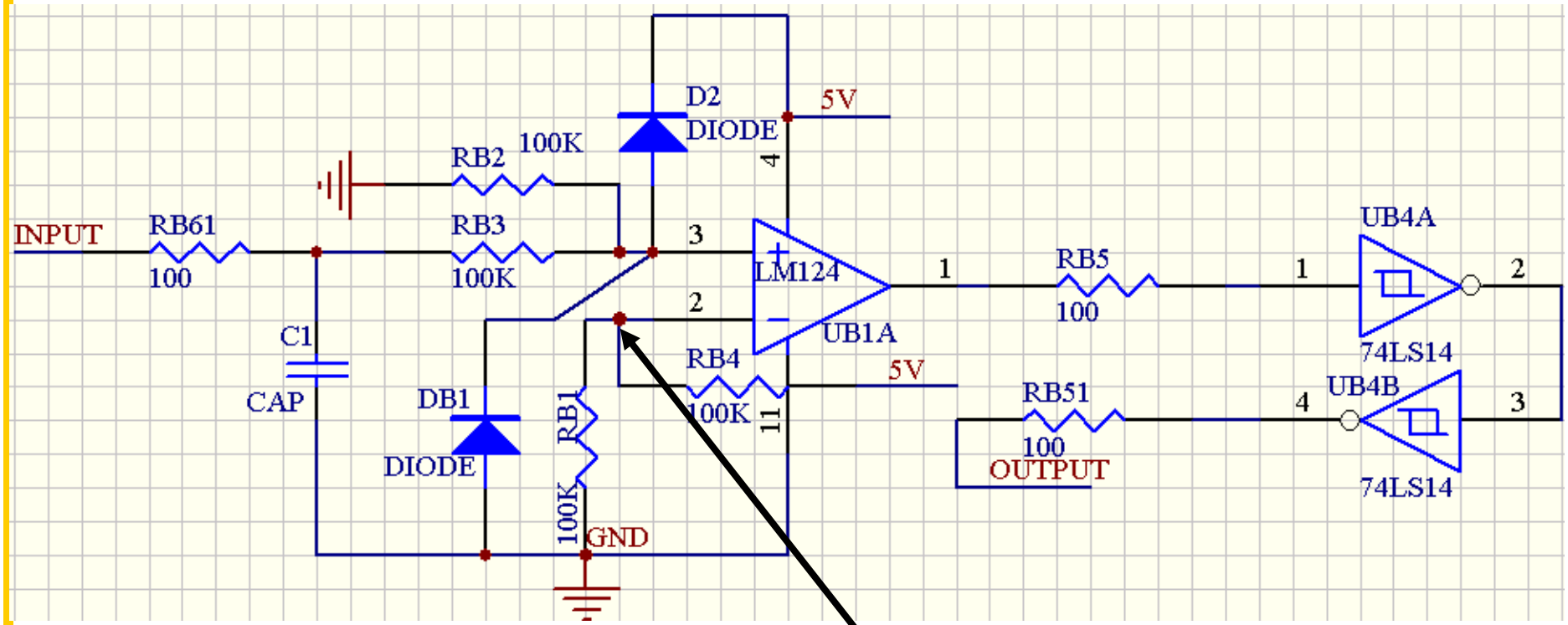
滤波/ 限幅 幅值调整 整形/

Filter Limits / Tunable gain Regular outputs

amplitude



§2 传感信号接口电路
/ Interface circuits of sensor signals

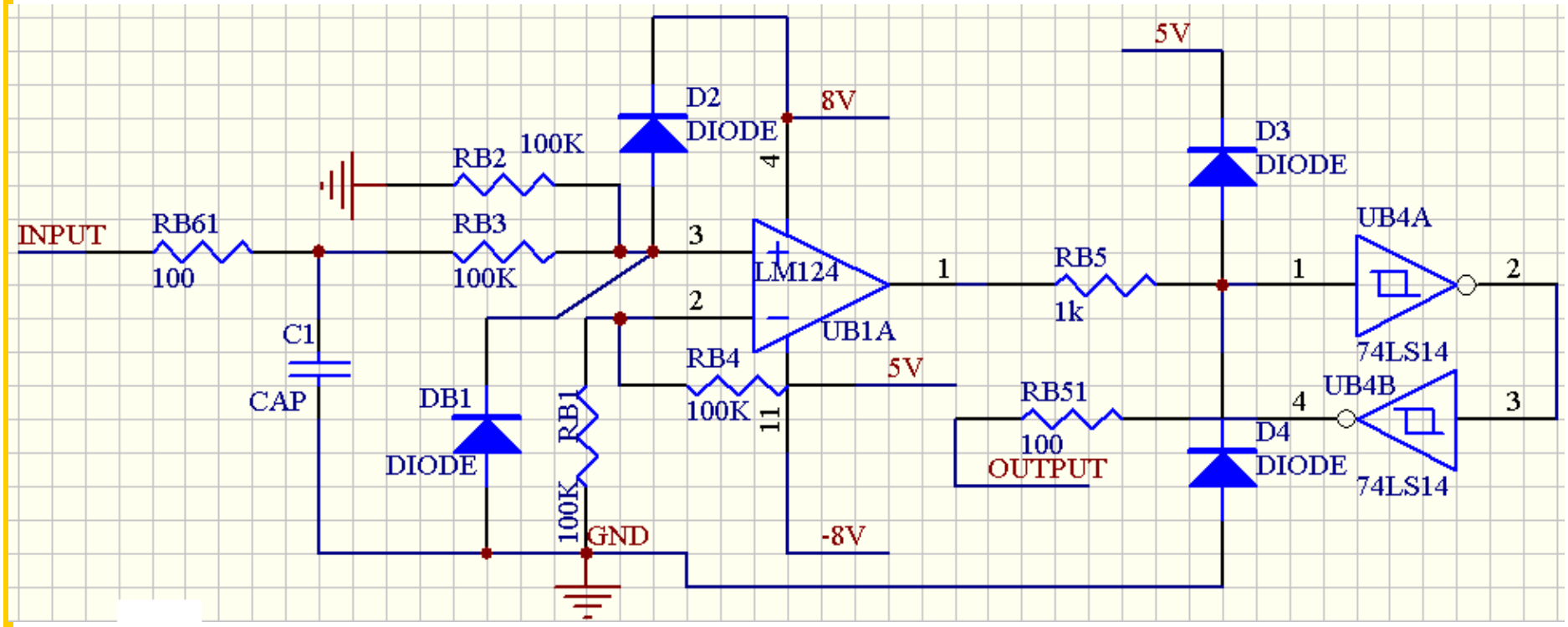


偏置电压/

Offset voltage



§2 传感信号接口电路
/ Interface circuits of sensor signals



采用对称正负电源供电处理电路

Conditioning circuit with Bipolar power supply



§目录 / Index

1. 传感器、信号接口电路及运算放大器
   - ① 传感器工作原理 （Principle of Sensors）
   - ② 传感信号接口电路 (Interface circuits of sensor signals)
   - ③ 典型运算放大器 (Typical Operational Amplifier)
2. 执行器、开关器件及驱动接口电路
   - ① 执行器及四种功率开关器件
   - ② 驱动信号接口电路：四拓扑结构
   - ③ 驱动电路的续流回路
   - ④ 驱动电路的故障诊断和容错控制
   - ⑤ 汽车专用功率驱动集成芯片介绍

重点：运算放大器(Operational Amplifier)和MOSFET的使用



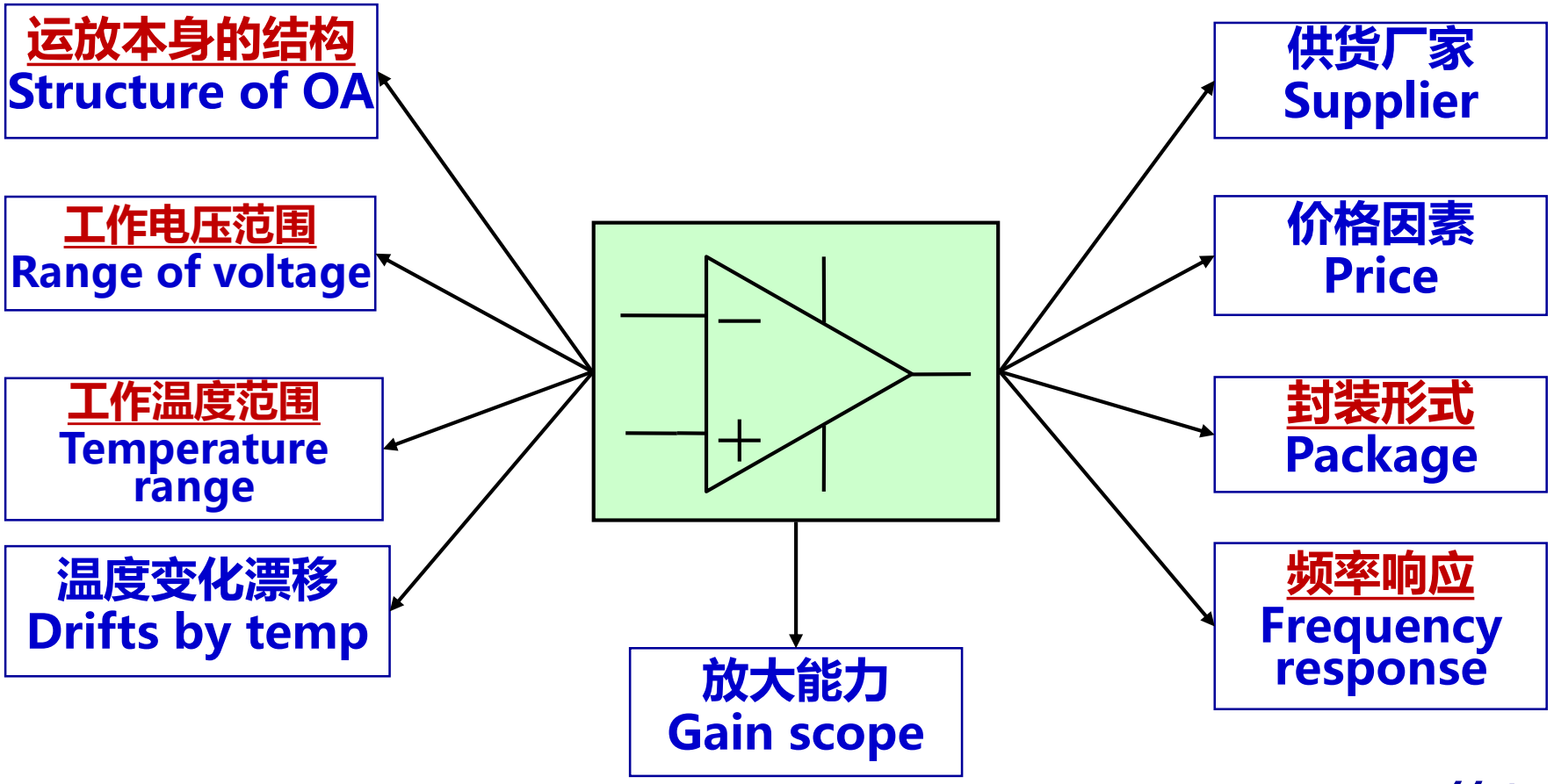
§3 典型运算放大器
/ Typical Operational Amplifier (Op–Amp, OA)

- http://www.ic–on–line.cn/ 器件查询/ IC Query
- http://www.analog.com Analog Devices
- http://www.national.com National Semiconductor

- **更多运放的应用举例：点击这里**
- For more applications, please click here.



工程上如何选择和实用运放/ How to select OA in engineering, 考虑哪些因素/ What facts need Considered?





§3 典型运算放大器
/ Typical Operational Amplifier (Op–Amp, OA)

常用运放的型号：LM系列

LM124： - 55°C ~ 125 °C

LM224： - 25°C ~ 85 °C

LM324： - 0°C ~ 70 °C

(National Semiconductor) →

Motorola:

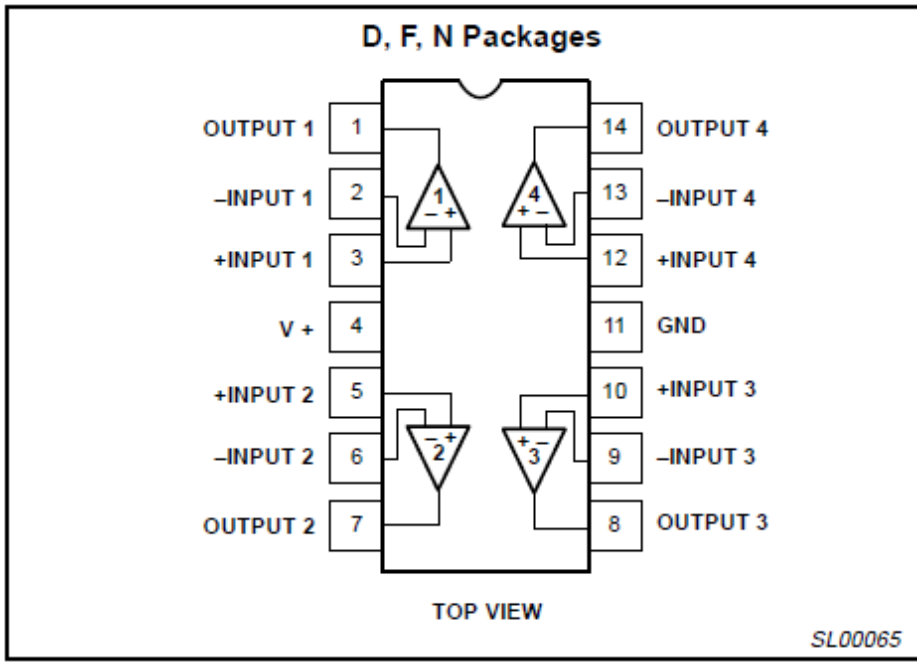
Device	Operating Temperature Range
LM2902D	T_A = –40° to +105°C
LM2902N	
LM2902VD	T_A = –40° to +125°C
LM2902VN	

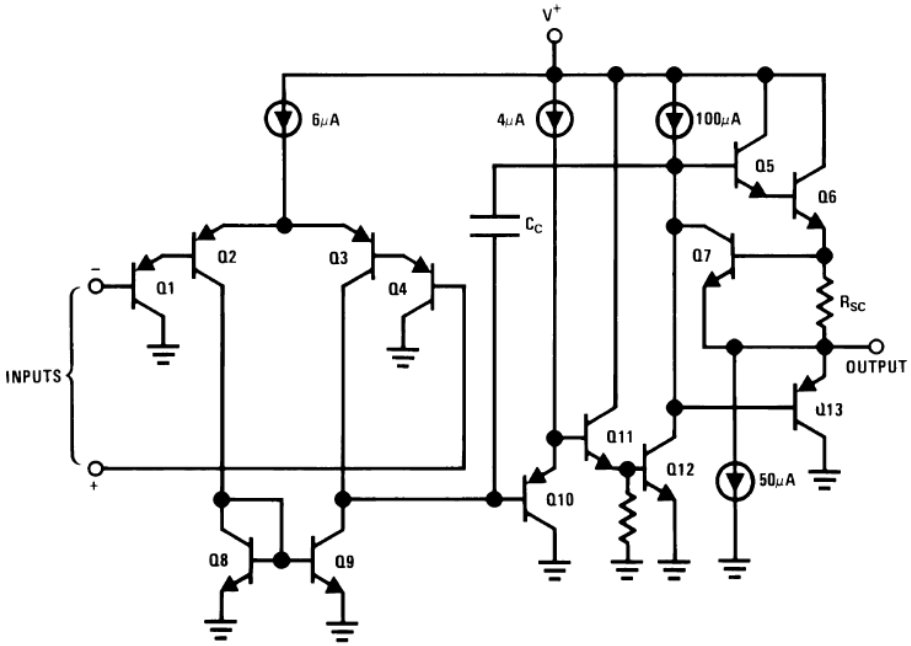
TEMPERATURE RANGE	ORDER CODE
-55°C to +125°C	LM124N
-55°C to +125°C	LM124F
-25°C to +85°C	LM224N
-25°C to +85°C	LM224F
-25°C to +85°C	LM224D
0°C to +70°C	LM324N
0°C to +70°C	LM324F
0°C to +70°C	LM324D
0°C to +70°C	LM324AN
0°C to +70°C	LM324AD
-40°C to +85°C	SA534N
-40°C to +85°C	SA534F
-40°C to +85°C	SA534D
-40°C to +125°C	LM2902D
-40°C to +125°C	LM2902N



LM324

LM324





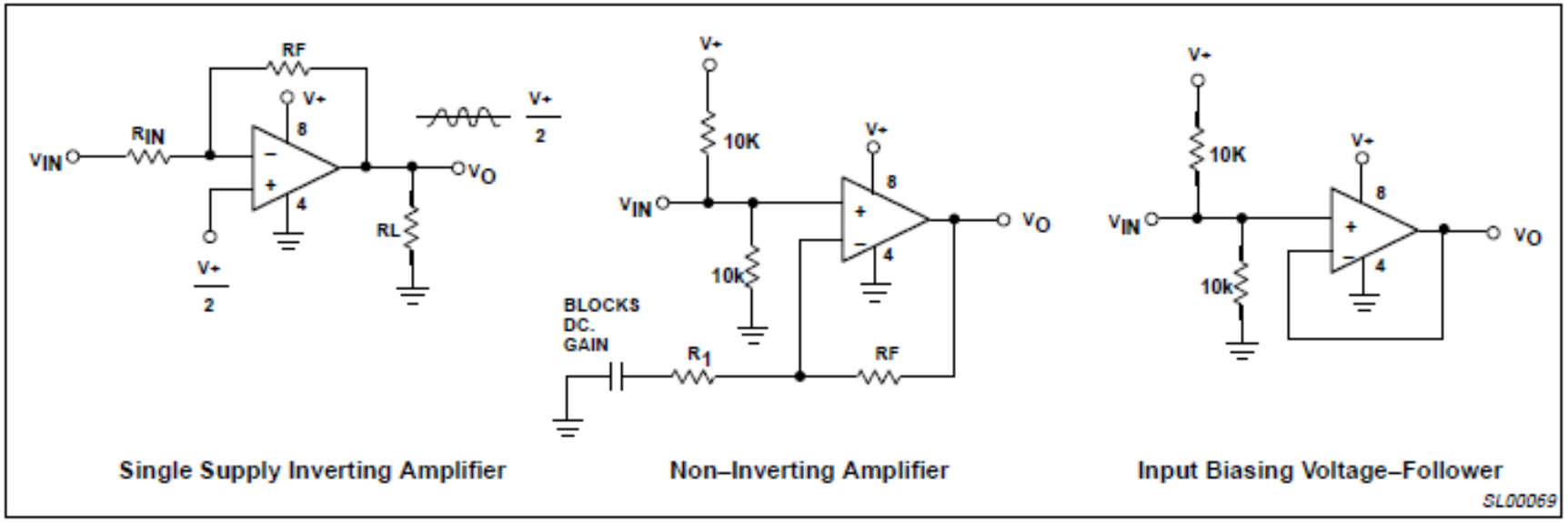
LM324工作电压：3～32V，
或者±1.5～16V；



LM324

LM324

TYPICAL APPLICATIONS

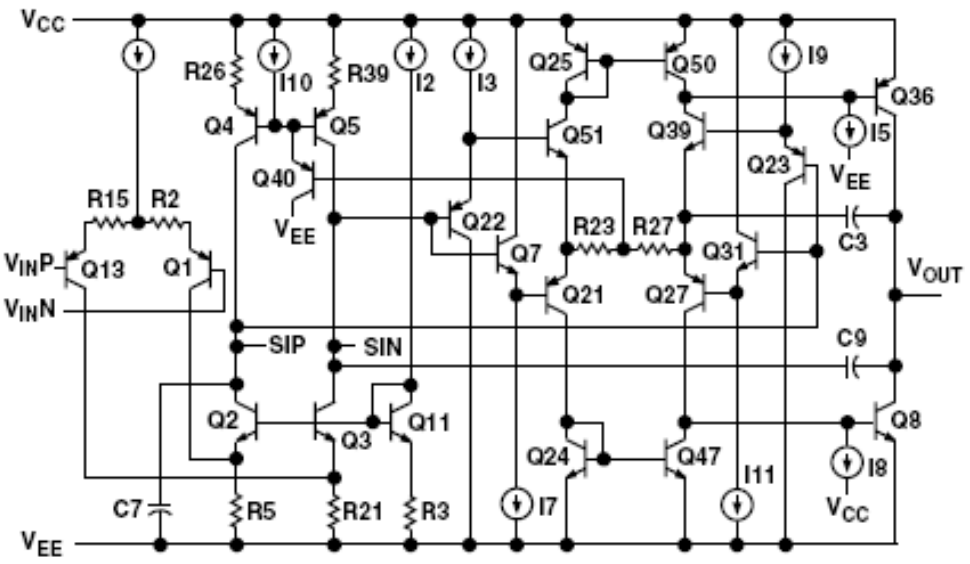


LM324工作电压：3～32V，
或者±1.5～16V；



AD8051

AD8051

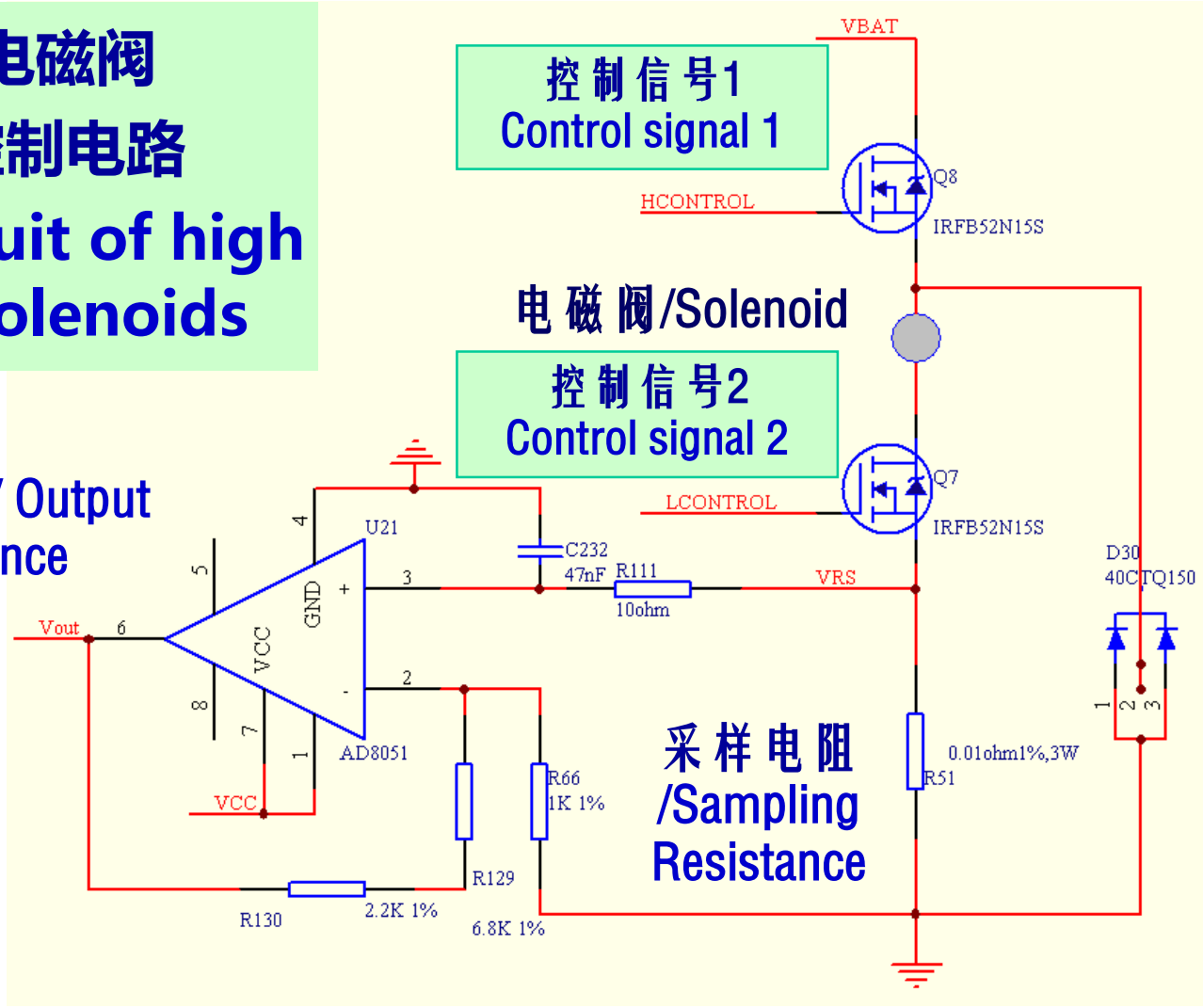


AD8051工作电压
3～12V，或者±1.5～6V。

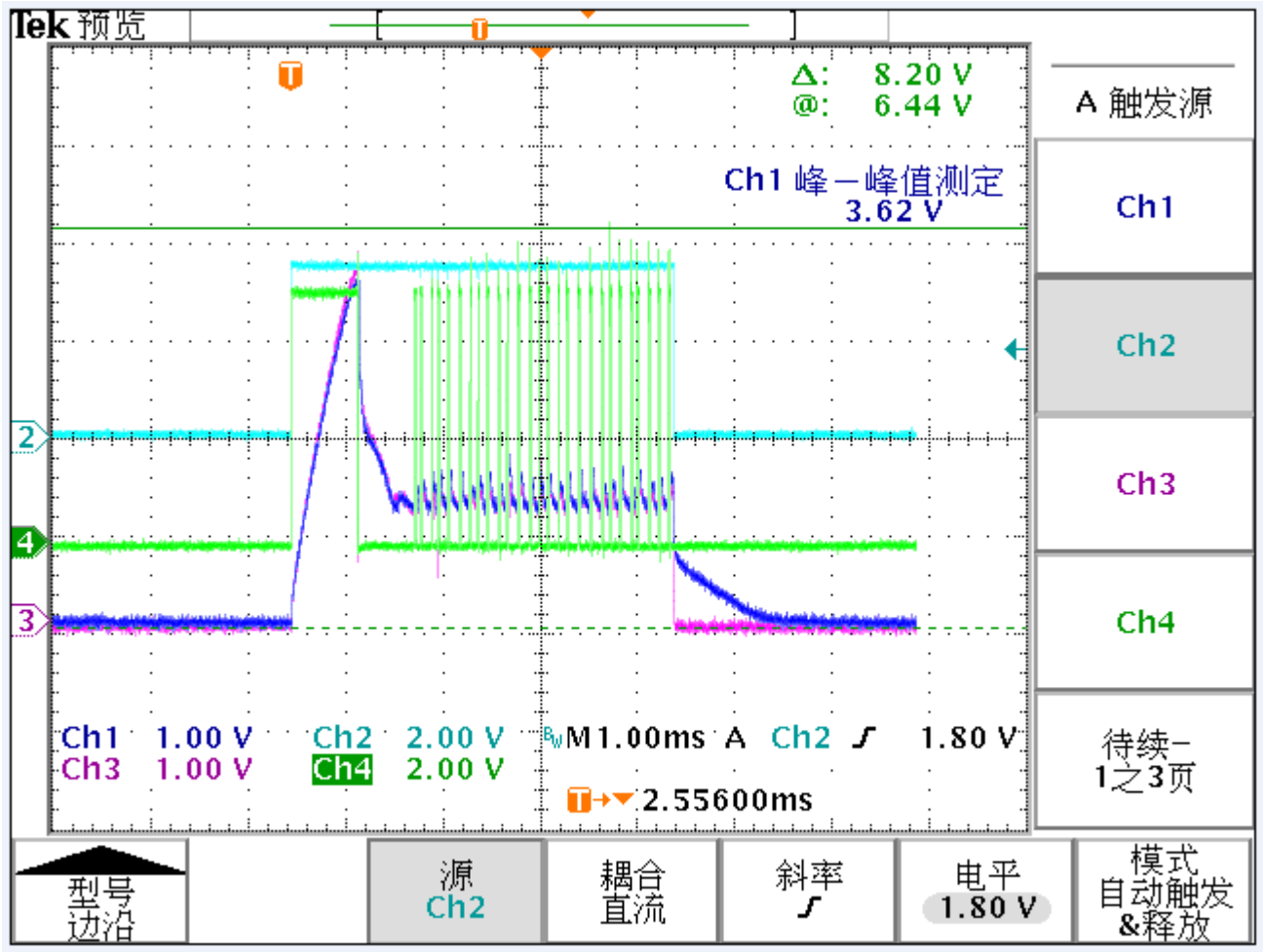
The rail-to-rail output range of the AD8051 is provided by a complementary common-emitter output stage. High output drive capability is provided by injecting all output stage pre-driver currents directly into the bases of the output devices Q8 and Q36. Biasing of Q8 and Q36 is accomplished by I8 and I5, along with a common-mode feedback loop (not shown). This circuit topology allows the AD8051 to drive 45 mA of output current and allows the AD8054 to drive 30 mA of output current with the outputs within 0.5 V of the supply rails.

AD8051的应用：高速电磁阀驱动控制电路

高速电磁阀
驱动控制电路
Drive circuit of high speed solenoids



AD8051的应用：高速电磁阀驱动控制电路



LM2902输出
LM2902 Output

控制信号1
Control signal1

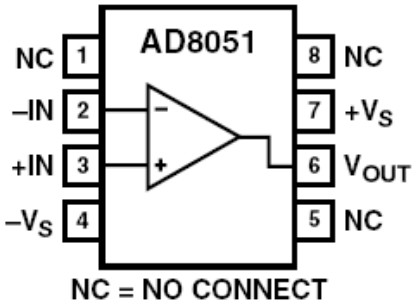
AD8051输出
AD8051 Output

控制信号2
Control Signal2

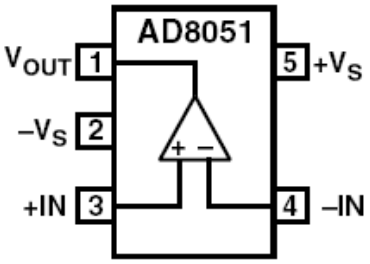


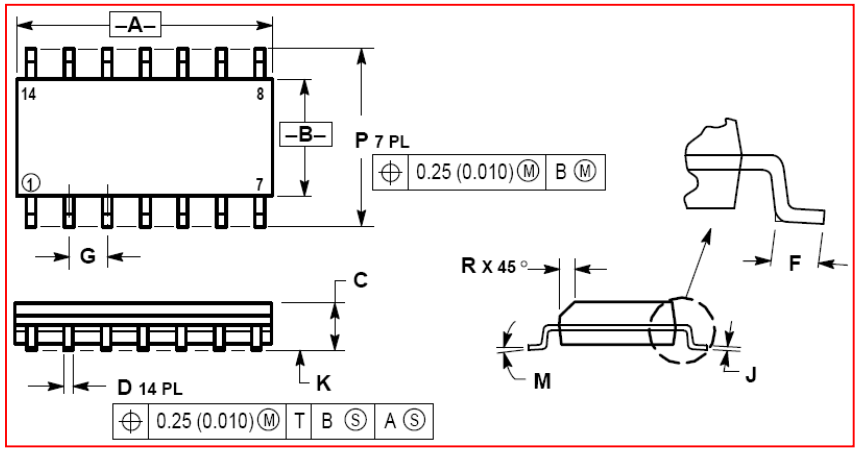
AD805X的封装形式/AD805x Package

RN-8

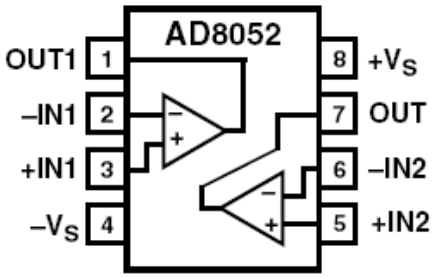


SOT-23-5 (RT)

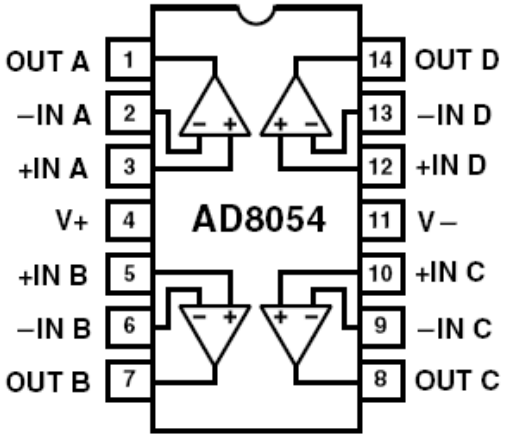


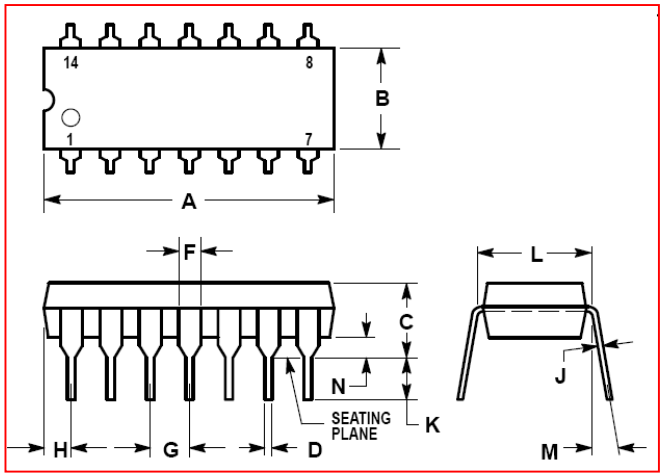


RN-8, MSOP (RM)



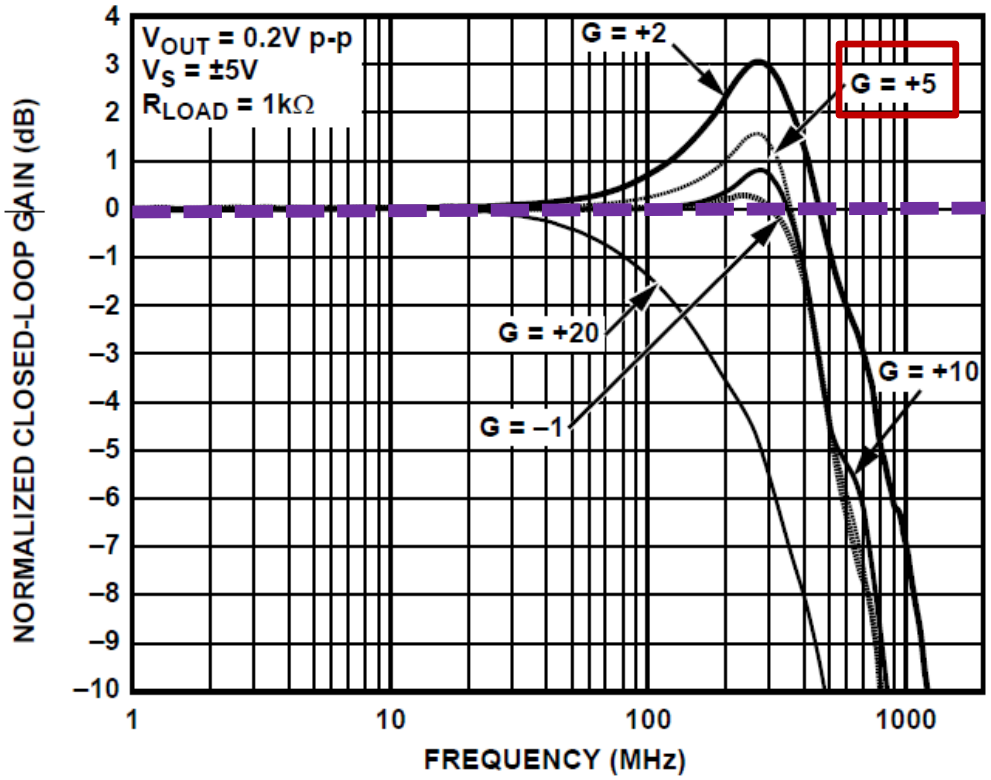
RN-14, TSSOP-14 (RU-14)







运放的频率响应的对比/ Comparison of Frequency Curve



G实质指的是G(0)，即直流电压增益

A(0)=0dB

理想的正则化闭环增益曲线为一条0dB的横线

AD8099

Figure 5. Small Signal Frequency Response for Various Gains (SOIC)

运放电路的实际闭环增益G会随着输入信号频率发生变化：G=G(f)
理想闭环增益：G=G(f)=G(0)
正则化闭环增益(dB)：$A(f)=20\lg(G_n(f))=20\lg(G(f)/G(0))$



运放的频率响应的对比/ Comparison of Frequency Curve

- 选择哪个合适/Which is better when f=200MHz，G=2?
- 选择哪个合适/Which is better when f=100MHz，G=10?

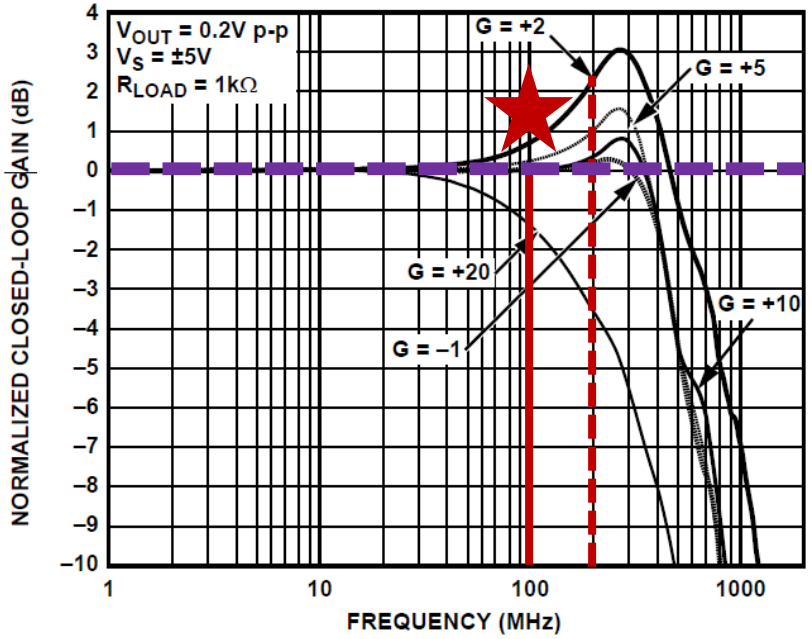


Figure 5. Small Signal Frequency Response for Various Gains (SOIC)

AD8099

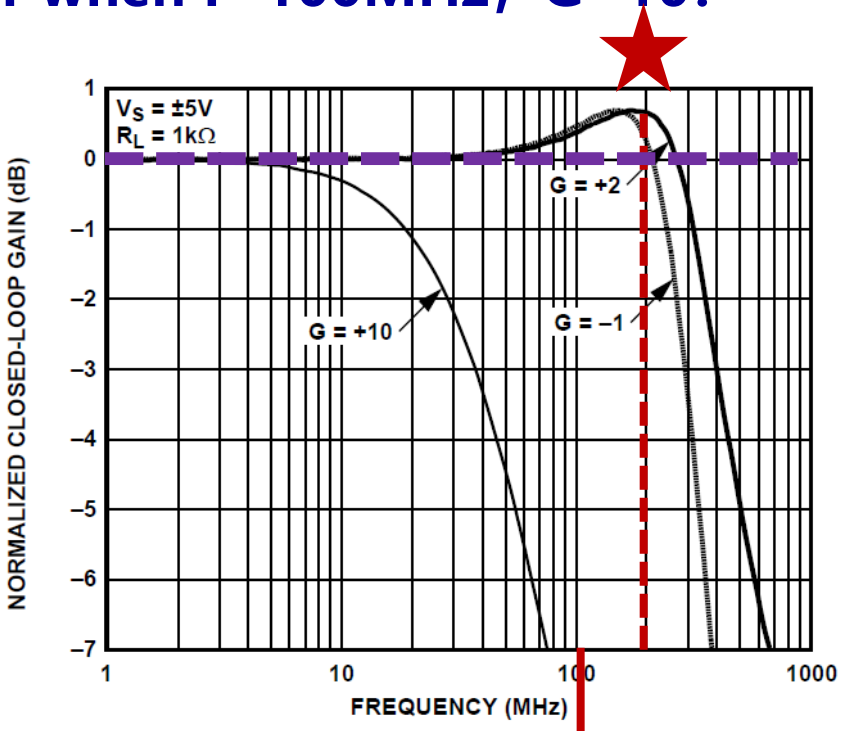


Figure 7. Small Signal Frequency Response for Various Gains

AD8045

闭环增益G
理想闭环增益：不随输入信号频率变化



带有数字接口的模拟滤波器
Configurable analogue filter with digital interface

- 优点/ Advantage

  可以实现复杂的滤波功能/ More complex filtering function

  可以由微控制器实时修改滤波参数/ Filter parameter can be modified by MCU online

- 缺点：代价较高，电路复杂

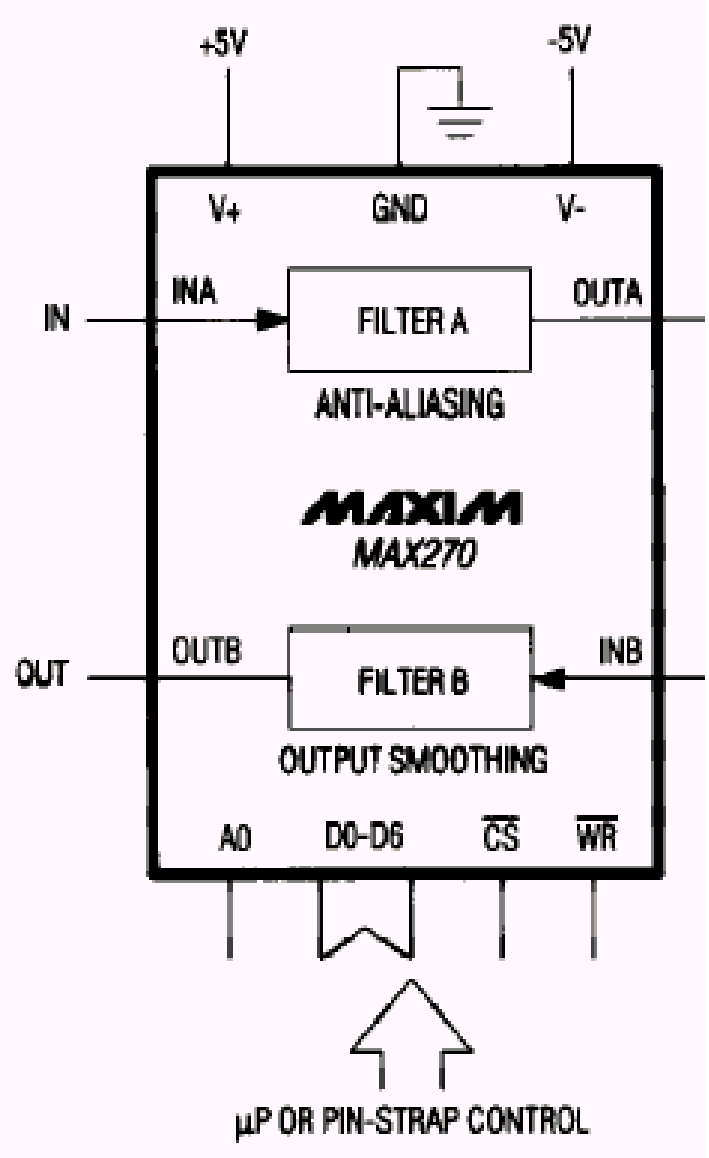
/ Drawbacks: High cost and high complex circuit

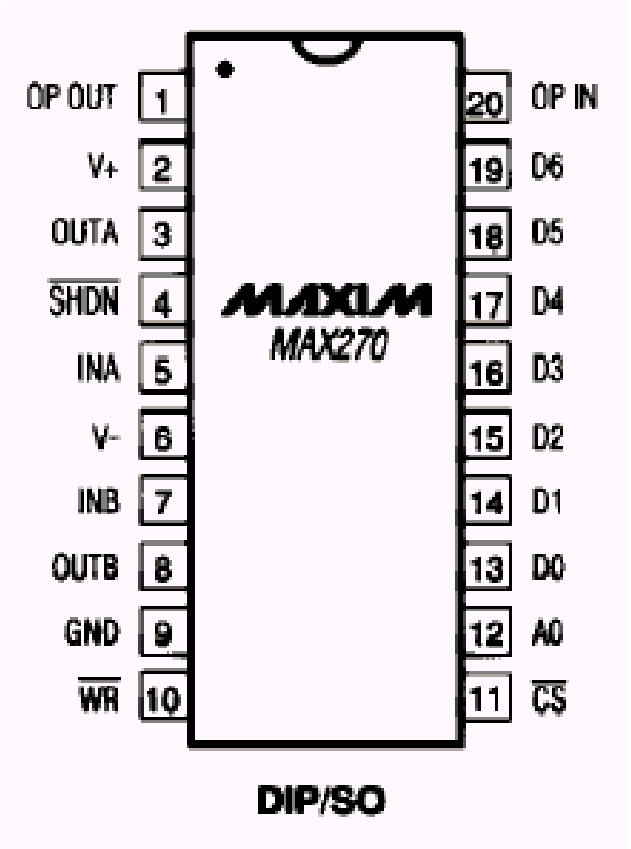
- 用于特殊的滤波器设计/Applied in special filter design
- 举例/Example: MAX270



带有数字接口的模拟滤波器

Configurable analogue filter with digital interface



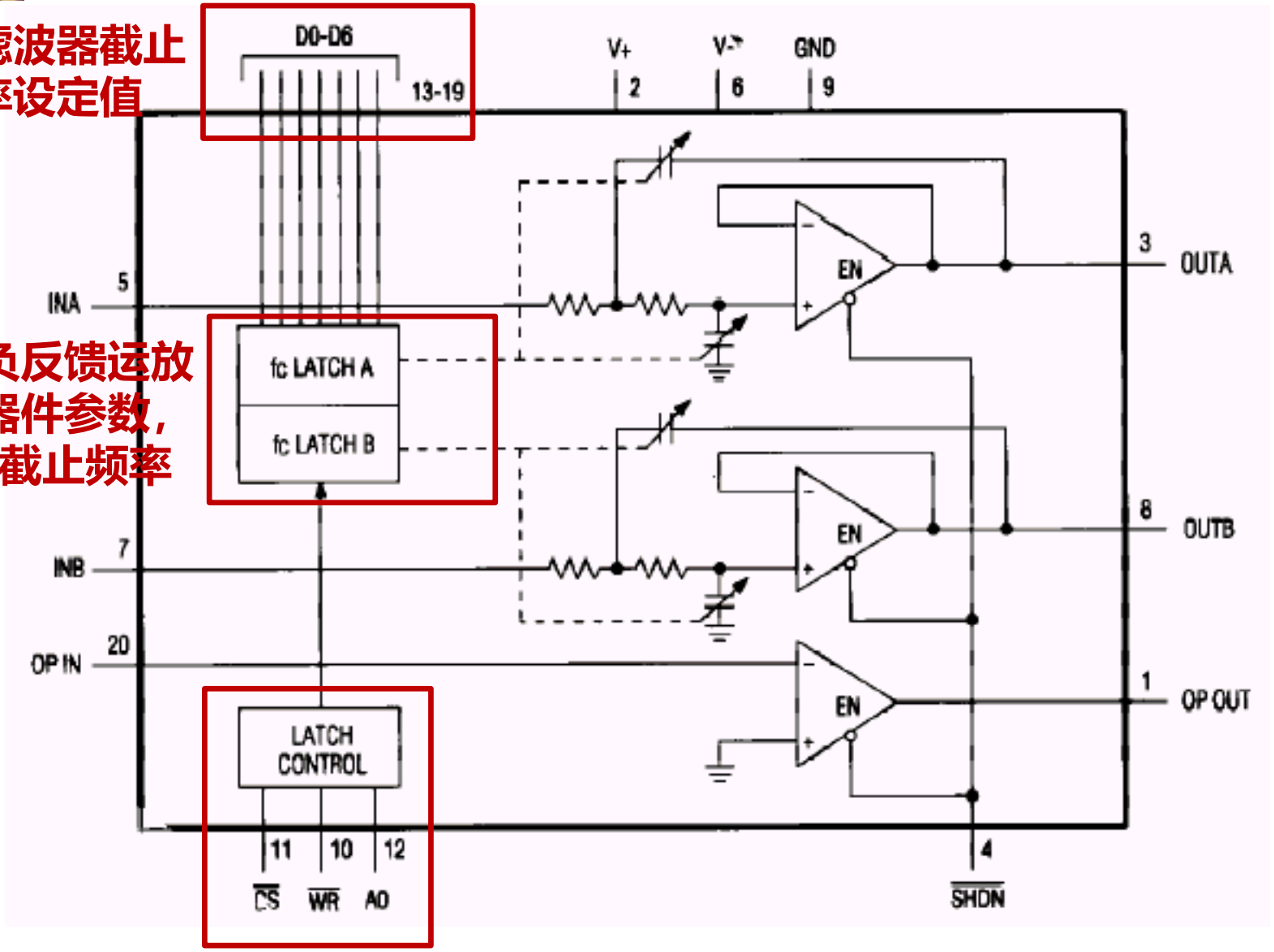




带有数字接口的模拟滤波器
Configurable analogue filter with digital interface

低通滤波器截止频率设定值

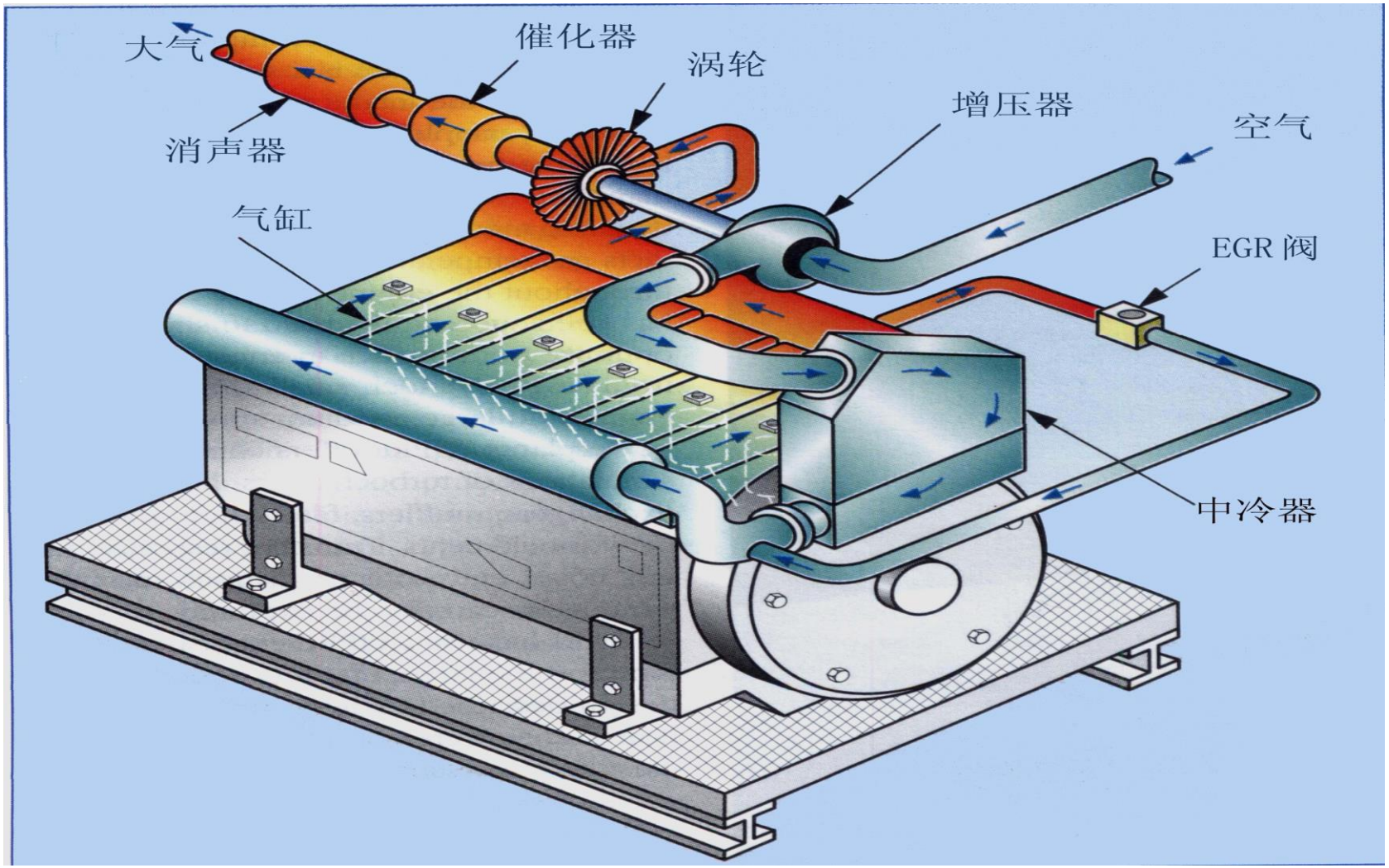
控制负反馈运放电路器件参数，修改截止频率



写入控制



带有数字接口的模拟滤波器
Configurable analogue filter with digital interface





带有数字接口的模拟滤波器
Configurable analogue filter with digital interface

- 增压压力传感器/Pressure sensor for IC——MAX4250
- 增压压力信号的成分/ Signal Characteristics

(1) 平均压力:需要反馈的量/ Average pressure: Component need detected.

(2) 循环内的压力波动:由进气的 脉动效应产生,要滤除/ Fluctuations in one engine cycle: generated by inlet pulse, is necessary to filter.

(3) 量测和传输噪声，要滤除/ Noise of measurement and transmitting, is necessary to filter.



小结

- 掌握双通道电位计的降噪和诊断原理
- 掌握霍尔效应原理、科里奥利力原理
- 理解开关式霍尔传感器、角位移霍尔传感器、磁电式转速传感器、磁电式位置反馈传感器（德国BOSCH）原理
- 掌握1/4、1/2惠斯通桥和惠斯通全桥
- 理解电容、电阻传感器、MEMS传感器（电容式加速度计、陀螺仪原理）
- 理解宽/窄范围氧传感器、氮氧化物传感器、热线式/热膜式空气流量计原理
- 了解基于光干涉法的缸内压力传感器、光电编码方向盘转角传感器、雨滴传感器原理
- 掌握集成运算放大器基本知识：同相反相、线性/饱和、正/负反馈电路、开环增益、闭环增益、虚短虚断等基本概念
- 掌握集成运算放大器同/反相比例电路、无源滤波电路、有缘滤波电路
- 掌握典型传感器信号接口电路分析
- 熟悉典型运算放大器芯片LM324、AD8051
- 了解带数字接口的模拟滤波器MAX270



本节结束，谢谢！

Thank you !